

數理統計

蘇南誠老師講義 · Quarto 教學整理版

蘇南誠

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數理統計講義

本講義由蘇南誠老師編授。感謝黃怡婷老師提供原始 講義檔案；本版保留原有公式、例題、圖形與英文統計術語，並加入一致的教學導覽架構。

使用方式

每章依序包含：

1. 學習目標
2. 必備的先備知識
3. 核心內容與原講義例題
4. 本章摘要
5. 常見錯誤
6. 章末練習

！ 編輯原則

目前版本以「忠實轉入原教材 + 建立一致骨架」為主。後續逐章修訂時，可補入中文導讀、完整證明、例題解答、課堂活動及難度標記。

課程主線

機率基礎 → 多變量分配 → 常用分配 → 抽樣推論 → 漸近理論 → 最大概似 → 充分性與完備性。

原始教材

Hogg, McKean, and Craig, *Introduction to Mathematical Statistics*, 2019。

1 機率與單變量分配

💡 本章定位

本章保留原有英文術語、公式與例題，並整理為可連續閱讀的教學形式。

學生在大二已使用 Hogg, Tanis & Zimmerman 的 *Probability and Statistical Inference* 學過機率基礎，因此本章**不是重新教授機率論**，而是採取

Recall → Deepen → Connect

的三階段安排：先快速喚醒已學內容，再深化數理統計所需的推導工具，最後連接抽樣分配、收斂與後續推論。標題保留 Hogg, McKean & Craig Chapter 1 的 section numbering，方便對照課本；各節不採等比例講解。

1.1 學習目標

完成本章後，學生應能：

- 以樣本空間與事件描述隨機實驗
- 運用條件機率、全機率公式與 Bayes 定理
- 辨認離散與連續隨機變數的分配
- 計算期望、變異數、動差與動差生成函數
- 使用 Markov、Chebyshev 與 Jensen 不等式
- 由 Chebyshev 不等式推導樣本平均數的依機率收斂

1.2 必備的先備知識

集合符號、基本微積分與有限和。機率法則、條件機率、隨機變數及 pmf / pdf / cdf 視為大二已學內容，課堂中以診斷題與例題快速喚醒。

i 教學節奏

- **Stage 1 — Recall (快速)** : Sections 1.1–1.5，以 60–90 分鐘的 example-driven review 喚醒基本概念；忘記的內容在例題中補回，不重走完整的 Definition–Theorem–Proof 流程。
- **Stage 2 — Deepen (主體)** : Sections 1.6–1.9。1.6 與 1.7 的 pmf / pdf 基礎快速帶過，把時間放在 transformation；1.8 與 1.9 深入處理 $E[g(X)]$ 、moments、variance 與 MGF 的推導。
- **Stage 3 — Connect (橋接)** : Section 1.10。以重要不等式連接 sample mean、Chebyshev、convergence in probability 與後續 mathematical statistics。

Stage 1 — Recall：快速喚醒

學過的，快速喚醒；忘記的，在例題中補回來。

本階段以概念辨識與短例題為主。學生應能迅速回答 $P(A | B)$ 、 $F_X(x) = P(X \leq x)$ 、pmf / pdf 的差異及 cdf 的基本性質；細部證明可依診斷結果選講。

1.3 1.1 Introduction

Definition of experiment

- Random experiments
 - Outcome cannot be predicted with certainty.

Sample space

- Outcome (Sample) space
 - Is the collection of all possible outcomes.
 - Is known.
 - Is denoted as $\mathcal{S}$.

Event

- Event
 - Is a part of the collection of outcomes in $\mathcal{S}$.
 - Is denoted as A .
 - $A \subset \mathcal{S}$.

Frequency and relative frequency

- The probability of an event A is called the chance of A occurring.
- For example, a fair coin is tossed repeatedly n times.
- Count the number of times that event A (Head) occurs.
- The number is called frequency and is denoted as $N(A)$.
- The relative frequency is $N(A)/n$.
 - $n = 10$
 - H T H H T T H H T H
 - H: 6/10; T: 4/10

Features of relative frequency

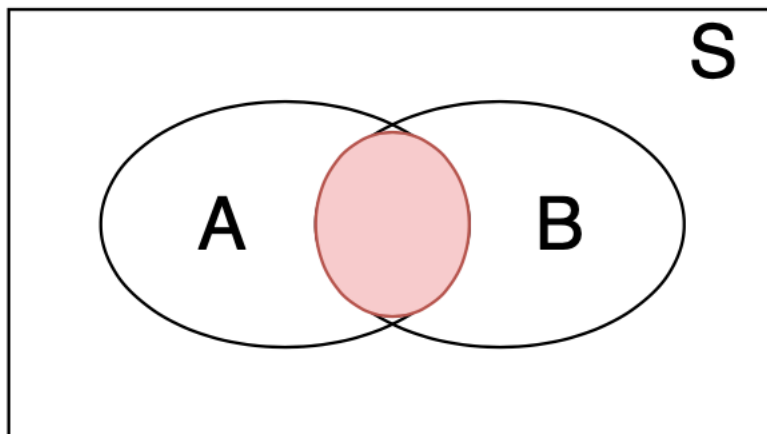
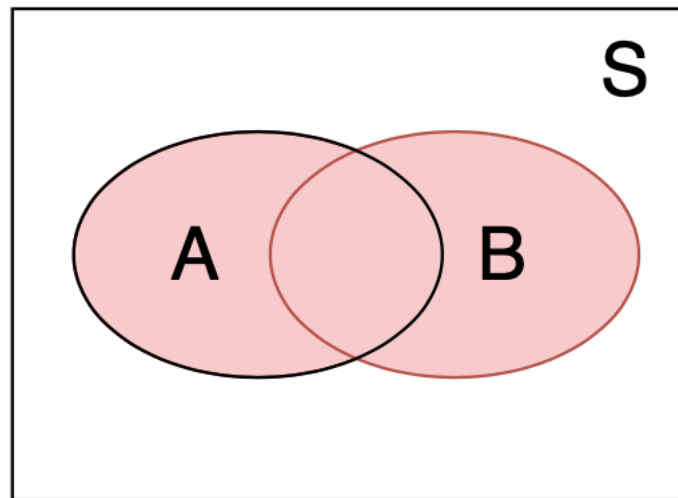
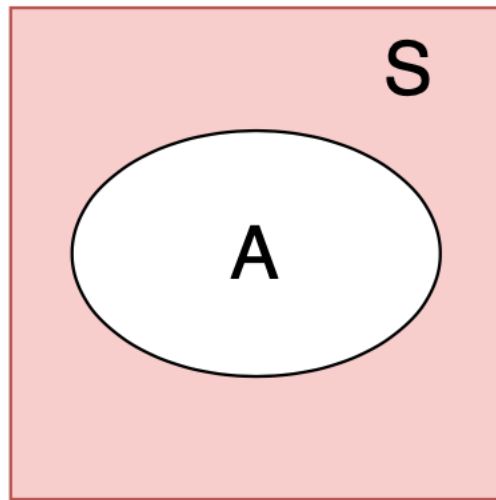
- As n tends to ∞ , relative frequency tends to be a fixed value.
- Suppose this fixed value is p .
- p is considered as the future performance of outcomes (event A) of the experiment.
- Tossing coin experiment n times
 - $S = \{H, T\}$
 - $N(S) = N(H) + N(T) = n$
 - $N(S)/n = 1$

1.4 1.2 Sets

Algebra of sets

- Null or empty set
 - $\emptyset$
- Subset
 - $A \subset B$
- Intersection
 - $A \cap B$
- Union
 - $A \cup B$
- Complement
 - A^C or A'

Venn Diagrams



Properties for set operations

- Commutative law

$$A \cup B = B \cup A \quad A \cap B = B \cap A$$

- Associative law

$$(A \cup B) \cup C = A \cup (B \cup C)$$

$$(A \cap B) \cap C = A \cap (B \cap C)$$

Properties for set operations

- Distributive law

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

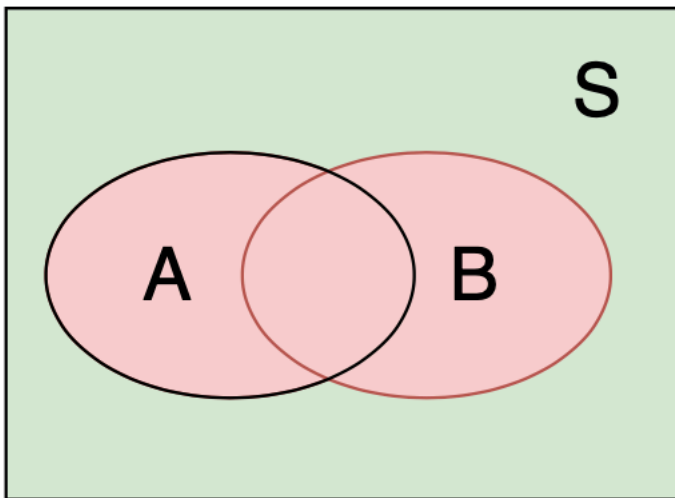
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

- De Morgan's law

$$(A \cup B)' = A' \cap B'$$

$$(A \cap B)' = A' \cup B'$$

Venn Diagrams



More than 2 sets

- Let $A_1, A_2, \dots, A_n$ be any sets.

- Define

$$- A_1 \cup A_2 \cup \dots \cup A_n = \{x : x \in A_i, \text{ for some } i = 1, 2, \dots, n\} = \bigcup_{i=1}^n A_i.$$

$$- A_1 \cap A_2 \cap \dots \cap A_n = \{x : x \in A_i, \text{ for all } i = 1, 2, \dots, n\} = \bigcap_{i=1}^n A_i.$$

Countable sets

- Let $A_1, A_2, \dots, A_n, \dots$ be a sequence of sets.
- Define
 - $A_1 \cup A_2 \cup \dots \cup A_n \cup \dots = \{x : x \in A_i, \text{ for some } i = 1, 2, \dots, n, \dots\} = \bigcup_{n=1}^{\infty} A_n.$
 - $A_1 \cap A_2 \cap \dots \cap A_n \cap \dots = \{x : x \in A_i, \text{ for all } i = 1, 2, \dots, n, \dots\} = \bigcap_{n=1}^{\infty} A_n.$

Supplementary Example 1.2-A

- Let $\mathcal{S} = \{1, 2, 3, \dots\}.$
- Let $A_n = \{1, 3, \dots, 2n - 1\}$ and $B_n = \{n, n + 1, \dots\}$ for $n = 1, 2, 3, \dots$
- Find $\bigcup_{n=1}^{\infty} A_n, \bigcap_{n=1}^{\infty} A_n, \bigcup_{n=1}^{\infty} B_n, \bigcap_{n=1}^{\infty} B_n.$

Monotone (nondecreasing) sets

- Let $A_1, A_2, \dots, A_n, \dots$ be a sequence of nondecreasing sets, i.e.,

$$A_n \subset A_{n+1}, n = 1, 2, 3, \dots$$

- Define

$$\lim_{n \rightarrow \infty} A_n = \bigcup_{n=1}^{\infty} A_n.$$

Monotone (nonincreasing) sets

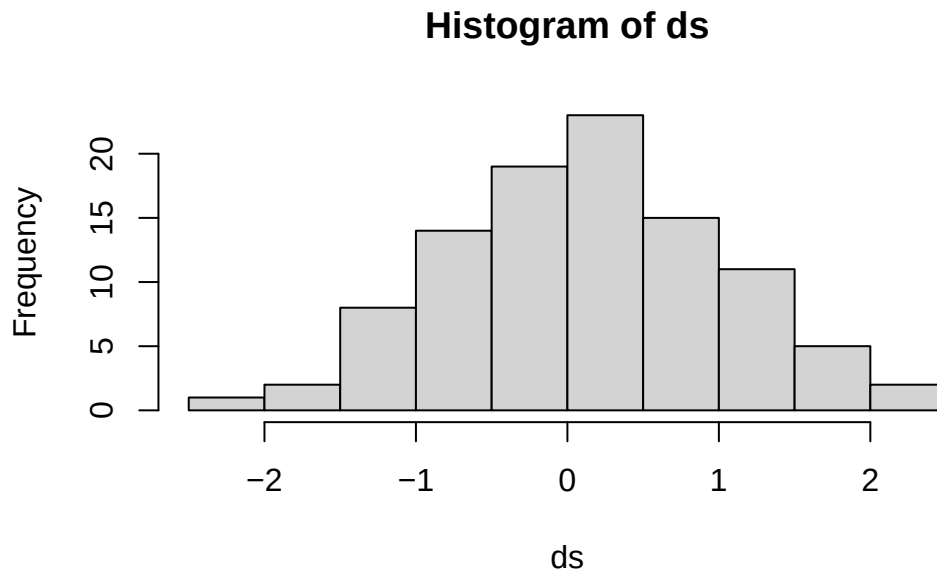
- Let $A_1, A_2, \dots, A_n, \dots$ be a sequence of nonincreasing sets, i.e.,

$$A_n \supset A_{n+1}, n = 1, 2, 3, \dots$$

- Define

$$\lim_{n \rightarrow \infty} A_n = \bigcap_{n=1}^{\infty} A_n.$$

```
ds <- rnorm(100)
hist(ds)
```



Set functions

- Set function (集合関数)
 - Is a function that map sets into real numbers.

Common set functions

- Suppose $\mathcal{S}$ be a set of real number.
 - $Q(A) = \int_A f(x)dx$
 - $Q(A) = \int \int_A g(x, y)dxdy$
- Suppose $\mathcal{C}$ be a set of integers.
 - $Q(A) = \sum_{x \in A} f(x)$
 - $Q(A) = \sum \sum_{(x, y) \in A} g(x, y)$

Supplementary Example 1.2-B

- Suppose $\mathcal{S}$ be the interval of positive real numbers, i.e., $\mathcal{S} = (0, \infty)$.
- Let A be a subset of $\mathcal{S}$.
- Define
 - $Q(A) = \int_A e^{-x}dx$, provided the integral exists.
- Find $Q((1, 3))$.

- Find $Q((5, \infty))$.
- Find $Q((1, 3) \cup [3, 5))$.
- Find $Q(\mathcal{S})$.

1.5 1.3 Probability Set Function

Recall frequency

- The probability of an event A equals p .
- Denote as $P[A]$.
- $P[A]$ is a set function that associates A with the relative frequency.

Definition of Probability

定義 1.1.

- Let $\mathcal{S}$ be a sample space.
- Let $\mathcal{B}$ be the set of events.
- Let P be a real-valued set function defined on $\mathcal{B}$.
- P is a probability set function if it satisfies
 - $P[A] \geq 0$, for all $A \in \mathcal{B}$.
 - $P[\Omega] = 1$
 - If $\{A_n\}$ is a sequence of events in $\mathcal{B}$ and $A_m \cap A_n = \emptyset$ for all $m \neq n$, then

$$P[A_1 \cup A_2 \cup A_3 \cup \dots] = \sum_{i=1}^{\infty} P[A_i]$$

Special terminology

- Mutually exclusive events
 - $A_i \cap A_j = \emptyset, \forall i \neq j$.
 - $A_1, \dots, A_k$ are disjoint sets.
 - The union is referred to as a disjoint union.
- Exhaustive events, if
 - $A_i \cap A_j = \emptyset, \forall i \neq j$.
 - $\cup_{n=1}^{\infty} A_n = \mathcal{S}$.

Special terminology

- Mutually exclusive and exhaustive events
 - $A_i \cap A_j = \emptyset, \forall i \neq j.$
 - $A_1, \dots, A_k$ are disjoint sets.
 - $A_1 \cup A_2 \cup \dots A_k = S.$
- It is a partition of $\mathcal{S}$.

Theorem

定理 1.1. For each event $A \in \mathcal{B}$,

$$P[A] = 1 - P[A']$$

Theorem

定理 1.2.

$$P[\emptyset] = 0$$

Theorem

定理 1.3. If events A and B are such that $A \subset B$, then

$$P[A] \leq P[B].$$

Theorem

定理 1.4. For each event $A \in \mathcal{B}$,

$$P[A] \leq 1.$$

Theorem

定理 1.5. If A and B are any two events in $\mathcal{S}$, then

$$P[A \cup B] = P[A] + P[B] - P[A \cap B].$$

Theorem

定理 1.6. If A , B and C are any three events, then

$$\begin{aligned} P[A \cup B \cup C] = & P[A] + P[B] + P[C] - P[A \cap B] - P[A \cap C] \\ & - P[B \cap C] + P[A \cap B \cap C]. \end{aligned}$$

Example 1.3.1

- Consider one red die and one white die.
- Let the outcome be the ordered pair.
- Assume these two dice is repeatedly cast under the same condition.
- The cast of this pair of dice is a random experiment.
- This is a random experiment.
- Find the sample space $\mathcal{S}$.

Example 1.3.1 (continued)

- Assume the probability set function assigns a probability of $1/36$ to each of the 36 points in $\mathcal{S}$.
- If $C_1 = \{(1, 1), (2, 1), (3, 1), (4, 1), (5, 1)\}$ and $C_2 = \{(1, 2), (2, 2), (3, 2), (4, 2), (5, 2)\}$.
- Find $P[C_1]$, $P[C_2]$, $P[C_1 \cup C_2]$ and $P[C_1 \cap C_2]$.

Inclusion exclusion formula

- Consider a sequence of events $\{C_n, n = 1, 2, \dots\}$.
- Let $p_1 = P[C_1] + P[C_2] + P[C_3]$.
- Let $p_2 = P[C_1 \cap C_2] + P[C_1 \cap C_3] + P[C_2 \cap C_3]$.
- Let $p_3 = P[C_1 \cap C_2 \cap C_3]$.
- Then

$$P[C_1 \cup C_2 \cup \dots \cup C_k] = p_1 - p_2 + p_3 - \dots + (-1)^{n+1} p_n.$$

Bonferroni's inequality

- Consider two events C_1 and C_2 .
- We have

$$P[C_1 \cap C_2] \geq P[C_1] + P[C_2] - 1$$

Boole's inequality

定理 1.7.

- Let $\{C_n\}$ be an arbitrary sequence of events.
- Then

$$P[\cup_{n=1}^{\infty} C_n] \leq \sum_{n=1}^{\infty} P[C_n].$$

Equally likely principle

- Let $x_i, i = 1, \dots, m$, be all possible outcomes of an experiment.
- Suppose m is the total number of ways that the random experiment can terminate.
- If each of e_i has the same probability of occurring, m outcomes are equally likely.
- We have

$$P[e_i] = \frac{1}{m}, i = 1, 2, \dots, m.$$

- Let $A \in \mathcal{S}$ and define

$$P[A] = \sum_{x_i \in A} \frac{1}{m} = \frac{\#(A)}{m}.$$

Example 1.3.2

- Toss a fair coin twice
- Possible outcomes: TT, HT, TH, HH
- Assume

$$P[TT] = P[HT] = P[TH] = P[HH] = \frac{1}{4}$$

- Consider $A = \{\text{at least one } H\}$

$$P[A] = P[HT + TH + HH] = P[HT] + P[TH] + P[HH] = \frac{3}{4}$$

Counting rule – Multiplication Principle

- Suppose an experiment E_1 has n_1 outcomes.
- Each of possible outcomes in E_1 , an experiment has n_2 outcomes
- The composite experiment $E_1 E_2$ consisting of performing E_1 and then E_2 has $n_1 n_2$ possible outcomes.

More on multiplication Principle

- Suppose experiment E_i has n_i outcomes, $i = 1, 2, \dots, m$.
- The composite experiment $E_1 E_2 \dots E_m$ has $n_1 n_2 \dots n_m$ possible outcomes.

Supplementary Example 1.3-A — Dinner choices

- Choices of dinner
 - E_1 : soup or tomato juice
 - E_2 : steak or shrimp
 - E_3 : French fried potatoes, mashed potatoes or a baked potatoes

- E_4 : corn or pea
- E_5 : jelly, tossed salad, cottage cheese or coleslaw
- E_6 : cake, cookies, pudding, brownie, vanilla ice cream, chocolate ice cream or orange sherbet
- E_7 : coffee, tea, milk or punch
- How many different dinner selections are possible?

Counting methods

- Permutation
- Combination
- Sampling with replacement
- Sampling without replacement
- Ordered sample
- Unordered sample

Common counting rules

- Let A be a set with n elements.
- A permutation of n objects equals $n!$.
- Suppose k -tuples whose components are elements of A , then there are n^k such k -tuples whose components are elements of A .
- Suppose $k < n$ and we are interest in k -tuples whose elements are distinct. Then, the number of k permutations taken from A equals

$$P_k^n = \frac{n!}{(n-k)!}.$$

Common counting rules

- Let A be a set with n elements.
- Suppose $k < n$ and we are interest in k -tuples whose elements are distinct.
- Suppose the order is not important.
- Then, the number of k permutations taken from A equals

$$C_k^n = \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Remark

- $C_r^n = \binom{n}{r}$ is called binomial coefficients.
- The binomial expansion

$$(a + b)^n = \sum_{r=1}^n \binom{n}{r} b^r a^{n-r}$$

Definition

定義 1.2. Each of the

$$C_k^n = \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

permutations of n objects, k of one type and $n - r$ of another type, is called a distinguishable permutation.

Remark

- Suppose there are n objects.
- n_1 are similar, n_2 are similar, $\dots$, n_s are similar.
- $n = \sum_{i=1}^s n_i$, $0 \leq n_i \leq n$.
- The number of distinguishable permutations of the n objects is

$$\binom{n}{n_1, n_2, \dots, n_s} = \frac{n!}{n_1! n_2! \dots n_s!}$$

- It is called multinomial coefficients.

Properties of probability

定理 1.8.

- Let $\{C_n\}$ be a nondecreasing sequence of events.
- Then

$$\lim_{n \rightarrow \infty} P[C_n] = P \left[\lim_{n \rightarrow \infty} C_n \right] = P \left[\bigcup_{n=1}^{\infty} C_n \right].$$

Properties of probability

定理 1.9.

- Let $\{C_n\}$ be a nonincreasing sequence of events.
- Then

$$\lim_{n \rightarrow \infty} P[C_n] = P \left[\lim_{n \rightarrow \infty} C_n \right] = P \left[\bigcap_{n=1}^{\infty} C_n \right].$$

1.6 1.4 Conditional Probability and Independence

Example 1.4.7

	Early (E)	Late (L)	Totals
Red (R)	5	8	13
Yellow (Y)	3	4	7
Totals	8	12	20

$$\bullet P[R|E] = \frac{5}{8} = \frac{N(R \cap E)}{N(E)} = \frac{N(R \cap E)/20}{N(E)/20} = \frac{P[R \cap E]}{P[E]}$$

Definition

定義 1.3. The conditional probability of an event A , given that event B has occurred, is defined by

$$P[A|B] = \frac{P[A \cap B]}{P[B]}$$

provided that $P[B] > 0$.

Properties

- $P[A|B] \geq 0$
- $P[B|B] = 1$
- $P[A \cap B] = P[B] \times P[A|B]$ (Multiplication rule)

Properties

- If $A_1, A_2, \dots$ are mutually exclusive events, then

$$P[\cup_{n=1}^{\infty} A_n | B] = \sum_{n=1}^{\infty} P[A_n | B]$$

Example 1.4.4

- Consider an ordinary deck of playing cards.
- Cards are to be drawn successively at random without replacement.
- Compute the probability that the third spade appears in the first six cards drawn.

Define

A = the event of two spades in the first five cards

B = the event of a spade on the sixth draw

Remark

- The probability that three events A , B and C , all occur is given by the multiplication rule,

$$P[A \cap B \cap C] = P[A]P[B|A]P[C|A \cap B]$$

Law of total probability

- Let $\{C_i, i = 1, 2, \dots, k\}$ be a partition of $\mathcal{S}$.
- Let C be an event.
- Then

$$P[C] = \sum_{i=1}^k P[C_i] \times P[C|C_i]$$

Bayes' theorem

- Let $\{C_i, i = 1, 2, \dots, k\}$ be a partition of $\mathcal{S}$.
- Let C be an event.
- Suppose $P[C] > 0$.
- Then

$$P[C_j|C] = \frac{P[C_j] \times P[C|C_j]}{\sum_{i=1}^k P[C_i] \times P[C|C_i]}.$$

Example 1.4.5

- The bowl A_1 contains three red and seven blue chips.
- The bowl A_2 contains eight red and two blue chips.
- All chips are identical in size and shape.
- A die is cast and bowl A_1 is selected if five or six spots show on the side that is up; otherwise, bowl A_2 is selected.
- Assign $P[A_1] = 2/6$ and $P[A_2] = 4/6$.
- The selected bowl is handed to another person and one chip is taken at random.
- Let B denote an event of a red chip is taken.
- Find $P[A_1|B]$ and $P[A_2|B]$.

Bayesian inference terminology

- The prior probability – the probability of $P[A_i]$.
- The posterior probability – the probability of $P[A_i|B]$.

Definition

定義 1.4. Events A and B are independent if and only if

$$P[A \cap B] = P[A]P[B]$$

Otherwise, A and B are called dependent events.

Remark

- Independent events are also called statistically independent, stochastically independent or independent in a probabilistic sense.
- The definition always holds If $P[A] = 0$ or $P[B] = 0$.

Example 1.4.8

- A red die and a white die are rolled.
- Let $A = \{4 \text{ on the red die}\}$.
- Let $B = \{\text{sum of dice is odd}\}$
- Show that A and B are independent events.

Example 1.4.9

- A red die and a white die are rolled.
- Let $A = \{5 \text{ on the red die}\}$.
- Let $B = \{\text{sum of dice is 11}\}$
- Show that A and B are dependent events.

Properties

- If A and B are independent events, then the following pairs of events are also independent:
 1. A and B'
 2. A' and B
 3. A' and B'

Supplementary Example 1.4-A

- An urn contains four balls numbered 1, 2, 3, and 4.
- One ball is to be drawn at random from the urn.
- Let $A = \{1, 2\}$, $B = \{1, 3\}$ and $C = \{1, 4\}$.
- Show that A and B , A and C , B and C are independent events, but not A , B and C .

Definition

定義 1.5. Events A , B and C are mutually independent if and only if the following two conditions hold:

1. A , B and C are pairwise independent; that is,

$$P[A \cap B] = P[A]P[B], P[A \cap C] = P[A]P[C], P[B \cap C] = P[B]P[C]$$

2. $P[A \cap B \cap C] = P[A]P[B]P[C]$

Properties

- If A , B and C are mutually independent events, then the following pairs of events are also independent:
 1. A and $B \cap C$
 2. A and $B \cup C$
 3. A' and $B \cap C'$
- A' , B' and C' are mutually independent.

Remark

- Many experiments consist of a sequence of n trials that are mutually independent.
- If the outcomes of the trials are not related to each other, then events, such that each is associated with a different trials, should be independent in the probability sense.
- If A_i is associated with the i th trials, $i = 1, 2, \dots, n$, then

$$P[A_1 \cap A_2 \cap \dots \cap A_n] = P[A_1]P[A_2] \dots P[A_n]$$

Example 1.4.11

- A compute system is built so that if component K_1 fails, it is bypassed and K_2 is used.
- If K_2 fails, then K_3 is used.
- Suppose that the probability that K_1 fails is 0.01, that K_2 fails is 0.03, and that K_3 fails is 0.08.
- Assume that the failures are mutually independent events.
- Find $P[\text{Failure of the system}]$.

1.7 1.5 Random Variables

Definition

定義 1.6.

- Consider a random experiment with a sample space $\mathcal{S}$.
- A function X assigns one and only one real number $X(c) = x$ to each element $c \in \mathcal{S}$.
- X is called a random variable.
- The space or range of X is the set of real numbers $\mathcal{D} = \{x : x = X(c), c \in \mathcal{S}\}$.

Types of random variables

- Discrete random variables – $\mathcal{D}$ is a countable set.
- Continuous random variables – $\mathcal{D}$ is an interval of real numbers.

Distribution of a random variable

- Given a random variable X , its range $\mathcal{D}$ becomes the sample space of interest.
- Besides inducing the sample space $\mathcal{D}$, X also induces a probability.
- We call the distribution of X .

Discrete distribution

- Let X be a discrete random variable with a finite space $\mathcal{D} = \{d_1, \dots, d_m\}$.
- Define

$$p_X(d_i) = P[\{c : X(c) = d_i\}], i = 1, 2, \dots, m.$$

which is called the probability mass function (pmf) of X .

- The probability of event D is

$$P_X(D) = \sum_{d_i \in D} p_X(d_i), D \subset \mathcal{D}.$$

Example 1.5.1

- Let X be the sum of the upfaces on a roll of a pair of fair 6-sided dice, each with the numbers 1 through 6 on it.
- The sample space is $\mathcal{S} = \{(i, j) : 1 \leq i, j \leq 6\}$.
- Find the pmf of X .

Example 1.5.1 (continued)

- Let B_1 denote the event of 7 and 11.
- Let B_2 denote the event of 2, 3 and 12.
- Find $P_X[B_1]$ and $P_X[B_2]$.

Continuous distribution

- Let X be a continuous random variable with an interval of real numbers as the space $\mathcal{D}$.
- For the continuous random variables, the simple events of interest are intervals.
- Determine a nonnegative function $f_X(x)$ such that for any interval of real numbers $(a, b) \in \mathcal{D}$, the induced probability distribution of X , $P_X(\cdot)$, is defined as

$$P_X[(a, b)] = P[\{c \in \mathcal{S} : a < X(c) < b\}] = \int_a^b f_X(x) dx.$$

which is the probability that X falls between a and b is the area under the curve $y = f_X(x)$ between a and b .

- The function f_X is formally defined as the probability density function (pdf) of X .

Example 1.5.2

- Let X denote the outcome when a point is selected at random from an interval $[0, 1]$.
- If the experiment is performed in a fair manner.
- Assume

$$P_X[(a, b)] = b - a, 0 < a < b < 1.$$

which is the length of the interval.

- The pdf of X is

$$f_X(x) = \begin{cases} 1 & 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}.$$

- Find $P[\{X < \frac{1}{8}\} \cup \{X > \frac{7}{8}\}]$.

Example 1.5.2 (continued)

- For any point a , $0 < a < 1$.
- Choose n_0 such that $0 < a - n_0^{-1} < a < a + n_0^{-1} < 1$.
- We have

$$P[X = a] \leq P\left[a - \frac{1}{n} < X < a + \frac{1}{n}\right] = \frac{2}{n}, \forall n \geq n_0.$$

- Since $2/n \rightarrow 0$ as $n \rightarrow \infty$ and a is arbitrary, we conclude that $P(X = a) = 0$ for all $a \in (0, 1)$.

Cumulative distribution function

- Let X be a random variable.

- Define

$$\begin{aligned} F_X(x) &= P_X[(-\infty, x)] = P[X \leq x] \\ &= P[\{c \in \mathcal{S} : X(c) \leq x\}], \quad -\infty < x < \infty. \end{aligned}$$

the cumulative distribution function of X .

- Use cdf as an abbreviate.

Example 1.5.4

- Let X denote the outcome when a point is selected at random from an interval $[0, 1]$.
- If the experiment is performed in a fair manner.
- Assume

$$P_X[(a, b)] = b - a, \quad 0 < a < b < 1.$$

which is the length of the interval.

- The pdf of X is

$$f_X(x) = \begin{cases} 1 & 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}.$$

- Find cdf of X .

Example 1.5.4 (continued)

- $F_X(x)$ and $f_X(x)$ are related, i.e.,

$$F_X(x) = \int_{-\infty}^x f_X(t) dt, \quad \forall x \in R$$

and

$$\frac{d}{dx} F_X(x) = f_X(x), \quad \forall x \in R, \text{ except for } x = 0 \text{ and } x = 1.$$

Remark

- X and Y are equal in distribution and write $X \stackrel{D}{=} Y$ if and only if $F_X(x) = F_Y(x)$, for all $x \in R$.

Supplementary Example 1.5-A

- Let X denote the outcome when a point is selected at random from an interval $[0, 1]$.
- Define the random variable Y as $Y = 1 - X$.
- $Y \neq X$.
- The space of Y is the interval $(0,1)$, the same as X .

- The cdf of Y is

$$F_Y(y) = \begin{cases} 0 & y < 0 \\ P[Y \leq y] = 1 - (1 - y) = y & 0 \leq y < 1 \\ 1 & y \geq 1 \end{cases}.$$

- Y has the same cdf as X , i.e., $X \stackrel{D}{=} Y$, but $Y \neq X$.

Properties of cdf

定理 1.10.

- Let X be a random variable with cumulative distribution function $F(x)$.
- Then
 - For all a and b , if $a < b$, then $F(a) \leq F(b)$ (F is nondecreasing).
 - $\lim_{x \rightarrow -\infty} F(x) = 0$
 - $\lim_{x \rightarrow \infty} F(x) = 1$
 - $\lim_{x \rightarrow x_0^-} F(x) = F(x_0)$ (F is right continuous).

Properties of cdf

定理 1.11.

- Let X be a random variable with cumulative distribution function $F(x)$.
- Then, for $a < b$,

$$P[a < X \leq b] = F_X(b) - F_X(a).$$

Properties of cdf

定理 1.12.

- Let X be a random variable with cumulative distribution function $F(x)$.
- Then,

$$P[X = x] = F_X(x) - F_X(x-),$$

for all $x \in \mathbb{R}$, where $F_X(x-) = \lim_{z \rightarrow x-} F_X(z)$.

Example 1.5.5

- Let X be the lifetime in years of a mechanical part.
- Assume that X has the cdf

$$F_X(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-x} & 0 \leq x \end{cases}.$$

- Find the pdf of X and $P[a < X \leq 3]$.

Properties of pdf or pmf

- The total probability associated with a random variable X of the discrete type with pmf $p_X(x)$ or of the continuous type with pdf $f_X(x)$ is 1.
- It must be true that

$$\sum_{x \in \mathcal{D}} p_X(x) = 1 \quad \text{and} \quad \int_{\mathcal{D}} f_X(x) dx = 1,$$

where $\mathcal{D}$ is the space of X .

Example 1.5.8

- Let X has the pdf

$$f_X(x) = \begin{cases} cx^3 & 0 < x < 2 \\ 0 & \text{elsewhere,} \end{cases}$$

for an appropriate constant c .

- Find c .

Stage 2 — Deepen : 從會算到會推導

本階段開始放慢。Sections 1.6–1.7 的前段定義是 Recall 與 Deepen 的橋接，課堂重點不是再次背誦 pmf / pdf，而是從

$$X \longrightarrow g(X) \longrightarrow \text{distribution of } g(X)$$

進一步走向

$$X_1, \dots, X_n \longrightarrow T(X_1, \dots, X_n),$$

為 sampling distribution 建立工具。接著以 Sections 1.8–1.9 深化 expectation、moments、variance 與 MGF。

1.8 1.6 Discrete Random Variables

Discrete random variables

定義 1.7. A random variable is a discrete random variable if its space is either finite or countable.

Probability mass function

定義 1.8.

- Let X be a discrete random variable with space $\mathcal{D}$.
- The probability mass function (pmf) of X given by

$$p_X(x) = P[X = x], \text{ for } x \in \mathcal{D}.$$

Properties of Probability mass function

1. $p_X(x) > 0, x \in \mathcal{D}$.
2. $\sum_{x \in \mathcal{D}} p_X(x) = 1$.
3. $P[X \in A] = \sum_{x \in A} p_X(x)$, where $A \subset \mathcal{D}$.

Support

- Usually define $p_X(x) = 0$ if $x \notin S$.
- The domain of $p_X(x)$ is the set of R .
- To distinguish, the **support** of X is the set of outcomes with positive probabilities.

Hypergeometric distribution

- Consider a collection of $N = N_1 + N_2$ similar objects.
 - N_1 objects belong to the first class (red chip).
 - N_2 objects belong to the second class (blue chip).
- n objects are randomly selected from N objects.
- Let X be the number of objects drawing from the first class.
- Find the pmf of X .

Supplementary Example 1.6-A — Hypergeometric distribution

- There are 50 fishes in a small pond.
- 10 of which have been tagged.
- Suppose 7 fishes are caught randomly without replacement.
- Find the probability that exactly 2 tagged fishes are caught.

Transformation

- X has a pmf $f(x)$.
- Let $Y = u(X)$ for a given function $u(\cdot)$.
- Find the pmf of Y

Example 1.6.1

- Consider a sequence of independent flips of a coin, each resulting in a head (H) or a tail (T).
- Let X be the flip number on which the first head appear.
- Let Y be the number of flips before the first head, i.e., $Y = X - 1$.

- Find the pmf of Y .

Example 1.6.3

- Consider a sequence of independent flops of a coin, each resulting in a head (H) or a tail (T).
- Let X be the flip number on which the first head appear.
- If the first head appears on an odd number of flips, we pay the house one dollar, while if it appears on an even number of flips, we win one dollar from the house.
- Let Y denote our net gain.
- The space of Y is $-1, 1$
- Find the pmf of Y .

1.9 1.7 Continuous Random Variables

Continuous random variables

定義 1.9. A random variable is a continuous random variable if the cdf $F_X(x)$ is a continuous function for all $x \in \mathbb{R}$.

Probability density function

- Most continuous random variables are absolutely continuous; that is,

$$F_X(x) = \int_{-\infty}^x f_X(t) dt,$$

for some function $f_X(t)$.

- The function $f_X(t)$ is called a probability density function (pdf) of X .
- If $f_X(t)$ is also continuous, then the Fundamental Theorem of Calculus implies that

$$\frac{d}{dx} F_X(x) = f_X(x).$$

Support

- The support of a continuous random variable X consists of all points x such that $f_X(x) > 0$.
- Denote the support of X by S .

Properties of pdf

- The pdf $f(x)$ of X of the continuous type with space S .
- S is an interval or union of intervals.

- $f(x)$ satisfies
 - $f(x) \geq 0, x \in S$.
 - $\int_S f(x)dx = 1$.
 - If $(a, b) \subseteq S$, then

$$P[a < X < b] = \int_a^b f(x)dx.$$

Uniform distribution

- X has a uniform distribution if

$$f(x) = \frac{1}{b-a}, a \leq x \leq b$$

and 0 elsewhere.

- Denote $U(a, b)$.

Example 1.7.1

- Suppose a point is selected at random in the interior of a circle of radius 1.
- Let X be the distance of the selected point from the origin.
- The sample space for the experiment is

$$\mathcal{S} = \{(w, y) : w^2 + y^2 < 1\}.$$

- Assume subsets of $\mathcal{S}$ which have equal area are equilikely.
- Hence, the probability of the selected point lying in a set $A \subset \mathcal{S}$ is proportional to the area of A ; i.e.,

$$P[A] = \frac{\text{area of } A}{\pi}$$

- Find cdf and pdf of X .

Quantiles

定義 1.10.

- Let $0 < p < 1$.
- The quantile of order p of the distribution of a random variable X is a value ξ_p such that $P(X < \xi_p) \leq p$ and $P(X \leq \xi_p) \geq p$.
- It is also known as the $(100p)$ th percentile of X .

Quartiles

定義 1.11.

- The quantile of order 0.25, 0.5, 0.75 of the distribution of a random variable X is called the first quartile (q_1), median and the third quartile (q_3).
- The difference $IQ = q_3 - q_1$ is called the interquartile range of X .

Note

- Quantiles need not be unique even for continuous random variables with pdfs.
- Any point in the interval (2,3) serves as a median for the following pdf:

$$f_X(x) = \begin{cases} 3(1-x)(x-2) & 1 < x < 2 \\ 3(3-x)(x-4) & 3 < x < 4 \\ 0 & \text{elsewhere} \end{cases}.$$

Note

- A quantile, say ξ_p , is in the support of an absolutely continuous random variable X with cdf $F_X(x)$ then ξ_p is the unique solution to the equation:

$$\xi_p = F_X^{-1}(p),$$

where $F_X^{-1}(u)$ is the inverse function of $F_X(x)$.

Distribution function technique

- Let X be a random variable of the continuous type.
- Let $Y = u(X)$, where $u(\cdot)$ is a known function.
- Then the pdf of Y can be derived from

$$G(y) = P[Y \leq y] = P[u(X) \leq y]$$

Supplementary Example 1.7-A — Log gamma distribution

- Let X have a gamma distribution with pdf

$$f(x) = \frac{1}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} e^{-x/\theta}, 0 < x < \infty,$$

where $\alpha > 0$ and $\theta > 0$.

- Let $Y = \ln(X)$.
- Find the pdf of Y .

Theorem

定理 1.13.

- Let X be a random variable of the continuous type with pdf $f(x)$ with support S_X .
- Let $Y = g(X)$, where $g(\cdot)$ is a one-to-one differentiable function on S_X .
- Assume the inverse function g by $x = g^{-1}(y)$ and let

$$J = \frac{dx}{dy} = \frac{dg^{-1}(y)}{dy} (\text{Jacobian})$$

- Then the pdf of Y is given by

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|, \text{ for } y \in S_Y,$$

where S_Y is the support of Y and $S_Y = \{y = g(x) : x \in S_X\}$.

Supplementary Example 1.7-A (continued)

- Let X have a gamma distribution with pdf

$$f(x) = \frac{1}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} e^{-x/\theta}, 0 < x < \infty,$$

where $\alpha > 0$ and $\theta > 0$.

- Let $Y = \ln(X)$.
- Find the pdf of Y .

Example 1.5.6 — Mixture type of distribution

- Let X have cdf

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{x+1}{2} & 0 \leq x < 1 \\ 1 & 1 \leq x \end{cases}.$$

- Find $P[-3 < X \leq \frac{1}{2}]$ and $P[X = 0]$.

1.10 1.8 Expectation of a Random Variable

Expectation – Continuous

定義 1.12.

- Let X be a random variable.

- If X with a continuous random variable with pdf $f(x)$ and

$$\int_{-\infty}^{\infty} |x|f(x)dx < \infty$$

then the expectation of X is

$$E[X] = \int_{-\infty}^{\infty} xf(x)dx.$$

Expectation – Discrete

定義 1.13.

- Let X be a random variable.
- If X with a pmf $p(x)$ and

$$\sum_{x \in S} |x|p(x)$$

then the expectation of X is

$$E[X] = \sum_{x \in S} xp(x).$$

Note

- The expectation $E(X)$ is called the mathematical expectation of X , the expected value of X , or the mean of X .
- When the mean designation is used, $E(X)$ is denoted by μ ; i.e., $\mu = E(X)$.

Supplementary Example 1.8-A

- Let X have a pmf

$$f(x) = \frac{1}{3}, x \in S_X,$$

where $S_X = \{-1, 0, 1\}$.

- Find $E[X]$.

Example 1.8.3

- Let X have a pdf

$$f(x) = \begin{cases} 4x^3 & 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}.$$

- Find $E[X]$.

Expectation of a function of random variable (continuous)

定理 1.14.

- Let X be a random variable and let $Y = g(X)$ for some function g .
- If X with a continuous random variable with pdf $f_X(x)$ and

$$\int_{-\infty}^{\infty} |g(x)|f_X(x)dx < \infty$$

then the expectation of X is

$$E[Y] = \int_{-\infty}^{\infty} g(x)f(x)dx.$$

Expectation of a function of random variable (discrete)

定理 1.15.

- Let X be a random variable and let $Y = g(X)$ for some function g .
- If X with a pmf $p(x)$ and

$$\sum_{x \in S_X} |g(x)|p(x)$$

then the expectation of X is

$$E[Y] = \sum_{x \in S_X} g(x)p(x).$$

Supplementary Example 1.8-B

- Consider a sequence of independent flips of a coin, each resulting in a head (H) or a tail (T).
- Let X was the flip number on which the first head appeared.
- Let Y be the number of flips before the first head.
- Let $Z = e^{-Y}$.
- Find $E[Z]$.

Supplementary Example 1.8-C

- Let X be a continuous random variable with the pdf $f(x) = 2x$ which has support on the interval $(0, 1)$.
- Let $Y = 1/(1 + X)$.
- Find $E[Y]$.

Theorem

定理 1.16. When the expectation exists, it satisfies the following properties:

- If c is a constant, then $E[c] = c$.
- If c is a constant and $u(\cdot)$ is a function, then

$$E[cu(X)] = cE[u(X)].$$

- If c_1 and c_2 are constants and u_1 and u_2 are functions, then

$$E[c_1u_1(X) + c_2u_2(X)] = c_1E[u_1(X)] + c_2E[u_2(X)].$$

Comment

- If $c_1, \dots$ and c_k are constants and $u_1, u_2, \dots$ and u_k are functions, then

$$E\left[\sum_{i=1}^k c_i u_i(X)\right] = \sum_{i=1}^k c_i E[u_i(X)].$$

Example 1.8.4

- Let X have the pdf

$$f(x) = \begin{cases} 2(1-x) & 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}.$$

- Find $E[X]$, $E[X^2]$ and $E[6X + 3X^2]$.

Example 1.8.5

- Let X have the pmf

$$f(x) = \begin{cases} \frac{x}{6} & x = 1, 2, 3 \\ 0 & \text{elsewhere} \end{cases}.$$

- Find $E[6X^3 + X]$.

Example 1.8.6

- Let us divide, at random, a horizontal line segment of length 5 into two parts.
- If X is the length of the left-hand part, it is reasonable to assume X has the pdf

$$f(x) = \begin{cases} \frac{1}{5} & 0 < x < 5 \\ 0 & \text{elsewhere} \end{cases}.$$

- Find $E[X(5 - X)]$.

1.11 1.9 Some Special Expectations

Settings

- Let X be a discrete random variable.
- Let the support of X be denoted as $S_X = \{u_1, u_2, \dots, u_k\}$.
- Assume

$$P[X = u_i] = f(u_i) = p_i, i = 1, 2, \dots, k$$

and

$$\sum_{i=1}^k p_i = 1.$$

Definition of mean

定義 1.14.

- Let X be a random variable whose expectation exists.
- The mean value μ of X is defined to be $\mu = E(X)$.

Definition of variance

定義 1.15.

- Let X be a random variable with finite mean μ and such that $E[(X - \mu)^2] < \infty$.
- The variance of X is defined to be $E[(X - \mu)^2]$.
- It is usually denoted by σ^2 or by $\text{Var}(X)$.

Supplementary Example 1.8-D

- Consider a discrete random variable X has a support $S_X = \{1, 2, 3\}$.
- Let $f(1) = \frac{3}{6}, f(2) = \frac{2}{6}, f(3) = \frac{1}{6}$.
- Find $E[X]$ and $\text{Var}(X)$.

Comment

- Show that

$$\text{Var}(X) = E[X^2] - \mu^2,$$

- The standard deviation of X is the positive square root of variance and is denoted as σ .

Class Example 1.9-A

- Let a and b be constants.
- Let $Y = aX + b$.
- Find $E[Y]$ and $\text{Var}(Y)$.

Example 1.9.1

- Suppose the random variable X has a uniform distribution with pdf

$$f_X(x) = 1/(2a), -a < x < a,$$

zero elsewhere.

- Find μ and σ^2 .

Example 1.9.2

- Let X has the pdf

$$f(x) = \begin{cases} \frac{1}{2}(x+1) & -1 < x < 1 \\ 0 & \text{elsewhere} \end{cases}.$$

- Find μ and σ^2 .

Example 1.9.5

- Let X has the pdf

$$f(x) = \begin{cases} \frac{1}{x^2} & 1 < x < \infty \\ 0 & \text{elsewhere} \end{cases}.$$

- Find μ .

Moment

- The r th moment is defined as

$$E[X^r] = \sum_{x \in S} x^r f(x).$$

- The r th moment about b is defined as

$$E[(X - b)^r] = \sum_{x \in S} (x - b)^r f(x).$$

- The r th factorial moment is defined as

$$E[(X)_r] = E[X(X - 1) \cdots (X - r + 1)].$$

Supplementary Example 1.9-A — Hypergeometric distribution

- Let X have a hypergeometric distribution where n objects are selected from $N = N_1 + N_2$.
- Find $\text{Var}(X)$.

Moment generating function

定義 1.16.

- Let X be a random variable.
- If there is a $h > 0$ such that

$$E[e^{tX}] = \sum_{x \in S} e^{tx} f(x)$$

exists and is finite for $-h < t < h$, then the function

$$M(t) = E[e^{tX}]$$

is called the moment-generating function (mgf) of X .

Comments

- If $t = 0$, $M(0) = 1$.
- If $S = \{b_1, b_2, \dots\}$, the mgf is

$$M(t) = e^{tb_1} f(b_1) + e^{tb_2} f(b_2) + e^{tb_3} f(b_3) + \dots,$$

- The coefficient of e^{tb_i} is

$$f(b_i) = P[X = b_i]$$

Comments

- Consider two random variables had two pmfs $f(x)$ and $g(y)$.
- Suppose the same space $S = \{b_1, b_2, \dots\}$ is defined.
- If

$$\begin{aligned} e^{tb_1} f(b_1) + e^{tb_2} f(b_2) + e^{tb_3} f(b_3) + \dots \\ = e^{tb_1} g(b_1) + e^{tb_2} g(b_2) + e^{tb_3} g(b_3) + \dots, \end{aligned}$$

for all t , $-h < t < h$, then

$$f(b_i) = g(b_i), \forall i$$

Example 1.9.3

- Suppose we have a fair spinner with the numbers 1, 2, and 3 on it.
- Let X be the number of spins until the first 3 occurs.

- Assuming that the spins are independent, the pmf of X is

$$p(x) = \frac{1}{3} \left(\frac{2}{3}\right)^{x-1}, x = 1, 2, 3, \dots$$

- Find the mgf of X .

Equivalence

定理 1.17.

- Let X and Y be random variables with moment generating functions $M_X(\cdot)$ and $M_Y(\cdot)$, respectively, existing in open intervals about 0.
- Then $F_X(z) = F_Y(z)$ for all $z \in R$ if and only if $M_X(t) = M_Y(t)$ for all $t \in (-h, h)$ for some $h > 0$.

Supplementary Example 1.9-B

- Suppose X has the mgf

$$M(t) = e^t \left(\frac{1}{10}\right) + e^{2t} \left(\frac{2}{10}\right) + e^{3t} \left(\frac{3}{10}\right) + e^{4t} \left(\frac{4}{10}\right), -\infty < t < \infty$$

- Find the pmf of X .

Class Example 1.9-B

- Show that $M'(0) = \mu$ and $M''(0) - [M'(0)]^2 = \sigma^2$.

Note

- If m is a positive integer and $M^{(m)}(t)$ denotes the m th derivative of $M(t)$.
- Then $M^{(m)}(0) = E[X^m]$.

Stage 3 — Connect : 由機率工具走向數理統計

本階段不把 inequalities 當作孤立公式。Markov、Chebyshev 與 Jensen 分別提供尾端機率控制、集中現象及凸性比較；其中 Chebyshev 不等式會直接連接樣本平均數與依機率收斂，為後續抽樣分配、漸近理論與統計推論鋪路。

1.12 1.10 Important Inequalities

Existence of moments

- Let X be a random variable and let m be a positive integer.

- Suppose $E[X^m]$ exists.
- If k is a positive integer and $k \leq m$, then $E[X^k]$ exists.

Markov's Inequality

定理 1.18.

- Let $u(X)$ be a nonnegative function of the random variable X .
- If $E[u(X)] < \infty$, random variable and let m be a positive integer.
- For every positive constant c ,

$$P[u(X) \geq c] \leq \frac{E[u(X)]}{c}.$$

Chebyshev's Inequality

定理 1.19.

- Let X be a random variable with finite variance σ^2 .
- For every positive constant k ,

$$P[|X - \mu| \geq k\sigma] \leq \frac{1}{k^2}.$$

Example 1.10.1

- Let X has the pdf

$$f(x) = \begin{cases} \frac{1}{2\sqrt{3}} & -\sqrt{3} < x < \sqrt{3} \\ 0 & \text{elsewhere} \end{cases}.$$

- Find $P[|X - \mu| \geq k\sigma]$ and the upper bound.

Example 1.10.2

- Let the random variable X of the discrete type have probabilities $1/8, 6/8, 1/8$ at the points $x = -1, 0, 1$, respectively.
- Find $P[|X - \mu| \geq k\sigma]$ and the upper bound.

Connect: sample mean and Chebyshev

令 $X_1, \dots, X_n$ 為獨立同分配 (iid) 的隨機變數，且

$$E(X_i) = \mu, \quad \text{Var}(X_i) = \sigma^2 < \infty.$$

樣本平均數為

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

由期望值與獨立性，

$$E(\bar{X}_n) = \mu, \quad \text{Var}(\bar{X}_n) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{\sigma^2}{n}.$$

因此對任意 $\varepsilon > 0$ ，Chebyshev 不等式給出

$$P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\text{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0.$$

這正是

$$\bar{X}_n \xrightarrow{P} \mu,$$

也就是樣本平均數依機率收斂到母體平均數；這個結果是弱大數法則的一個版本。

💡 往後連接

這裡的邏輯鏈是

$$\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n} \implies \text{Chebyshev bound} \rightarrow 0 \implies \bar{X}_n \xrightarrow{P} \mu.$$

後續將把 $T(X_1, \dots, X_n)$ 視為 statistic，研究其 sampling distribution，並在漸近理論中系統整理 convergence in probability。參見 [Sampling and Inference](#) 與 [Asymptotics](#)。

Convex function

定義 1.17.

- A function ϕ defined on an interval (a, b) , $-\infty < a < b < \infty$, is said to be a convex function if for all x, y in (a, b) and for all $0 < \gamma < 1$,

$$\phi(\gamma x + (1 - \gamma)y) \leq \gamma\phi(x) + (1 - \gamma)\phi(y).$$

We say ϕ is strictly convex if the above inequality is strict.

Properties – Convex function

定理 1.20.

- If ϕ is differentiable on (a, b) , then
 - ϕ is convex if and only if $\phi'(x) \leq \phi'(y)$, for all $a < x < y < b$.

- ϕ is strictly convex if and only if $\phi'(x) < \phi'(y)$, for all $a < x < y < b$.
- If ϕ is twice differentiable on (a, b) , then
 - ϕ is convex if and only if $\phi''(x) \geq 0$, for all $a < x < b$.
 - ϕ is strictly convex if and only if $\phi''(x) > 0$, for all $a < x < b$.

Jensen's Inequality

定理 1.21.

- If ϕ is convex on an open interval I and X is a random variable whose support is contained in I and has finite expectation, then

$$\phi(E[X]) \leq E[\phi(X)]$$

- If ϕ is strictly convex, then the inequality is strict unless X is a constant random variable.

Example 1.10.3

- Let X be a nondegenerate random variable with mean μ and a finite second moment.
- Show that $(E[X])^2 \leq E[X^2]$.

Example 1.10.4

- Let $\{a_1, \dots, a_n\}$ be a set of positive numbers.
- Create a distribution for a random variable X by placing weight $1/n$ on each of the numbers $a_1, \dots, a_n$.
- The arithmetic mean (AM) equals

$$E[X] = \frac{1}{n} \sum_{i=1}^n a_i.$$

- The geometric mean (GM) equals $(a_1 a_2 \cdots a_n)^{1/n}$.
- The harmonic mean (HM) equals

$$\frac{1}{\frac{1}{n} \sum_{i=1}^n \frac{1}{a_i}}$$

- Show that $HM \leq GM \leq AM$.

1.13 本章摘要

本章依 **Recall** $\rightarrow$ **Deepen** $\rightarrow$ **Connect** 建立後續數理統計所需的共同語言。Stage 1 快速喚醒機率模型、條件機率、隨機變數與分配；Stage 2 深化 transformation、 $E[g(X)]$ 、moments、variance 與 MGF；Stage 3 以重要不等式連接樣本平均數、依機率收斂及後續推論理論。

1.14 常見錯誤

- 把事件與隨機變數混為一談
- 使用密度值代替區間機率
- 套用條件機率時忘記分母事件
- 引用不等式時沒有檢查非負性或凸性條件

1.15 章末練習

1. 從本章選出三個關鍵定義，分別寫出數學形式、成立條件及一個例子。
2. 選擇一個主要定理或方法，重做推導並標記每一步使用的假設。
3. 從原講義例題挑選一題，完整寫出解題目標、方法選擇、計算與結果解釋。
4. 設計一個會違反本章方法條件的反例，說明錯誤會出現在哪一步。

授課提醒

本章保留完整材料供學生查閱，但「保留」不等於「逐頁等速講授」。Stage 1 可按班級診斷結果選題，Stage 2 應保留推導時間，Stage 3 則務必讓學生說明 Chebyshev 如何導出 $\overline{X}_n \xrightarrow{P} \mu$ 。

2 多變量分配與變數轉換

本章定位

學生在大二已使用 Hogg, Tanis & Zimmerman 的 *Probability and Statistical Inference* 學過雙變量分配。該書 Chapter 4 已涵蓋 joint、marginal 與 conditional distributions、independence、covariance 與 correlation。Chapter 5.2 亦已介紹雙變量變數轉換。因此本章不將所有內容等量重教，而採取

Recall → Deepen → Connect

的三階段安排：先快速喚醒二維隨機變數的基本工具，再推廣到 n 維隨機向量、共變異矩陣與多維變數轉換，最後連接線性組合、random sample、sample mean 與 sample variance。標題依 Hogg, McKean & Craig Chapter 2 的內容順序安排，各節不採等比例講解。

第 8 版編號原則

本章以 Hogg, McKean & Craig, *Introduction to Mathematical Statistics*, 8th Global Edition (以下簡稱 HMC8) Chapter 2, pp. 99–168 為主要對照。原書的 section、definition、theorem、corollary 與 example 編號均照錄；講義自行增加或延伸的內容標為 **Supplementary Example**，不占用課本編號。講義未選入的課本項目保留編號缺口，不重新連號。

2.1 學習目標

完成本章後，學生應能：

- 由聯合分配求邊際與條件分配
- 處理二維與多維隨機變數轉換
- 計算共變異數、相關係數與共變異矩陣
- 判斷隨機變數的獨立性
- 使用向量與矩陣表示多維隨機變數的平均與共變異結構
- 計算隨機變數線性組合的期望、變異數與共變異數
- 說明 random sample、sample mean 與 sample variance 如何連接後續統計推論

2.2 必備的先備知識

第 1 章的分配函數、期望與單變量變數轉換，以及偏微分與基本矩陣運算。joint pmf / pdf、marginal distribution、conditional distribution、independence、covariance、correlation 與二維 Jacobian 視為大二已學內容，課堂中以診斷題與代表性例題快速喚醒。

i 教學節奏

- **Stage 1 — Recall (快速)** : Sections 2.1–2.5 · 對照大二課本 Chapter 4 與 Section 5.2 · 以 60–90 分鐘的 example-driven review 喚醒 joint、marginal、conditional、independence、covariance、correlation 與二維 transformation；不重走完整的 Definition–Theorem–Proof 流程。
- **Stage 2 — Deepen (主體)** : Sections 2.6–2.7 · 把二維概念推廣到 n 維 random vector · 深入處理 covariance matrix、線性代數表示、多維 Jacobian · 以及 one-to-one / many-to-one transformations。
- **Stage 3 — Connect (橋接)** : Section 2.8 · 以線性組合、random sample、 $\bar{X}$ 與 S^2 連接後續的 sampling distributions、sufficiency、likelihood 與 statistical inference。

Stage 1 — Recall : 二維基本款快速喚醒

大二學過的，先確認能辨認、能設支持集、能完成積分；忘記的，再由代表性例題補回。

本階段的重點不是重複所有計算，而是確認學生能在離散與連續情形中，從 joint distribution 正確取得 marginal 與 conditional distributions，判斷 independence，計算 covariance / correlation，並在變數轉換時同時處理 Jacobian 與轉換後的支持集。

2.3 2.1 Distributions of Two Random Variables

Definition 2.1.1 — Random Vector

定義 2.1.

- Consider a random experiment with a sample space $\mathcal{C}$.
- Define consider two random variables X_1 and X_2 .
- Assign to each element c of $\mathcal{C}$ one and only one ordered pair of numbers $X_1(c) = x_1$, $X_2(c) = x_2$.
- Then,
 - (X_1, X_2) is a random vector.
 - The space of (X_1, X_2) is the set of ordered pairs

$$\mathcal{D} = \{(x_1, x_2) : x_1 = X_1(c), x_2 = X_2(c), c \in \mathcal{C}\}.$$

Cumulative distribution function (cdf)

- Let $A \subset \mathcal{D}$.
- Denote the probability of the event A as $P_{X_1, X_2}[A]$.
- $P_{X_1, X_2}[A]$ is uniquely determined by the cumulative distribution function (cdf), which is given by

$$F_{X_1, X_2}(x_1, x_2) = P[\{X_1 \leq x_1\} \cap \{X_2 \leq x_2\}],$$

for all $(x_1, x_2) \in \mathbb{R}^2$.

Properties of cdf

- For all $a_1 < b_1$ and $a_2 < b_2$, show that

$$P[a_1 < X_1 \leq b_1, a_2 < X_2 \leq b_2] = F_{X_1, X_2}(b_1, b_2) - F_{X_1, X_2}(a_1, b_2) - F_{X_1, X_2}(b_1, a_2) + F_{X_1, X_2}(a_1, a_2)$$

Discrete random vector

定義 2.2.

- Suppose $\mathcal{D}$ is finite or countable.
- Let S denote the two-dimensional space of X_1 and X_2 .
- The joint probability mass function of X_1 and X_2 .

$$p_{X_1, X_2}(x_1, x_2) = P[X_1 = x_1, X_2 = x_2]$$

for all $(x_1, x_2) \in \mathcal{D}$.

Conditions for joint pmf

- $0 \leq p_{X_1, X_2}(x_1, x_2) \leq 1$.
- $\sum \sum_{(x_1, x_2) \in \mathcal{D}} p_{X_1, X_2}(x_1, x_2) = 1$.
- $P[(X, Y) \in A] = \sum \sum_{(x_1, x_2) \in A} p_{X_1, X_2}(x_1, x_2)$.

Supplementary Example 2.1-A

- Roll a pair of fair dice.
- Let X denote the smaller and Y the larger outcome on the dice.
- Find the joint pmf.

Note

- The support of a discrete random vector (X_1, X_2) is defined as

$$\{(x_1, x_2) : p_{X_1, X_2}(x_1, x_2) > 0\}.$$

- The definition of a pmf is extended to a convenient set by using zero elsewhere.

Continuous random vector

定義 2.3.

- A random vector (X_1, X_2) with space $\mathcal{D}$ is of the continuous type if its cdf $F_{X_1, X_2}(x_1, x_2)$ is continuous.
- $F_{X_1, X_2}(x_1, x_2)$ can be expressed as

$$F_{X_1, X_2}(x_1, x_2) = \int_{-\infty}^{x_1} \int_{-\infty}^{x_2} f_{X_1, X_2}(w_1, w_2) dw_1 dw_2,$$

for all $(x_1, x_2) \in \mathcal{D}$.

- The integrand is called the joint probability density function (pdf) of (X_1, X_2) .

Note

- We have

$$\frac{\partial^2 F_{X_1, X_2}(x_1, x_2)}{\partial x_1 \partial x_2} = f_{X_1, X_2}(x_1, x_2)$$

except possibly on events that have probability zero.

Conditions for joint pdf

- $0 \leq f_{X_1, X_2}(x_1, x_2)$.
- $\int \int_{(x_1, x_2) \in \mathcal{D}} f_{X_1, X_2}(x_1, x_2) dx_1 dx_2 = 1$.
- For an event A ,

$$P[(X, Y) \in A] = \int \int_{(x_1, x_2) \in A} f_{X_1, X_2}(x_1, x_2) dx_1 dx_2.$$

Note

- The support of a continuous random vector (X_1, X_2) is defined as

$$\{(x_1, x_2) : f_{X_1, X_2}(x_1, x_2) > 0\}.$$

- The support is denoted as $\mathcal{S}$.
- The definition of a pdf is extended to R^2 by using zero elsewhere.

Special event

- Let (X_1, X_2) be a random vector.
- Consider the event $\{X_1 \leq x_1\}$.
- We have

$$\{X_1 \leq x_1\} = \{X_1 \leq x_1\} \cap \{-\infty < X_2 < \infty\}$$

- The probability of this event is

$$F_{X_1}(x_1) = P[X_1 \leq x_1, -\infty < X_2 < \infty]$$

for all $x_1 \in \mathbb{R}$.

2.1.1 Marginal Distributions

Marginal pmf

- Let (X_1, X_2) be a discrete random vector.
- Find the probability of $\{X_1 \leq x_1\}$ and the pmf of X_1 .

Example 2.1.4 — Marginal pmfs

- Let the joint pmf of X and Y be defined by

$$f(x, y) = \frac{x + y}{21}, x = 1, 2, 3; y = 1, 2.$$

- Find $f_X(x)$ and $f_Y(y)$.

Marginal pdf

- Let (X_1, X_2) be a continuous random vector.
- Find the probability of $\{X_1 \leq x_1\}$ and the pdf of X_1 .

Supplementary Example 2.1-C — Marginal pdfs

- Let X and Y have the joint pdf

$$f(x, y) = \begin{cases} x + y & 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}.$$

- Find the marginal pdf of X and Y , respectively.
- Find $P[X + Y \leq 1]$.

2.1.2 Expectation

Expectation – Discrete type

- Let X_1 and X_2 be random variables of the discrete type with the joint pmf $f(x_1, x_2)$ on the space S .
- If $u(X_1, X_2)$ is a function of X_1 and X_2 , then

$$E[u(X_1, X_2)] = \sum_{(x_1, x_2) \in S} u(x_1, x_2) f(x_1, x_2),$$

if it exists, is called the mathematical expectation of $u(X_1, X_2)$.

Remark

- If

$$\sum \sum_{(x_1, x_2) \in S} |u(x_1, x_2)| f(x_1, x_2),$$

converges, then $E[u(X_1, X_2)]$ exists.

Expectation – Continuous type

- Let X_1 and X_2 be random variables of the discrete type with the joint pdf $f(x_1, x_2)$ on the space S .
- If $u(X_1, X_2)$ is a function of X_1 and X_2 , then

$$E[u(X_1, X_2)] = \int \int_{(x_1, x_2) \in S} u(x_1, x_2) f(x_1, x_2) dx_1 dx_2,$$

if it exists, is called the mathematical expectation of $u(X_1, X_2)$.

Remark

- If

$$\int \int_{(x_1, x_2) \in S} |u(x_1, x_2)| f(x_1, x_2) dx_1 dx_2 < \infty,$$

converges, then $E[u(X_1, X_2)]$ exists.

Theorem 2.1.1 — Linearity of Expectation**定理 2.1.**

- Let (X_1, X_2) be a random vector.
- Let $Y_1 = g_1(X_1, X_2)$ and $Y_2 = g_2(X_1, X_2)$ be random variables whose expectations exist.
- For all real numbers k_1 and k_2 , we have

$$E[k_1 Y_1 + k_2 Y_2] = k_1 E[Y_1] + k_2 E[Y_2].$$

Definition 2.1.2 — Moment Generating Function of a Random Vector**定義 2.4.**

- Let $X = (X_1, X_2)'$ be a random vector.
- If $E[e^{t_1 X_1 + t_2 X_2}]$ exists for $|t_1| < h_1$ and $|t_2| < h_2$, where h_1 and h_2 are positive.
- It is denoted by $M_{X_1, X_2}(t_1, t_2)$ and is called the moment generating function (mgf) of X .

Note

- Let $t = (t_1, t_2)'$.
- Then we can write the mgf of X as

$$M_{X_1, X_2}(t_1, t_2) = E[e^{t'X}].$$

- The mgfs of X_1 and X_2 are immediately seen to be $M_{X_1, X_2}(t_1, 0)$ and $M_{X_1, X_2}(0, t_2)$.

Example 2.1.10 — Joint MGF

- Let X and Y have the joint pdf

$$f(x, y) = \begin{cases} e^{-y} & 0 < x < y < \infty \\ 0 & \text{elsewhere} \end{cases}.$$

- Find $M(t_1, t_2)$, $M(t_1, 0)$ and $M(0, t_2)$.

Definition 2.1.3 — Expected Value of a Random Vector**定義 2.5.**

- Let $X = (X_1, X_2)'$ be a random vector.
- Then the expected value of X exists if the expectations of X_1 and X_2 exist.
- If it exists, then the expected value is given by

$$E[X] = \begin{bmatrix} E[X_1] \\ E[X_2] \end{bmatrix}$$

2.4 2.2 Transformations: Bivariate Random Variables**Transformations – Discrete**

- Let $p_{X_1, X_2}(x_1, x_2)$ be the joint pmf of two discrete-type random variables X_1 and X_2 with the support $\mathcal{S}$.
- Let $y_1 = u_1(x_1, x_2)$ and $y_2 = u_2(x_1, x_2)$ define a one-to-one transformation that maps $\mathcal{S}$ onto $\mathcal{T}$.
- The joint pmf of the two new random variables $Y_1 = u_1(X_1, X_2)$ and $Y_2 = u_2(X_1, X_2)$ is given by

$$p_{Y_1, Y_2}(y_1, y_2) = \begin{cases} p_{X_1, X_2}(\omega_1(y_1, y_2), \omega_2(y_1, y_2)), & (y_1, y_2) \in \mathcal{T} \\ 0 & \text{elsewhere,} \end{cases}$$

where $x_1 = \omega_1(y_1, y_2)$, $x_2 = \omega_2(y_1, y_2)$ is the single-valued inverse of $y_1 = u_1(x_1, x_2)$, $y_2 = u_2(x_1, x_2)$.

Example 2.2.2

- Consider an experiment in which a person chooses at random a point (X_1, X_2) from the unit square $\mathcal{S} = \{(x_1, x_2) : 0 < x_1 < 1, 0 < x_2 < 1\}$.
- Suppose that our interest is not in X_1 or in X_2 , but in $Z = X_1 + X_2$.
- Find the pdf of Z .

Continuous transformation

- Let (X_1, X_2) have a jointly continuous distribution with pdf $f_{X_1, X_2}(x_1, x_2)$ and support set $\mathcal{S}$.
- Consider the transformed random vector $(Y_1, Y_2) = T(X_1, X_2)$ where T is a one-to-one continuous transformation.
- Let $\mathcal{T} = T(\mathcal{S})$ denote the support of (Y_1, Y_2) .
- Rewrite the transformation in terms of its components as

$$(Y_1, Y_2) = T(X_1, X_2) = (u_1(X_1, X_2), u_2(X_1, X_2)),$$

where the functions $y_1 = u_1(x_1, x_2)$ and $y_2 = u_2(x_1, x_2)$ define T .

Continuous transformation

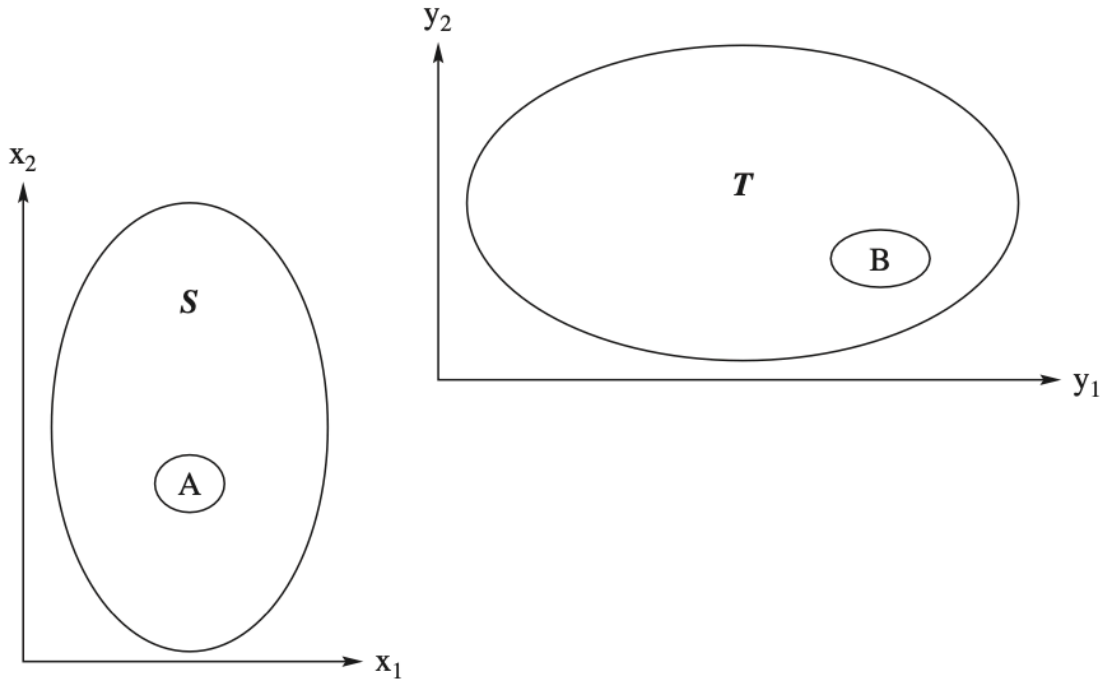
- Since the transformation is one-to-one, the inverse transformation T^{-1} exists.
- Write $x_1 = w_1(y_1, y_2)$, $x_2 = w_2(y_1, y_2)$.
- Define the Jacobian of the transformation which is the determinant of order 2 given by

$$J = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} \end{vmatrix}$$

Continuous transformation

- Let B be any region in $\mathcal{T}$.
- Let $A = T^{-1}(B)$.
- Because the transformation T is one-to-one,

$$\begin{aligned} P[(X_1, X_2) \in A] &= P[T(X_1, X_2) \in T(A)] \\ &= P[(Y_1, Y_2) \in B]. \end{aligned}$$



Continuous transformation

- Based on the change-in-variable technique, we have

$$\begin{aligned}
 P[(X_1, X_2) \in A] &= \int \int_A f_{X_1, X_2}(x_1, x_2) dx_1 dx_2 \\
 &= \int \int_{T(A)} f_{X_1, X_2}(T^{-1}(y_1, y_2)) |J| dy_1 dy_2 \\
 &= \int \int_B f_{X_1, X_2}(w_1(y_1, y_2), w_2(y_1, y_2)) |J| dy_1 dy_2
 \end{aligned}$$

- The pdf of (Y_1, Y_2) is

$$f_{Y_1, Y_2}(y_1, y_2) = \begin{cases} f_{X_1, X_2}(w_1(y_1, y_2), w_2(y_1, y_2)) |J|, & (y_1, y_2) \in \mathcal{T} \\ 0 & \text{elsewhere} \end{cases}$$

Example 2.2.3

- Consider an experiment in which a person chooses at random a point (X_1, X_2) from the unit square $\mathcal{S} = \{(x_1, x_2) : 0 < x_1 < 1, 0 < x_2 < 1\}$.
- Find the joint pdf of $Y_1 = X_1 + X_2$ and $Y_2 = X_1 - X_2$.

Example 2.2.5

- Let X_1 and X_2 have the joint pdf

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} 10x_1x_2^2, & 0 < x_1 < x_2 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

- Find the pdf of $Y_1 = X_1/X_2$.

2.5 2.3 Conditional Distributions and Expectations**Conditional distribution for discrete r.v.**

- Let X_1 and X_2 denote random variables of the discrete type.
- Let the joint pmf be denoted as $p_{X_1, X_2}(x_1, x_2)$ that is positive on the support set $\mathcal{S}$ and is zero elsewhere.
- Let $p_{X_1}(x_1)$ and $p_{X_2}(x_2)$ denote, respectively, the marginal probability mass functions of X_1 and X_2 .
- Let x_1 be a point in the support of X_1 , i.e., $p_{X_1}(x_1) > 0$.
- Using the definition of conditional probability, we have

$$\begin{aligned} p_{X_2|X_1}(x_2 | x_1) &= P[X_2 = x_2 | X_1 = x_1] \\ &= \frac{P[X_1 = x_1, X_2 = x_2]}{P[X_1 = x_1]} \\ &= \frac{p_{X_1, X_2}(x_1, x_2)}{p_{X_1}(x_1)} \end{aligned}$$

for all $x_2 \in \mathcal{S}_{X_2}$.

Properties

- $\sum_{x_2} p_{X_2|X_1}(x_2|x_1) = 1$.
- $p_{X_2|X_1}(x_2|x_1) > 0$ for all $x_2 \in \mathcal{S}_{X_2}$.
- $p_{X_2|X_1}(x_2|x_1)$ is called the conditional pmf of the discrete type of random variable.

Conditional distribution for continuous r.v.

- Let X_1 and X_2 denote random variables of the continuous type.
- Let the joint pdf be denoted as $f_{X_1, X_2}(x_1, x_2)$ that is positive on the support set $\mathcal{S}$ and is zero elsewhere.
- Let $f_{X_1}(x_1)$ and $f_{X_2}(x_2)$ denote, respectively, the marginal probability density functions of X_1 and X_2 .

- When $f_{X_1}(x_1) > 0$, we define

$$f_{X_2|X_1}(x_2 | x_1) = \frac{f_{X_1, X_2}(x_1, x_2)}{f_{X_1}(x_1)}.$$

Properties

- $\int_{-\infty}^{\infty} f_{X_2|X_1}(x_2|x_1)dx_2 = 1$.
- $f_{X_2|X_1}(x_2 | x_1) \geq 0$ for all $x_2 \in \mathcal{S}_{X_2}$.
- $f_{X_2|X_1}(x_2|x_1)$ is called the conditional pdf of the continuous type of random variable.

Conditional probability of continuous r.v.

- If the random variables are of the continuous type, the conditional probability that $a < X_2 < b$, given that $X_1 = x_1$, is

$$P[a < X_2 < b | X_1 = x_1] = \int_a^b f_{X_2|X_1}(x_2|x_1)dx_2$$

Conditional expectation of r.v.

- If $u(X_2)$ is a function of X_2 , the conditional expectation of $u(X_2)$, given that $X_1 = x_1$, if it exists, is given by

$$E[u(X_2)|X_1 = x_1] = \begin{cases} \sum_{x_2} u(x_2)p_{X_2|X_1}(x_2|x_1), & \text{Discrete} \\ \int_{x_2} u(x_2)f_{X_2|X_1}(x_2|x_1)dx_2, & \text{Continuous} \end{cases}$$

- The conditional variance of X_2 , given that $X_1 = x_1$, if it exists, is given by

$$\begin{aligned} \text{Var}(X_2 | X_1 = x_1) &= E[(X_2 - E[X_2 | X_1 = x_1])^2 | X_1 = x_1] \\ &= E[X_2^2 | X_1 = x_1] - (E[X_2 | X_1 = x_1])^2 \end{aligned}$$

Example 2.3.1

- Let X_1 and X_2 have the joint pdf

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} 2, & 0 < x_1 < x_2 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

- Find $f_{X_1|X_2}(x_1|x_2)$ and $f_{X_2|X_1}(x_2|x_1)$.

Example 2.3.1 (continued)

- Let X_1 and X_2 have the joint pdf

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} 2, & 0 < x_1 < x_2 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

- Find $E[X_1|x_2]$ and $\text{Var}(X_1|x_2)$.

Supplementary Example 2.3-A — Extension of Example 2.3.1

- Let X_1 and X_2 have the joint pdf

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} 2, & 0 < x_1 < x_2 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

- Find $P[0 < X_1 < \frac{1}{2} | X_2 = \frac{3}{4}]$ and $P[0 < X_1 < \frac{1}{2}]$

Theorem 2.3.1 — Conditional Mean and Variance Identities

定理 2.2.

- Let (X_1, X_2) be a random vector such that the variance of X_2 is finite.
- We have
 - $E[E[X_2|X_1]] = E[X_2]$.
 - $\text{Var}(E[X_2|X_1]) \leq \text{Var}(X_2)$.

2.6 2.4 Independent Random Variables**Definition 2.4.1 — Independence**

定義 2.6.

- Let the random variables X_1 and X_2 have the joint pdf $f(x_1, x_2)$ [joint pmf $p(x_1, x_2)$].
- Let the marginal pdfs [pmfs] $f_1(x_1)[p_1(x_1)]$ and $f_2(x_2)[p_2(x_2)]$.
- X_1 and X_2 are said to be independent if, and only if,

$$f(x_1, x_2) \equiv f_1(x_1)f_2(x_2) [p(x_1, x_2) \equiv p_1(x_1)p_2(x_2)].$$

- Random variables that are not independent are said to be dependent.

Remark

- The product of two positive functions $f_1(x_1)f_2(x_2)$ means a function that is positive on the product space.
- There may be certain points $(x_1, x_2) \in \mathcal{S}$ at which $f(x_1, x_2) \neq f_1(x_1)f_2(x_2)$. However, if A is the set of points (x_1, x_2) at which the equality does not hold, then $P(A) = 0$.

Example 2.4.1 — Sampling Without Replacement

- Suppose an urn contains 10 blue, 8 red, and 7 yellow balls that are the same except for color.
- Suppose 4 balls are drawn without replacement.
- Let X and Y be the number of red and blue balls drawn.
- Are X and Y independent?

Theorem 2.4.1 — Factorization Criterion

定理 2.3.

- Let the random variables X_1 and X_2 have supports $\mathcal{S}_1$ and $\mathcal{S}_2$, respectively, and have the joint pdf $f(x_1, x_2)$.
- Then X_1 and X_2 are independent if and only if $f(x_1, x_2)$ can be written as a product of a nonnegative function of x_1 and a nonnegative function of x_2 .
- That is,

$$f(x_1, x_2) \equiv g(x_1)h(x_2),$$

where $g(x_1) > 0$, $x_1 \in \mathcal{S}_1$, zero elsewhere, and $h(x_2) > 0$, $x_2 \in \mathcal{S}_2$, zero elsewhere.

Supplementary Example 2.4-A — Product-form Density

- Let X_1 and X_2 have the joint pdf

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} 4xe^{-x^2}ye^{-y^2}, & x > 0, y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

- Are X_1 and X_2 independent?

Example 2.4.3 — A Non-product Support

- Let X_1 and X_2 have the joint pdf

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} 8x_1x_2 & 0 < x_1 < x_2 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

- Are X and Y independent?

Theorem 2.4.2 — CDF Criterion

定理 2.4.

- Let X_1 and X_2 have the joint cdf $F(x_1, x_2)$.
- Let X_1 and X_2 have the marginal cdfs $F_1(x_1)$ and $F_2(x_2)$.

- Then X_1 and X_2 are independent if and only if

$$F(x_1, x_2) = F_1(x_1)F_2(x_2)$$

for all $(x_1, x_2) \in R^2$.

Theorem 2.4.3 — Rectangle Criterion

定理 2.5.

- X_1 and X_2 are independent if and only if

$$P[a < X_1 \leq b, c < X_2 \leq d] = P[a < X_1 \leq b]P[c < X_2 \leq d]$$

for every $a < b$ and $c < d$, where a, b, c, d are constants.

Theorem 2.4.4 — Products of Functions

定理 2.6.

- Suppose X_1 and X_2 are independent.
- Suppose $E[u(X_1)] < \infty$ and $E[\nu(X_2)] < \infty$.
- We have

$$E[u(X_1)\nu(X_2)] = E[u(X_1)]E[\nu(X_2)].$$

Theorem 2.4.5 — MGF Factorization

定理 2.7.

- Suppose $M(t_1, t_2) < \infty$.
- Then X_1 and X_2 are independent if and only if

$$M(t_1, t_2) = M(t_1, 0)M(0, t_2).$$

Example 2.4.5 — Dependence from the Joint MGF

- Let X_1 and X_2 have the joint pdf

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} e^{-x_2} & 0 < x_1 < x_2 < \infty \\ 0 & \text{elsewhere} \end{cases}$$

- Find $M(t_1, t_2)$.
- Show that $M(t_1, t_2) \neq M(t_1, 0)M(0, t_2)$.

2.7 2.5 The Correlation Coefficient

Covariance

定義 2.7.

- Let (X, Y) have a joint distribution.
- Denote the means of X and Y by μ_1 and μ_2 and their respective variances by σ_1^2 and σ_2^2 .
- The covariance of (X, Y) is denoted by $\text{cov}(X, Y)$ and is defined by the expectation

$$\text{cov}(X, Y) = E[(X - \mu_1)(Y - \mu_2)].$$

Note

- The covariance of X and Y can also be expressed as

$$\text{cov}(X, Y) = E[XY] - \mu_1\mu_2.$$

Correlation coefficient

定義 2.8.

- If each of σ_1 and σ_2 is positive, then the correlation coefficient between X and Y is defined by

$$\rho = \frac{\text{cov}(X, Y)}{\sigma_1\sigma_2}.$$

Note

- The expected value of the product of two random variables is equal to the product of their expectations plus their covariance;

$$E[XY] = \mu_1\mu_2 + \text{Cov}(X, Y) = \mu_1\mu_2 + \rho\sigma_1\sigma_2.$$

Example 2.5.1 — Coin-flip Random Vector

- A fair coin is flipped three times.
- X_1 is the number of heads on the first two flips while X_2 is the number of heads on all three flips.
- Find the correlation coefficient of (X_1, X_2) .

Theorem 2.5.1 — Bounds for Correlation¹

定理 2.8.

- For all jointly distributed random variables (X, Y) whose correlation coefficient ρ exists, $-1 \leq \rho \leq 1$.

¹Hint: Define $h(v) = E\{[(X - \mu_1) + v(Y - \mu_2)]^2\}$

Note

- A quadratic equation is said positive when its discriminant is negative.
- Define the quadratic equation as $y = ax^2 + bx + c$.
- The discriminant is $b^2 - 4ac$.

Theorem 2.5.2 — Independence Implies Zero Covariance

定理 2.9.

- If X and Y are independent random variables, then

$$\rho = 0.$$

Example 2.5.3 — Uncorrelated but Dependent

- Let X and Y be jointly discrete random variables whose distribution has mass 1/4 at each of the four points $(-1, 0)$, $(0, -1)$, $(1, 0)$ and $(0, 1)$.
- Find the correlation coefficient of (X_1, X_2) .
- Are X and Y independent?

Theorem 2.5.3 — Linear Conditional Means

定理 2.10.

- Suppose (X, Y) have a joint distribution with the variances of X and Y finite and positive.
- Denote the means and variances of X and Y by μ_1 , μ_2 and σ_1^2 , σ_2^2 .
- Let ρ be the correlation coefficient between X and Y .
- If $E[Y|X]$ is linear in X then

$$E[Y|X] = \mu_2 + \rho \frac{\sigma_2}{\sigma_1} (X - \mu_1)$$

and

$$E[\text{Var}(Y|X)] = \sigma_2^2(1 - \rho^2).$$

Example 2.5.4 — Linear Conditional Means

- Let the random variables X and Y have the linear conditional means $E(Y|x) = 4x + 3$ and $E(X|y) = \frac{1}{16}y - 3$.
- Can μ_1 , μ_2 , ρ , σ_1 and σ_2 be obtained?

Stage 2 — Deepen：由二維推廣到多維

從兩個變數的逐項計算，提升為 **random vector**、**matrix** 與一般維度的表示。

大二教材主要集中在雙變量情形；本階段是本章的主要新內容。學習重點是用向量與矩陣統整平均數、變異數與共變異數，理解 **covariance matrix** 的結構，並將二維 **transformation** 的方法推廣到多維與 many-to-one 情形。

2.8 2.6 Extension to Several Random Variables

Random vector

定義 2.9.

- Consider a random experiment with a sample space $\mathcal{C}$.
- Let the random variable X_i assign to each element $c \in \mathcal{C}$ one and only one real number $X_i(c) = x_i, i = 1, 2, \dots, n$.
- Then,
 - $X = (X_1, X_2, \dots, X_n)$ is a n -dimensional random vector.
 - The space of X is the set of ordered n -tuples

$$\mathcal{D} = \{(x_1, x_2, \dots, x_n) : x_1 = X_1(c), \dots, x_n = X_n(c), c \in \mathcal{C}\}.$$

- Furthermore, let A be a subset of the space $\mathcal{D}$. Then $P[(X_1, \dots, X_n) \in A] = P[C]$, where $C = \{c : c \in \mathcal{C} \text{ and } (X_1(c), X_2(c), \dots, X_n(c)) \in A\}$.

Cumulative distribution function (cdf)

- Let $A \subset \mathcal{D}$.
- Denote the probability of the event A as $P_{X_1, X_2, \dots, X_n}[A]$.
- $P_{X_1, X_2, \dots, X_n}[A]$ is uniquely determined by the cumulative distribution function (cdf), which is given by

$$F_X(x_1, x_2, \dots, x_n) = P[X_1 \leq x_1, \dots, X_n \leq x_n],$$

for all $(x_1, x_2, \dots, x_n) \in \mathbb{R}^n$.

Cumulative distribution function (cdf)

- Discrete random vector

$$F_X(x_1, x_2, \dots, x_n) = \sum \cdots \sum_{\omega_1 \leq x_1, \dots, \omega_n \leq x_n} p(\omega_1, \dots, \omega_n)$$

- Continuous random vector

$$F_X(x_1, x_2, \dots, x_n) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} f(\omega_1, \dots, \omega_n) d\omega_1 \cdots d\omega_n.$$

Moreover,

$$\frac{\partial^n}{\partial x_1 \cdots \partial x_n} F_X(x) = f(x).$$

Example 2.6.1

- Let the random variables X, Y and Z have the joint pdf defined as

$$f(x, y, z) = \begin{cases} e^{-x+y+z} & 0 < x, y, z < \infty \\ 0 & \text{elsewhere} \end{cases}$$

- Find the distribution function of X, Y and Z .

Expectation for discrete random vector

- Let $X = (X_1, X_2, \dots, X_n)$ be a random vector.
- Let $Y = u(X_1, X_2, \dots, X_n)$ for some function u .
- Then

$$E[u(X_1, X_2, \dots, X_n)] = \sum \cdots \sum_{(x_1, \dots, x_n) \in \mathcal{D}} u(x_1, \dots, x_n) f(x_1, \dots, x_n),$$

if it exists, is called the mathematical expectation of $u(X_1, \dots, X_n)$.

Expectation for continuous random vector

- Let $X = (X_1, X_2, \dots, X_n)$ be a random vector.
- Let $Y = u(X_1, X_2, \dots, X_n)$ for some function u .
- Then

$$E[u(X_1, \dots, X_n)] = \int \cdots \int_{(x_1, \dots, x_n) \in \mathcal{D}} u(x_1, \dots, x_n) f(x_1, \dots, x_n) dx_1 \cdots dx_n,$$

if it exists, is called the mathematical expectation of $u(X_1, \dots, X_n)$.

Marginal pmf

- Let $(X_1, \dots, X_n)$ be a random vector.
- Find the pmf of X_1 .

Note

- Take any group of $k < n$ of these random variables and find the joint pmf of them.
- This joint pmf is called the marginal pmf of this particular group of k variables.

Marginal pdf

- Let $(X_1, \dots, X_n)$ be a random vector.
- Find the probability of $\{X_1 \leq x_1\}$ and the pdf of X_1 .

Note

- Take any group of $k < n$ of these random variables and find the joint pdf of them.
- This joint pdf is called the marginal pdf of this particular group of k variables.

Conditional distribution for discrete type

- Let x_1 be a point in the support of X_1 , i.e., $p_{X_1}(x_1) > 0$.
- Using the definition of conditional probability, we have

$$\begin{aligned} p_{X_2, \dots, X_n | X_1}(x_2, \dots, x_n | x_1) &= P[X_2 = x_2, \dots, X_n = x_n | X_1 = x_1] \\ &= \frac{p_X(x_1, x_2, \dots, x_n)}{p_{X_1}(x_1)}. \end{aligned}$$

Conditional distribution for continuous type

- When $f_{X_1}(x_1) > 0$, we define

$$f_{X_2, \dots, X_n | X_1}(x_2, \dots, x_n | x_1) = \frac{f_X(x_1, x_2, \dots, x_n)}{f_{X_1}(x_1)}.$$

- The joint conditional pdf of any $n - 1$ random variables, say $X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n$, given $X_i = x_i$, is defined as the joint pdf of $X_1, \dots, X_n$ divided by the marginal pdf $f_{X_i}(x_i)$, provided that $f_{X_i}(x_i) > 0$.

Conditional expectation of a discrete random vector

- If $u(X_2, \dots, X_n)$ is a function of $X_2, \dots, X_n$, the conditional expectation of $u(X_2, \dots, X_n)$, given that $X_1 = x_1$, if it exists, is given by

$$E[u(X_2, \dots, X_n) | X_1 = x_1] = \sum \cdots \sum_{x_2, \dots, x_n} u(x_2, \dots, x_n) p_{X_2, \dots, X_n | X_1}(x_2, \dots, x_n | x_1),$$

Conditional expectation of a continuous random vector

- If $u(X_2, \dots, X_n)$ is a function of $X_2, \dots, X_n$, the conditional expectation of $u(X_2, \dots, X_n)$, given that $X_1 = x_1$, if it exists, is given by

$$E[u(X_2, \dots, X_n) | X_1 = x_1] = \int \cdots \int_{\mathcal{D}_{2:n}} u(x_2, \dots, x_n) f_{X_2, \dots, X_n | X_1}(x_2, \dots, x_n | x_1) dx_2 \cdots dx_n,$$

Independence

定義 2.10.

- Let the random variables $X_1, \dots, X_n$ have the joint pdf $f(x_1, \dots, x_n)$ [joint pmf $p(x_1, \dots, x_n)$].
- Let the marginal pdfs [pmfs] $f_1(x_1)[p_1(x_1)], \dots, f_n(x_n)[p_n(x_n)]$.
- $X_1, \dots, X_n$ are said to be mutually independent if, and only if,

$$f(x_1, \dots, x_n) \equiv f_1(x_1) \cdots f_n(x_n) [p(x_1, \dots, x_n) \equiv p_1(x_1) \cdots p_n(x_n)].$$

- Random variables that are not independent are said to be dependent.

Product Expectation for Independent Variables

HMC8 unnumbered result

- Suppose $X_1, \dots, X_n$ are independent.
- Suppose $E[u(X_i)] < \infty, i = 1, 2, \dots, n$.
- We have

$$E[u(X_1) \cdots u(X_n)] = E[u(X_1)] \cdots E[u(X_n)] = \prod_{i=1}^n E[u(X_i)]$$

Moment generating function

定義 2.11.

- Let $X = (X_1, X_2, \dots, X_n)$ be a random vector.
- If $E[e^{\sum_{i=1}^n t_i X_i}]$ exists for $|t_i| < h_i, i = 1, \dots, n$ where $h_i > 0, i = 1, \dots, n$.
- It is denoted by $M_X(t_1, \dots, t_n)$ and is called the moment generating function (mgf) of X .

Theorem 2.6.1 — MGF of a Linear Combination

定理 2.11.

- Suppose $X_1, X_2, \dots, X_n$ are n mutually independent random variables.
- Suppose, for all $i = 1, 2, \dots, n$, X_i has mgf $M_i(t)$, for $|t| < h_i$, where $h_i > 0$.
- Let $T = \sum_{i=1}^n k_i X_i$, where $k_1, k_2, \dots, k_n$ are constants.
- Then T has the mgf given by

$$M_T(t) = \prod_{i=1}^n M_i(k_i t),$$

where $-\min_i(h_i) < t < \min_i h_i$.

Example 2.6.2 — Maximum of Independent Variables

- Let the random variables X_1, X_2 and X_3 be three mutually independent random variable and let each have the pdf

$$f(x) = \begin{cases} 2x & 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

- Find the distribution function and pdf of $Y = \max_i X_i$.

Note

- Unless there is a possible misunderstanding between mutual and pairwise independence, we usually drop the modifier mutual.
- If several random variables are mutually independent and have the same distribution, we say that they are independent and identically distributed, which we abbreviate as iid.

Corollary 2.6.1 — MGF of an iid Sum

推論 2.1.

- Suppose $X_1, X_2, \dots, X_n$ are iid random variable with the common mgf $M(t)$, for $|t| < h$, where $h > 0$.
- Let $T = \sum_{i=1}^n X_i$.
- Then T has the mgf given by

$$M_T(t) = [M(t)]^n,$$

where $|t| < h$.

2.6.1 Multivariate Variance-Covariance Matrix

Random matrix

- Let $X = (X_1, \dots, X_n)'$ be an n -dimensional random vector.
- Define $E(X) = (E[X_1], \dots, E[X_n])'$.
- Suppose W is an $m \times n$ matrix of random variables with the ij th component W_{ij} , $1 \leq i \leq m$ and $1 \leq j \leq n$.
- Define the expectation of a random matrix

$$E[W] = [E(W_{ij})].$$

Theorem 2.6.2 — Linearity for Random Matrices

定理 2.12.

- Let W_1 and W_2 be $m \times n$ matrices of random variables.
- Let A_1 and A_2 be $k \times m$ matrices of constants.

- Let B be an $n \times l$ matrix of constants.
- Then, we have
 - $E[A_1 W_1 + A_2 W_2] = A_1 E[W_1] + A_2 E[W_2]$.
 - $E[A_1 W_1 B] = A_1 E[W_1] B$

Variance-covariance matrix

- Let $X = (X_1, \dots, X_n)'$ be an n -dimensional random vector, such that $\sigma_i^2 = \sigma_{ii} = \text{Var}(X_i) < \infty$.
- The mean of X is $\mu = E[X]$, and we define its variance-covariance matrix as

$$\text{Cov}(X) = E[(X - \mu)(X - \mu)'] = [\sigma_{ij}]$$

whose the (i, j) th off diagonal entry is $\text{Cov}(X_i, X_j)$ and the (i, i) th diagonal entry is $\sigma_i^2 = \text{Var}(X_i)$.

Example 2.6.3

- Let the random variables X and Y have the joint pdf of X and Y is

$$f_{X,Y}(x, y) = \begin{cases} e^{-y} & 0 < x < y < \infty \\ 0 & \text{elsewhere} \end{cases}$$

- Let $Z = (X, Y)'$.
- Find $E[Z]$ and $\text{Cov}(Z)$.

Theorem 2.6.3 — Covariance Matrix Identities

定理 2.13.

- Let $X = (X_1, \dots, X_n)'$ be an n -dimensional random vector, such that $\sigma_i^2 = \sigma_{ii} = \text{Var}(X_i) < \infty$.
- Let A be an $m \times n$ matrices of constants.
- Then, we have
 - $\text{Cov}(X) = E[XX'] - \mu\mu'$
 - $\text{Cov}(AX) = A \text{Cov}(X) A'$.

Note

- All variance-covariance matrices are positive semidefinite; that is,

$$a' \text{Cov}(X) a \geq 0,$$

for all vectors $a \in \mathbb{R}^n$.

- Let X be a random vector.
- Let a be any $n \times 1$ vector of constants.
- Then $Y = a'X$ is a random variable and, hence, has nonnegative variance; i.e.,

$$0 \leq \text{Var}(Y) = \text{Var}(a'X) = a' \text{Cov}(X)a.$$

2.9 2.7 Transformations for Several Random Variables

Continuous transformation

- Let $(X_1, \dots, X_n)$ have a jointly continuous distribution with pdf $f(x_1, \dots, x_n)$ and support set $\mathcal{S}$.
- Consider a one-to-one transformation that maps $\mathcal{S}$ onto $\mathcal{T}$.
- Let $y_1 = u_1(x_1, x_2, \dots, x_n)$, $y_2 = u_2(x_1, x_2, \dots, x_n)$, $\dots$ and $y_n = u_n(x_1, x_2, \dots, x_n)$.
- Let the inverse function be denoted as

$$\begin{aligned} x_1 &= w_1(y_1, y_2, \dots, y_n) \\ x_2 &= w_2(y_1, y_2, \dots, y_n) \\ &\vdots \\ x_n &= w_n(y_1, y_2, \dots, y_n) \end{aligned}$$

Continuous transformation

- Define the Jacobian of the transformation as

$$J = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} & \dots & \frac{\partial x_1}{\partial y_n} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} & \dots & \frac{\partial x_2}{\partial y_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x_n}{\partial y_1} & \frac{\partial x_n}{\partial y_2} & \dots & \frac{\partial x_n}{\partial y_n} \end{vmatrix}$$

- J can not be identically zero in $\mathcal{T}$.

Continuous transformation

- Based on the change-in-variable technique, we have

$$\int \dots \int_A f(x_1, x_2, \dots, x_n) dx_1 \dots dx_n = \int \dots \int_B f(w_1(y), \dots, w_n(y)) |J| dy_1 \dots dy_n$$

- The pdf of $(Y_1, Y_2, \dots, Y_n)$ is

$$f_Y(y_1, y_2, \dots, y_n) = \begin{cases} f(w_1(y_1, y_2, \dots, y_n), \dots, w_n(y_1, y_2, \dots, y_n)) |J|, & y \in \mathcal{T} \\ 0 & \text{elsewhere} \end{cases}$$

Example 2.7.1

- Let X_1, X_2 and X_3 have the joint pdf

$$f(x_1, x_2, x_3) = \begin{cases} 48x_1x_2x_3 & 0 < x_1 < x_2 < x_3 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

- Find the joint pdf of Y_1, Y_2 and Y_3 .
- Find the marginal pdf of Y_1 and that of Y_2 .

Class Example 2.7-A — Many to one

- Let X have the Cauchy pdf

$$f(x) = \begin{cases} \frac{1}{\pi(1+x^2)}, & -\infty < x < \infty \\ 0 & \text{elsewhere} \end{cases}$$

- Let $Y = X^2$.
- Find the pdf of Y .

Continuous transformation

- Let $(X_1, \dots, X_n)$ have a jointly continuous distribution with pdf $f(x_1, \dots, x_n)$ and support set $\mathcal{S}$.
- Consider the transformation $y_1 = u_1(x_1, x_2, \dots, x_n), y_2 = u_2(x_1, x_2, \dots, x_n), \dots$ and $y_n = u_n(x_1, x_2, \dots, x_n)$, which maps $\mathcal{S}$ to $\mathcal{T}$.
- To each point of $\mathcal{S}$ there corresponds, of course, only one point in $\mathcal{T}$; but to a point in $\mathcal{T}$ there may correspond more than one point in $\mathcal{S}$.

Continuous transformation

- Suppose that we can represent $\mathcal{S}$ as the union of a finite number, say k , of mutually disjoint sets $A_1, A_2, \dots, A_k$ so that

$$y_1 = u_1(x_1, x_2, \dots, x_n), \dots, y_n = u_n(x_1, x_2, \dots, x_n)$$

define a one-to-one transformation of each A_i onto $\mathcal{T}$.

- To each point in $\mathcal{T}$, there corresponds exactly one point in each of $A_1, A_2, \dots, A_k$.
- For $i = 1, \dots, k$, let the inverse function be denoted as

$$\begin{aligned} x_1 &= w_{1i}(y_1, y_2, \dots, y_n) \\ x_2 &= w_{2i}(y_1, y_2, \dots, y_n) \\ &\vdots \\ x_n &= w_{ni}(y_1, y_2, \dots, y_n) \end{aligned}$$

Continuous transformation

- Define the Jacobian of the transformation as

$$J_i = \begin{vmatrix} \frac{\partial x_{1i}}{\partial y_1} & \frac{\partial x_{1i}}{\partial y_2} & \cdots & \frac{\partial x_{1i}}{\partial y_n} \\ \frac{\partial x_{2i}}{\partial y_1} & \frac{\partial x_{2i}}{\partial y_2} & \cdots & \frac{\partial x_{2i}}{\partial y_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x_{ni}}{\partial y_1} & \frac{\partial x_{ni}}{\partial y_2} & \cdots & \frac{\partial x_{ni}}{\partial y_n} \end{vmatrix}, i = 1, \dots, k.$$

- $J_i, i = 1, \dots, k$, can not be identically zero in $\mathcal{T}$.
- The pdf of $(Y_1, Y_2, \dots, Y_n)$ is

$$f_Y(y_1, y_2, \dots, y_n) = \sum_{i=1}^k f(w_{1i}(y_1, y_2, \dots, y_n), \dots, w_{ni}(y_1, y_2, \dots, y_n)) |J_i|$$

provided $y \in \mathcal{T}$.

Example 2.7.5

- Let X_1, X_2, X_3, X_4 be independent random variables with common pdf

$$f(x) = \begin{cases} e^{-x} & 0 < x \\ 0 & \text{elsewhere} \end{cases}$$

- Find the distribution function of $Y = \sum_{i=1}^4 X_i$ using the mgf technique.

Stage 3 — Connect : 連接抽樣分配與統計推論

把 **random vector** 的工具放回樣本 $X_1, \dots, X_n$ ，開始研究由樣本形成的統計量。

本階段以線性組合與 **random sample** 收束本章。學生應能計算線性統計量的期望、變異數與共變異數，辨認 $\bar{X}$ 與 S^2 是樣本的函數，並理解 $E(S^2) = \sigma^2$ 為何是後續估計理論的重要起點。

2.10 2.8 Linear Combinations of Random Variables

Linear combination

- Let $X = (X_1, \dots, X_n)'$ and $Y = (Y_1, \dots, Y_m)'$ denote two random vectors.
- Consider two linear combinations of these variables as

$$T = \sum_{i=1}^n a_i X_i$$

$$W = \sum_{i=1}^m b_i Y_i$$

for specified constants a_i and $b_i, i = 1, 2, \dots, n$.

- Suppose $E[X_i] = \mu_{X_i}, i = 1, 2, \dots, n$ and $E[Y_i] = \mu_{Y_i}, i = 1, 2, \dots, m$.

Theorem 2.8.1 — Mean of a Linear Combination

定理 2.14.

$$E[T] = \sum_{i=1}^n a_i \mu_{X_i}.$$

Theorem 2.8.2 — Covariance of Linear Combinations

定理 2.15.

$$\text{Cov}(T, W) = \sum_{i=1}^n \sum_{j=1}^m a_i b_j \text{Cov}(X_i, Y_j).$$

Corollary 2.8.1 — Variance of a Linear Combination

推論 2.2.

$$\text{Var}(T) = \sum_{i=1}^n a_i^2 \text{Var}(X_i) + 2 \sum_{i < j} a_i a_j \text{Cov}(X_i, X_j).$$

Corollary 2.8.2 — Independent Components

推論 2.3. *If $X_1, \dots, X_n$ are independent and $\text{Var}(X_i) = \sigma_i^2$, then*

$$\text{Var}(T) = \sum_{i=1}^n a_i^2 \sigma_i^2.$$

Definition 2.8.1 — Random Sample

定義 2.12.

- If the random variables $X_1, X_2, \dots, X_n$ are independent and identically distributed.
- Each X_i has the same distribution.
- These random variables constitute a random sample of size n from that common distribution.
- Abbreviate independent and identically distributed by iid.

Example 2.8.1 — Sample Mean

Let $X_1, \dots, X_n$ be iid with common mean μ and variance σ^2 . For

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i,$$

Theorem 2.8.1 and Corollary 2.8.2 give

$$E(\bar{X}) = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

Example 2.8.2 — Sample Variance

- Let $X_i, i = 1, 2, \dots, n$, be a random sample with mean μ and variance σ^2 .
- Define the sample variance as

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

- Show $E[S^2] = \sigma^2$.

2.11 本章摘要

本章把單變量分配推廣到隨機向量，核心工具是邊際化、條件化、Jacobian 與共變異結構。

2.12 常見錯誤

- 將零相關直接解讀為獨立
- Jacobian 使用方向錯誤
- 只求聯合密度而未確認轉換後支持集
- 忽略條件分配中的正規化常數

2.13 章末練習

1. 從本章選出三個關鍵定義，分別寫出數學形式、成立條件及一個例子。
2. 選擇一個主要定理或方法，重做推導並標記每一步使用的假設。
3. 從原講義例題挑選一題，完整寫出解題目標、方法選擇、計算與結果解釋。
4. 設計一個會違反本章方法條件的反例，說明錯誤會出現在哪一步。

待逐章補強

建議後續為本章補入：中文概念導讀、完整解答、課堂練習難度標記，以及與下一章的銜接說明。

3 常用機率分配

! 第 8 版對照與教學安排

本章以 Hogg, McKean & Craig, *Introduction to Mathematical Statistics*, 8th Global Edition (以下簡稱 HMC8) Chapter 3 為編號依據。保留原講義的公式與例題，但不按篇幅平均講授：先喚醒大二已學過的常用分配，再深化分配之間的結構與推導，最後銜接後續抽樣分配與統計推論。

Recall → Deepen → Connect

💡 本章定位

本講義由蘇南誠老師編授。感謝黃怡婷老師提供原始數理統計講義檔案；本章保留原有英文術語與公式，並整理為可連續閱讀的教學形式。

3.1 學習目標

完成本章後，學生應能：

- 辨認二項、Poisson 與相關離散分配
- 掌握 Gamma、卡方與 Beta 分配
- 運用常態與多變量常態分配
- 說明 t 與 F 分配的生成方式
- 比較各分配的參數、支持集、期望與變異數

3.2 必備的先備知識

第 1 至 2 章的機率質量函數、密度函數與變數轉換。

以下依原講義順序整理核心定義、定理、公式與例題。

Stage 1 — Recall：常用離散分配快速複習

本階段以例題帶回 Bernoulli、binomial、negative binomial、geometric、multinomial、hypergeometric 與 Poisson 分配；定義與基本動差不重新等比例講解。

3.3 3.1 The Binomial and Related Distributions

Bernoulli experiment distribution

- A Bernoulli experiment is a random experiment.
- The outcome only has two mutually exclusive and exhaustive ways, say, success and failure.

Bernoulli distribution

- Let X denote a random variable associated with a Bernoulli experiment.
- Define

$$X(\text{success}) = 1 \quad \text{and} \quad X(\text{failure}) = 0$$

- The pmf of X is

$$f(x) = p^x(1-p)^{1-x}, \quad x = 0, 1.$$

Moment of Bernoulli distribution

- The expected value of X is

$$\mu = p.$$

- The variance of X is

$$\sigma^2 = pq.$$

Binomial distribution

- A sequence of n Bernoulli trials occurs.
 - A Bernoulli experiment is performed several independent times.
 - The probability of success remains the same from trial to trial.
- Let X denote the number of successes in n Bernoulli trials.
- The pmf of X is

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, \dots, n.$$

Moment generating function

- Show that the mgf of a binomial distribution is

$$M(t) = [(1-p) + pe^t]^n.$$

Example 1

- Let Y be equal to the number of successes throughout n independent repetitions of a random experiment with probability p of success.
- Find the upper bound of $P[|\frac{Y}{n} - p| \geq \epsilon]$.

Example 2

- Let the independent random variables X_1, X_2 and X_3 have the same cdf $F(x)$.
- Let Y be the middle value of X_1, X_2 and X_3 .
- Find $F_Y(y)$.

Negative binomial distribution

- A sequence of Bernoulli trials occurs
 - A Bernoulli experiment is performed several independent times.
 - The probability of success remains the same from trial to trial.
- Let X denote the number of trials where the r th success is observed.
- The pmf of X is

$$f(x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \quad x = r, r+1, r+2, \dots$$

Recall

- Let $h(x) = (1-x)^{-r}$.
- Based on Maclaurin's series expansion, we have

$$(1-x)^{-r} = \sum_{k=0}^{\infty} \frac{h^{(k)}(0)}{k!} x^k = \sum_{k=0}^{\infty} \binom{r+k-1}{r-1} x^k, \quad -1 < x < 1.$$

Probability of mass function

- Show that

$$\sum_{x=r}^{\infty} \binom{x-1}{r-1} p^r (1-p)^{x-r} = 1$$

Moment generating function

- Show that

$$M(t) = E[e^{tX}] = \frac{(pe^t)^r}{[1 - (1-p)e^t]^r}, \quad (1-p)e^t < 1.$$

Moment of Negative binomial distribution

- The expected value of X is

$$\mu = \frac{r}{p}.$$

- The variance of X is

$$\sigma^2 = \frac{rq}{p}.$$

Geometric distribution

- Let X denote the number of trials where the $r = 1$ th success is observed.
- The pmf of X is

$$f(x) = p(1-p)^{x-1}, \quad x = 1, 2, 3, \dots$$

Recall

- For a geometric series, the sum equals

$$\sum_{k=0}^{\infty} ax^k = \sum_{k=1}^{\infty} ax^{k-1} = \frac{a}{1-x}, \quad -1 < x < 1.$$

Probability of mass function of a geometric distribution

- Show that

$$\sum_{x=1}^{\infty} p(1-p)^{x-1} = 1$$

Distribution function of a geometric distribution

- Find $P[X > k]$ and $P[X \leq k]$.

Additive properties

定理 3.1.

- Let $X_1, X_2, \dots, X_m$ be independent random variables such that $X_i \sim \text{Bin}(n_i, p)$, for $i = 1, 2, \dots, m$.
- The distribution of $Y = \sum_{i=1}^m X_i$ is a $\text{Bin}(\sum_{i=1}^m n_i, p)$ distribution.

Settings for multinomial distribution

- Let a random experiment be repeated n independent times.
- On each repetition, there is one and only one outcome from one of k categories.
- Call the categories $C_1, C_2, \dots, C_k$.
- For $i = 1, \dots, k$, let p_i be the probability that the outcome is an element of C_i .
- Assume that p_i remains constant throughout the n independent repetitions.

PMF of multinomial distribution

- Define the random variable X_i to be equal to the number of outcomes that are elements of C_i , $i = 1, 2, \dots, k$, where $X_k = n - \sum_{i=1}^{k-1} X_i$.

- The joint pmf of $(X_1, \dots, X_{k-1})$ is

$$P[X_1 = x_1, X_2 = x_2, \dots, X_{k-1} = x_{k-1}] = \frac{n!}{x_1! \dots x_{k-1}! x_k!} p_1^{x_1} \dots p_k^{x_k},$$

for $0 \leq x_i \leq n, i = 1, \dots, k, \sum_{i=1}^k x_i = n$ and $\sum_{i=1}^k p_i = 1$.

Moment generating function

- Show that

$$M(t_1, \dots, t_{k-1}) = E[e^{t'X}] = \left(\sum_{i=1}^{k-1} p_i e^{t_i} + p_k \right)^n.$$

Example 3

- Suppose (X_1, X_2, X_3) has a multinomial distribution (n, p_1, p_2, p_3) .
- Find the marginal pmf of X_i .
- Find the conditional pmf of X_1 given X_2 .

Example 4

- Suppose (X_1, X_2, X_3) has a multinomial distribution (n, p_1, p_2, p_3) .
- Find the correlation coefficient of X_1 and X_2 .

Example 5 — Binomial distribution

- Suppose we have a lot of N items of which D are defective.
- Let X denote the number of defective items in a sample of size n .
- If the sampling is done with replacement and the items are chosen at random, then X has a binomial distribution with parameters n and D/N .
 - Mean $E[X] = n(D/N)$
 - Variance $\text{Var}(X) = n(D/N)[(N - D)/N]$

Example 6 — Hypergeometric distribution

- Suppose we have N items of which D are defective.
- Let X denote the number of defective items in a sample of size n .
- If the sampling is done without replacement and the items are chosen at random, then X has a hypergeometric distribution with pmf

$$p(x) = \frac{\binom{N-D}{n-x} \binom{D}{x}}{\binom{N}{n}}, x = 0, 1, \dots, n.$$

Example 7 — Hypergeometric distribution

- Show that

$$E[X] = n \frac{D}{N}$$

Example 8 — Hypergeometric distribution

- Show that

$$\text{Var}(X) = n \frac{D}{N} \frac{N-D}{N} \frac{N-n}{N-1}$$

3.4 3.2 The Poisson Distribution

Motivation

- Suppose the number of cell phone calls passing through a relay tower between 9 and 10 A.M. is counted.
- Suppose the number of flaws in 100 feet of wire is counted.
- Suppose the number of customers that arrive at a ticket window between 12 noon and 2 P.M. is counted.

Definition

- Let X_t denote the number of occurrences of such a process over the interval $(0, t]$.
- The range of X_t is the set of nonnegative integers $\{0, 1, 2, \dots\}$.
- Let $g(x, t)$ denote the probability of x changes in each interval of length t , i.e.

$$g(x, t) = P[X_t = x]$$

Definition

- An approximate Poisson process with parameter $\lambda > 0$ if the following conditions are satisfied:
 1. The number of occurrences in nonoverlapping subintervals are independent.
 2. The probability of exactly one occurrence in a sufficiently short subinterval of length h is approximately λh .

$$g(1, h) = \lambda h + o(h)$$

3. The probability of two or more occurrences in a sufficiently short subinterval is essentially zero.

$$\sum_{t=2}^{\infty} g(t, h) = o(h).$$

Note

- $o(h)$ notation means that $o(h)/h \rightarrow 0$ as $h \rightarrow 0$.
- For instance, $h^2 = o(h)$ and $o(h) + o(h) = o(h)$.

Proof I

- Show $g(0, t) = e^{-\lambda t}$.

Proof II

- Show $g(x, t) = \frac{(\lambda t)^x e^{-\lambda t}}{x!}$, $x = 1, 2, 3, \dots$

Recall

$$\begin{aligned}\lim_{n \rightarrow \infty} \frac{n(n-1) \cdots (n-x+1)}{n^x} &= \lim_{n \rightarrow \infty} \left[(1) \left(1 - \frac{1}{n}\right) \cdots \left(1 - \frac{x-1}{n}\right) \right] = 1 \\ \lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^n &= e^{-\lambda} \\ \lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^x &= 1\end{aligned}$$

Poisson distribution

- X has a Poisson distribution if its pmf is of the form

$$f(x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x = 0, 1, 2, \dots$$

- Show $\sum_{x=0}^{\infty} f(x) = 1$.

Moment

- Show that $E[X] = \lambda$ and $\text{Var}(X) = \lambda$.

Moment generating function

- Show that

$$M(t) = e^{\lambda(e^t - 1)}, \quad t \in \mathbb{R}.$$

Additive properties

定理 3.2.

- Let $X_1, X_2, \dots, X_n$ be independent random variables such that $X_i \sim \text{Poisson}(m_i)$, for $i = 1, 2, \dots, n$.
- The distribution of $Y = \sum_{i=1}^n X_i$ is a $\text{Poisson}(\sum_{i=1}^n m_i)$ distribution.

Stage 2 — Deepen : 連續分配的結構與推導

本階段著重由 transformation、Gamma function、MGF 與加法性質建立分配間的關係，並把單變量結果推廣到常態向量。

3.5 3.3 The Γ , χ^2 , and β Distributions

Recall

- Poisson distribution with mean λ
 - Record the number of occurrence in a given interval.
- Exponential distribution
 - Denote the waiting time W between the occurrences.

Derivation of the exponential distribution function

- The cdf of the waiting time is

$$\begin{aligned}F(w) &= P[W \leq w] = 1 - P[W > w] \\&= 1 - P[\text{no occurrence in } [0, w]] \\&= 1 - e^{-\lambda w}\end{aligned}$$

for $w > 0$.

The pdf of an exponential distribution

- The pdf of the waiting time is

$$f(w) = \lambda e^{-\lambda w}$$

for $w > 0$.

An alternative representation of pdf

- The pdf of the waiting time is

$$f(w) = \frac{1}{\theta} e^{-w/\theta}$$

for $w > 0$.

Mgf of exponential distribution

- The mgf of the exponential distribution is

$$M(t) = \frac{1}{1 - \theta t}, t < \frac{1}{\theta}.$$

Special moments of exponential distribution

- The expected value is $\mu = \theta$.
- The variance is $\sigma^2 = \theta^2$.

Remark

- The exponential distribution has a forgetfulness properties, i.e.,

$$P[X > x + y | X > x] = P[X > y].$$

Derivation of the Gamma distribution function

- Let W denote the waiting time till the α th occurrence.
- The cdf of W is

$$\begin{aligned} F(w) &= P[W \leq w] = 1 - P[W > w] \\ &= 1 - P[\text{fewer than } \alpha \text{ occurrences in } [0, w]] \\ &= 1 - \sum_{k=0}^{\alpha-1} \frac{(\lambda w)^k e^{-\lambda w}}{k!} \end{aligned}$$

for $w > 0$.

The pdf of gamma distribution

- The pdf of W is

$$f(w) = \frac{\lambda^\alpha}{(\alpha - 1)!} w^{\alpha-1} e^{-\lambda w}$$

for $w > 0$.

The gamma function

- The gamma function is defined by

$$\Gamma(t) = \int_0^\infty y^{t-1} e^{-y} dy, t > 0.$$

for $w > 0$.

- When $t = n$ is a positive integer, then $\Gamma(n) = (n - 1)!$.

The gamma distribution

- The gamma distribution is defined as

$$f(x) = \frac{1}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} e^{-x/\theta}, 0 \leq x < \infty.$$

The gamma distribution

- Show that

$$\int_0^{\infty} f(x)dx = \int_0^{\infty} \frac{1}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} e^{-x/\theta} dx = 1.$$

Mgf of the gamma distribution

- Show that

$$M(t) = \frac{1}{(1 - \theta t)^\alpha}, \quad t < \frac{1}{\theta}.$$

Special moments of the gamma distribution

- Show that $\mu = \alpha\theta$ and $\sigma^2 = \alpha\theta^2$.

Chi-square distribution

- Let X have a gamma distribution with $\theta = 2$ and $\alpha = r/2$, where r is a positive integer.
- The pdf of X is

$$f(x) = \frac{1}{\Gamma(r/2)2^{r/2}} x^{r/2-1} e^{-x/2},$$

for $0 < x < \infty$.

Properties of the chi-square distribution

- The mgf is

$$M(t) = (1 - 2t)^{-r/2}, \quad t < \frac{1}{2}.$$

- $E[X] = r$ and $\sigma^2 = 2r$.

Percentiles

- Let α be a positive probability (usually less than 0.05).
- $X \sim \chi^2(r)$.
- Then $\chi_\alpha^2(r)$ is a number such that

$$P[X \geq \chi_\alpha^2(r)] = \alpha.$$

- $\chi_{1-\alpha}^2(r)$ is the $100(1 - \alpha)$ th percentile.
- The 100α percentile is the number such that

$$P[X \leq \chi_{1-\alpha}^2(r)] = 1 - \alpha.$$

Example 9

- Let X have a gamma distribution with $\alpha = r/2$, where r is a positive integer, and $\beta > 0$.
- Define the random variable $Y = 2\beta X$.
- Find the distribution of Y .

Beta distribution

- Let X_1 and X_2 be two independent random variables that have Γ distributions with parameters α and β .
- Let $Y_1 = X_1 + X_2$ and $Y_2 = X_1/(X_1 + X_2)$.
- Find the distribution of Y_2 .

Special moments of beta distribution

- $E[Y_2] = \frac{\alpha}{\alpha+\beta}$
- $\text{Var}(Y_2) = \frac{\alpha\beta}{(\alpha+\beta+1)(\alpha+\beta)^2}$

Example 10 — Dirichlet Distribution

- Let $X_1, X_2, \dots, X_{k+1}$ be independent random variables that have Γ distributions with parameters α_i and $\beta = 1, i = 1, 2, \dots, k + 1$.
- Let $Y_i = X_i/(X_1 + X_2 + \dots + X_{k+1}), i = 1, 2, \dots, k$.
- Find the joint distribution of $Y_i, i = 1, 2, \dots, k$.

3.6 3.4 The Normal Distribution

Normal distribution

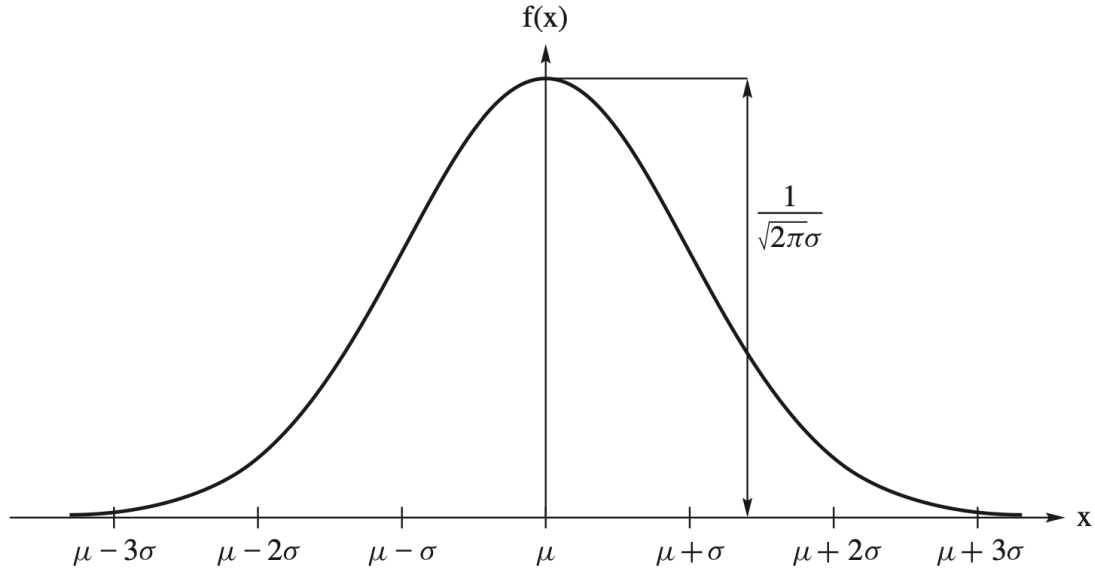
- Let X have a normal distribution.
- The pdf of X is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{(x - \mu)^2}{2\sigma^2} \right], -\infty < x < \infty,$$

where

- $-\infty < \mu < \infty$
- $0 < \sigma < \infty$
- $X \sim N(\mu, \sigma^2)$.

Pdf of normal distribution



Derivation

- Show that

$$\int_{-\infty}^{\infty} f(x) dx = 1.$$

Mgf of normal distribution

- Show that

$$M(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$$

Special moments of normal distribution

- Show that $E[X] = \mu$ and $\text{Var}(X) = \sigma^2$.

Standard normal distribution

- Let $Z \sim N(0, 1)$.
- Z has a standard normal distribution.
- The cdf of Z is

$$\Phi(z) = P[Z \leq z] = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-w^2/2} dw.$$

- Table values are provided.

Properties

- Show that $\Phi(z) = 1 - \Phi(-z)$.

Example 11

- Find
 - $P[Z \leq 1.24]$
 - $P[1.24 \leq Z \leq 2.37]$
 - $P[-2.37 \leq Z \leq -1.24]$

Example 12

- Find a and b such that
 - $P[Z \leq a] = 0.9147$ and
 - $P[Z \geq b] = 0.0526$.

Percentiles

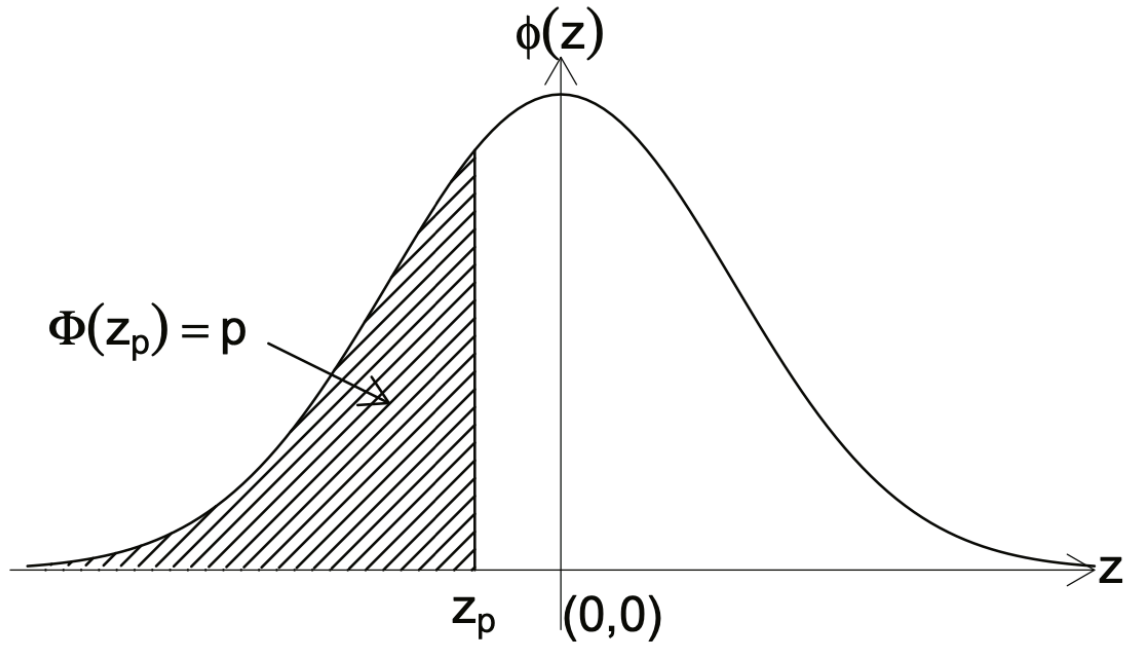
- Let α be a positive probability (usually less than 0.05).
- $Z \sim N(0, 1)$.
- Then z_α is a number such that

$$P[X \geq z_\alpha] = \alpha.$$

- z_α is the $100(1 - \alpha)$ th percentile.
- The 100α percentile is the number such that

$$P[X \leq z_\alpha] = 1 - \alpha.$$

Standard normal pdf



Example 13

- Show that $z_{1-\alpha} = -z_\alpha$.

Theorem

定理 3.3.

- If $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$.

Remark

- Let X is any random variable with $E[X] = \mu$ and $\text{Var}(X) = \sigma^2$.
- Let $Z = \frac{X-\mu}{\sigma}$.
- We have

$$\begin{aligned}\mu_Z &= E[Z] \\ \sigma_Z^2 &= E\left[\left(\frac{X-\mu}{\sigma}\right)^2\right]\end{aligned}$$

- Z is often called the standard score associated with X

Theorem

定理 3.4.

- If $X \sim N(\mu, \sigma^2)$, then $V = \left(\frac{X-\mu}{\sigma}\right)^2 \sim \chi^2(1)$.

Additive properties of normal distribution

定理 3.5.

- Let $X_1, \dots, X_n$ be independent random variables such that, for $i = 1, \dots, n$,
- X_i has a $N(\mu_i, \sigma_i^2)$ distribution.
- Let $Y = \sum_{i=1}^n a_i X_i$, where $a_1, \dots, a_n$ are constants.
- Then the distribution of Y is $N(\sum_{i=1}^n a_i \mu_i, \sum_{i=1}^n a_i^2 \sigma_i^2)$

3.7 3.5 The Multivariate Normal Distribution

Bivariate normal distribution

- Assume $X \sim N(\mu_X, \sigma_X^2)$.
- The joint pdf of X and Y is

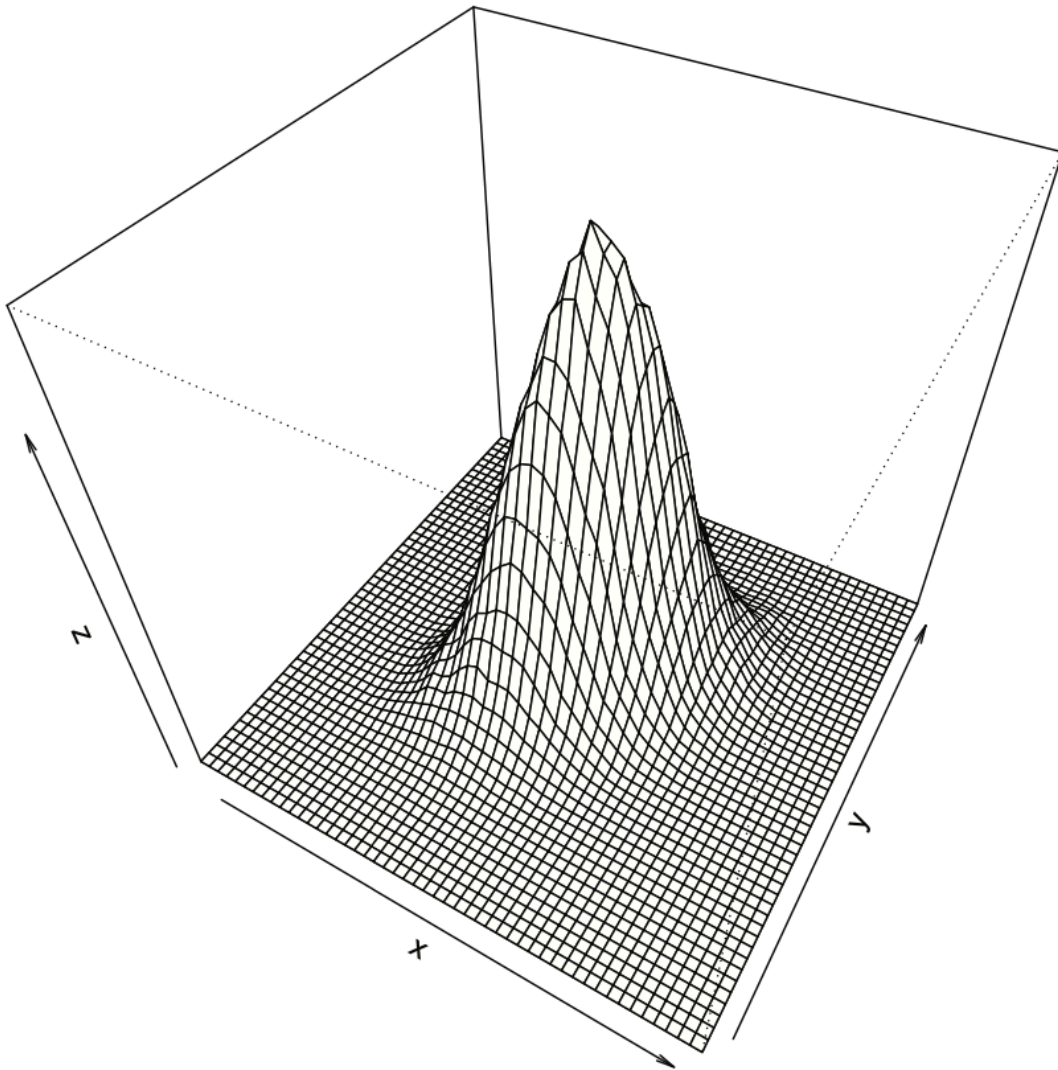
$$f(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left[-\frac{q(x, y)}{2}\right]$$

where

$$q(x, y) = \frac{1}{1-\rho^2} \left[\left(\frac{x-\mu_X}{\sigma_X} \right)^2 - 2\rho \left(\frac{x-\mu_X}{\sigma_X} \right) \left(\frac{y-\mu_Y}{\sigma_Y} \right) + \left(\frac{y-\mu_Y}{\sigma_Y} \right)^2 \right]$$

- It is called a bivariate normal pdf.

Bivariate normal with $\rho = 0.5$



Mgf

- Show that the joint mgf is

$$M(t_1, t_2) = \exp \left\{ t_1 \mu_1 + t_2 \mu_2 + \frac{1}{2} (t_1^2 \sigma_1^2 + 2t_1 t_2 \rho \sigma_1 \sigma_2 + t_2^2 \sigma_2^2) \right\}$$

Theorem

定理 3.6. *If X and Y have a bivariate normal distribution with correlation coefficient ρ , then X and Y are independent if and only if $\rho = 0$.*

Multivariate normal distribution

- Consider the random vector $Z = (Z_1, \dots, Z_n)'$, where $Z_1, \dots, Z_n$ are iid $N(0, 1)$.
- Then the joint density of Z is

$$f(z) = \left(\frac{1}{2\pi}\right)^{n/2} \exp\left\{-\frac{1}{2}z'z\right\}$$

for $z \in R^n$.

- Z has a multivariate normal distribution with mean vector 0 and covariance matrix I_n .
- We abbreviate this by saying that Z has an $N_n(0, I_n)$ distribution

Moments

- We have

$$\begin{aligned} E[Z] &= 0 \\ \text{Var}[Z] &= I_n \end{aligned}$$

Moment generating function

- We have

$$M(t) = \exp\left\{\frac{1}{2}t't\right\}$$

Spectral Decomposition of Covariance matrix

- Suppose Σ is a $n \times n$, symmetric, and positive semi-definite matrix.
- Σ can be decomposed as

$$\Sigma = \Gamma' \Lambda \Gamma,$$

where

- Λ is the diagonal matrix $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$,
- $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0$ are the eigenvalues of Σ ,
- the columns of Γ' , $v_1, v_2, \dots, v_n$, are the corresponding eigenvectors

Properties of orthogonal matrix

- Γ is an orthogonal matrix.
- $\Gamma^{-1} = \Gamma'$
- Thus, we have

$$\Gamma \Gamma' = I$$

Another representation of Σ

- Σ can be expressed as

$$\begin{aligned}\Sigma &= \Gamma' \Lambda \Gamma \\ &= [\Gamma' \Lambda^{1/2} \Gamma] [\Gamma' \Lambda^{1/2} \Gamma]\end{aligned}$$

- Denote

$$\Sigma^{1/2} = [\Gamma' \Lambda^{1/2} \Gamma]$$

- $\Sigma^{1/2}$ is symmetric and positive semi-definite.
- Suppose Σ is positive definite; that is, all of its eigenvalues are strictly positive. Then

$$(\Sigma^{1/2})^{-1} = [\Gamma' \Lambda^{-1/2} \Gamma]$$

Transformation of multivariate normal distribution

- Suppose Z has an $N_n(0, I_n)$ distribution.
- Let Σ be a positive semi-definite, symmetric matrix and let μ be an $n \times 1$ vector of constants.
- Define

$$X = \Sigma^{1/2} Z + \mu$$

Moments

- We have

$$\begin{aligned}E[X] &= \mu \\ \text{Var}[X] &= \Sigma\end{aligned}$$

Moment generating function

- We have

$$M(t) = \exp \left\{ t' \mu + \frac{1}{2} t' \Sigma t \right\}$$

Multivariate normal distribution

定義 3.1.

- X has a multivariate normal distribution if its mgf is

$$M(t) = \exp \left\{ t' \mu + \frac{1}{2} t' \Sigma t \right\}$$

for all $t \in R^n$.

- We say X has a $N_m(\mu, \Sigma)$.

Multivariate normal density function

- We have

$$Z = \Sigma^{-1/2}(X - \mu)$$

- The Jacobian equals $|\Sigma^{-1/2}| = |\Sigma|^{-1/2}$
- Then the joint density of X is

$$f(x) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2}(x - \mu)' \Sigma^{-1}(x - \mu) \right\}$$

for $x \in R^n$.

Distribution of quadratic term

定理 3.7.

- X has a $N_n(\mu, \Sigma)$, where Σ is positive definite.
- Then $Y = (X - \mu)' \Sigma^{-1}(X - \mu)$ has a χ_n^2 distribution.

Distribution of linear combination

定理 3.8.

- X has a $N_n(\mu, \Sigma)$, where Σ is positive definite.
- Let $Y = AX + b$, where A is an $m \times n$ matrix and $b \in R^m$.
- Then Y has a $N_m(A\mu + b, A\Sigma A')$ distribution.

Partition

- Let

$$X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

where X_1 is a subvector of dimension $m < n$ and X_2 is a subvector of dimension $p = n - m$.

- Let

$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$$

and

$$\Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$$

Distribution of X_1

- Let

$$A = [I_m : O_{m \times p}]$$

where $O_{m \times p}$ an $m \times p$ matrix of zeroes.

- Then $X_1 = AX$
- Suppose X has a $N_n(\mu, \Sigma)$ distribution.
- Let

$$X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

- Then X_1 has a $N_n(\mu_1, \Sigma_{11})$ distribution.

Independent partition

定理 3.9.

- Suppose X has a $N_n(\mu, \Sigma)$ distribution.
- Let

$$X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

- X_1 and X_2 are independent if and only if $\Sigma_{12} = 0$.

Conditional distribution

定理 3.10.

- Suppose X has a $N_n(\mu, \Sigma)$ distribution.
- Let

$$X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

- Assume Σ is positive definite.
- Then the conditional distribution of X_1 given X_2 is

$$N_m(\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(X_2 - \mu_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})$$

Special case $p = 2$

定理 3.11.

- Suppose X has a $N_2(\mu, \Sigma)$ distribution.
- Let

$$X = \begin{bmatrix} X \\ Y \end{bmatrix}$$

- Assume Σ is positive definite.
- Then the conditional distribution of Y given $X = x$ is

$$N(\mu_2 + \rho \frac{\sigma_2}{\sigma_1}(x - \mu_1), \sigma_2^2(1 - \rho^2))$$

Stage 3 — Connect : 抽樣分配與後續推論

本階段把常態、卡方、 t 與 F 分配連結到 sample mean、sample variance、Student's theorem、以及後續信賴區間與檢定。

3.8 3.6 t - and F -Distributions

Student's t distribution

定理 3.12.

- Let

$$T = \frac{Z}{\sqrt{U/r}}$$

where

- $Z \sim N(0, 1)$
- $U \sim \chi_r^2$
- Z and U are independent
- Then T has a t distribution with pdf

$$f(t) = \frac{\Gamma((r+1)/2)}{\sqrt{\pi r} \Gamma(r/2)} \frac{1}{(1 + t^2/r)^{(r+1)/2}}, -\infty < t < \infty.$$

Moment

- Show that

$$E[T] = 0$$

$$\text{Var}(T) = \frac{r}{r-2}, r > 2$$

Example 14 — F distribution

- Let U and V be independent chi-square random variables, with r_1 and r_2 degrees of freedom.

$$f(x_1) = \frac{1}{\Gamma(r_1/2) 2^{r_1/2}} x_1^{r_1/2-1} e^{-x_1/2}, 0 < x_1 < \infty$$

and

$$f(x_2) = \frac{1}{\Gamma(r_2/2) 2^{r_2/2}} x_2^{r_2/2-1} e^{-x_2/2}, 0 < x_2 < \infty$$

- Find the pdf of $F = \frac{U/r_1}{V/r_2}$.

Moment

- Show that

$$E[F] = \frac{r_2}{r_2 - 2}, r_2 > 2$$

Student's theorem

- Let $X_1, \dots, X_n$ be a random sample from $N(\mu, \sigma^2)$.
- Define $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ and $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$.
- Then

- $\bar{X} \sim N(\mu, \sigma^2/n)$
- $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$
- $\bar{X}$ and S^2 are independent.

–

$$T = \frac{\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}}{\sqrt{\frac{(n-1)S^2}{\sigma^2}/(n-1)}} = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

3.9 本章摘要

本章整理統計推論最常出現的分配家族及其生成關係，為抽樣分配、區間估計與檢定奠定基礎。

3.10 常見錯誤

- 混淆分配的參數化方式
- 忘記標示支持集
- 套用常態近似時未檢查條件
- 將卡方、t、F 的自由度視為一般尺度參數

3.11 章末練習

1. 從本章選出三個關鍵定義，分別寫出數學形式、成立條件及一個例子。
2. 選擇一個主要定理或方法，重做推導並標記每一步使用的假設。
3. 從原講義例題挑選一題，完整寫出解題目標、方法選擇、計算與結果解釋。
4. 設計一個會違反本章方法條件的反例，說明錯誤會出現在哪一步。

待逐章補強

建議後續為本章補入：中文概念導讀、完整解答、課堂練習難度標記，以及與下一章的銜接說明。

4 抽樣、統計量與信賴區間

! 第 8 版對照與教學安排

本章以 HMC8 Chapter 4 為編號依據。現有講義涵蓋 Sections 4.1、4.2 與 4.4；其他課本小節不為了形式完整而硬加內容。教學上先喚醒統計量與估計量，再深化 order statistics，最後連到 confidence intervals。

Recall → Deepen → Connect

💡 本章定位

本講義由蘇南誠老師編授。感謝黃怡婷老師提供原始數理統計講義檔案；本章保留原有英文術語與公式，並整理為可連續閱讀的教學形式。

4.1 學習目標

完成本章後，學生應能：

- 區分母體、樣本、統計量與估計值
- 推導常見統計量的抽樣分配
- 分析次序統計量
- 使用樞紐量建構信賴區間
- 正確解釋信賴水準與區間涵蓋率

4.2 必備的先備知識

常態、卡方、t、F 分配及聯合分配。

以下依原講義順序整理核心定義、定理、公式與例題。

教學路線

Stage 1 — Recall：樣本、統計量與估計量

4.3 4.1 Sampling and Statistics

Statistical Inference

- A random variable X of interest, but its pdf $f(x)$ or pmf $p(x)$ is not known.

- $f(x)$ or $p(x)$ can roughly be classified in one of two ways:
 - Pdf $f(x)$ or $p(x)$ is completely unknown.
 - The form of $f(x)$ or $p(x)$ is known down to a parameter θ , where θ may be a vector.

Example 1 — Parametric Inference

- X has an exponential distribution, where θ is unknown.
- X has a binomial distribution $b(n, p)$, where n is known but p is unknown.

Parameter space

- X has a density or mass function of the form $f(x; \theta)$ or $p(x; \theta)$, where $\theta \in \Omega$ for a specified set Ω .
- For example, an exponential distribution $\Omega = \{\theta | \theta > 0\}$.
- θ a parameter of the distribution.
- Because θ is unknown, we want to estimate it.

Random sample

- Consider the sample observations that have the same distribution as X .
- Denote them as the random variables $X_1, X_2, \dots, X_n$, where n denotes the sample size.
- Use lower case letters $x_1, x_2, \dots, x_n$ as the values or realizations of the sample.
- Assume the sample observations $X_1, X_2, \dots, X_n$ are mutually independent.
- It is called the random sample.

Statistics

定義 4.1.

- Let $X_1, X_2, \dots, X_n$ denote a sample on a random variable X .
- Let $T = T(X_1, X_2, \dots, X_n)$ be a function of the sample.
- Then T is called a statistic.

Estimator and Estimate

- Let $X_1, X_2, \dots, X_n$ denote a random sample on a random variable X with a density or mass function of the form $f(x; \theta)$ or $p(x; \theta)$, where $\theta \in \Omega$ for a specified set Ω .
- Consider a statistic T , which is an estimator of θ .
- T is formally called a point estimator of θ .
- Call its realization t an estimate of θ .

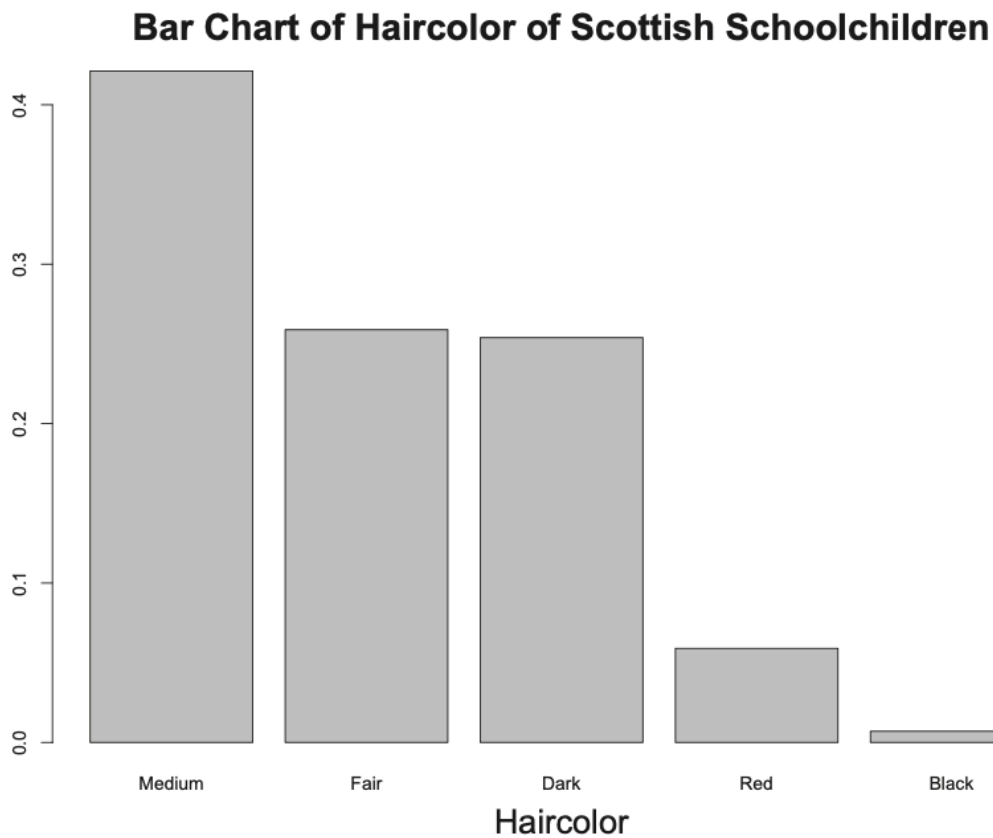
Estimator

定義 4.2.

- Let $X_1, X_2, \dots, X_n$ denote a sample on a random variable X with pdf $f(x; \theta)$, $\theta \in \Omega$.
- Let $T = T(X_1, X_2, \dots, X_n)$ be a statistic.
- T is an unbiased estimator of θ if $E(T) = \theta$.

Nonparametric estimator

- Let $X_1, X_2, \dots, X_n$ be a random sample on a random variable X with cdf $F(x)$.
- Make no assumptions on the form of the distribution of X .
- Often called a nonparametric estimator.
- Discrete – Bar chart
- Continuous – Histogram



Stage 2 — Deepen：次序統計量與樣本分位數

4.4 Order Statistics

Order statistics

- Let $X_1, X_2, \dots, X_n$ denote a random sample from a distribution of the continuous type having a pdf $f(x)$ that has support $S = (a, b)$, where $-\infty \leq a < b \leq \infty$.
- Let Y_1 be the smallest of these X_i , Y_2 the next X_i in order of magnitude, ..., and Y_n the largest of X_i .
- That is, $Y_1 < Y_2 < \dots < Y_n$ represent $X_1, X_2, \dots, X_n$ when the latter are arranged in ascending order of magnitude.
- Call $Y_i, i = 1, \dots, n$, the i th order statistic of the random sample $X_1, X_2, \dots, X_n$.

Joint pdf of order statistics

定理 4.1.

- Let $Y_1 < Y_2 < \dots < Y_n$ denote the n order statistics based on the random sample $X_1, X_2, \dots, X_n$ from a continuous distribution with pdf $f(x)$ and support (a, b) .
- Then the joint pdf of $Y_i, i = 1, \dots, n$, is given by

$$g(y_1, \dots, y_n) = \begin{cases} n!f(y_1)f(y_2)\cdots f(y_n) & a < y_1 < \dots < y_n < b \\ 0 & \text{elsewhere} \end{cases}$$

Example 2

- Let X denote a random variable of the continuous type with a pdf $f(x)$ that is positive and continuous, with support $S = (a, b)$, $-\infty \leq a < b < \infty$.
- Let m denote a unique median.
- Let X_1, X_2, X_3 denote a random sample from this distribution.
- Let $Y_1 < Y_2 < Y_3$ denote the order statistics of the sample.
- Find $P[Y_2 \leq m]$.

Example 3

- Let $Y_1 < Y_2 < \dots < Y_n$ denote the n order statistics based on the random sample $X_1, X_2, \dots, X_n$ from a continuous distribution with pdf $f(x)$ and support (a, b) .
- Find the marginal pdf of Y_k .

Example 4

- Let $Y_1 < Y_2 < Y_3 < Y_4$ denote the order statistics of a random sample of size 4 from a distribution having pdf

$$f(x) = \begin{cases} 2x & 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

- Find the marginal pdf of Y_3 .

Example 5

- Let $Y_1 < Y_2 < \dots < Y_n$ denote the n order statistics based on the random sample $X_1, X_2, \dots, X_n$ from a continuous distribution with pdf $f(x)$ and support (a, b) .
- Find the marginal pdf of Y_i and $Y_j, i < j$.

Special order statistics

- Sample range of the random sample is given by $Y_n - Y_1$.
- Sample midrange is given by $(Y_1 + Y_n)/2$.
- Sample median of the random sample is defined by

$$Q_2 = \begin{cases} Y_{(n+1)/2} & \text{if } n \text{ is odd} \\ (Y_{n/2} + Y_{(n/2)+1})/2 & \text{if } n \text{ is even} \end{cases}$$

Example 6

- Let $Y_1 < Y_2 < Y_3$ denote the order statistics of a random sample of size 4 from a distribution having pdf

$$f(x) = \begin{cases} 1 & 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

- Find the pdf of $Z_1 = Y_3 - Y_1$.

Recall – Quantiles

- Let X be a random variable with a continuous cdf $F(x)$.
- For $0 < p < 1$, define the p th quantile of X to be $\xi_p = F^{-1}(p)$.

Sample quantiles

- Let $Y_1 < Y_2 < \dots < Y_n$ denote the n order statistics based on the random sample $X_1, X_2, \dots, X_n$ from a continuous distribution with pdf $f(x)$.
- Let k be the greatest integer less than or equal to $[p(n+1)]$.
- Define Y_k as an estimator of the quantile ξ_p .

Reason – Sample quantiles

- The area under the pdf $f(x)$ to the left of Y_k is $F(Y_k)$.
- Find $E[F(Y_k)] = \frac{k}{n+1}$.
- On the average, there is $k/(n+1)$ of the total area to the left of Y_k .

Remark

- Some statisticians define sample quantiles slightly differently from what we have.
- For one modification with $1/(n+1) < p < n/(n+1)$, if $(n+1)p$ is not equal to an integer, then the p th quantile of the sample may be defined as follows.
- Write $(n+1)p = k + r$, where $k = [(n+1)p]$ and r is a proper fraction, using the weighted average.
- Then the p th quantile of the sample is the weighted average

$$(1-r)Y_k + rY_{k+1},$$

which is an estimator of the p th quantile.

- As n becomes large, however, all these modified definitions are essentially the same.

Five-number summary

- A five-number summary of the data consists of the following five sample quantiles:
 - Minimum (Y_1)
 - The first quartile ($Q_1 = Y_{.25(n+1)}$)
 - The median (Q_2)
 - The third quartile ($Q_3 = Y_{.75(n+1)}$)
 - The maximum (Y_n)
- Q_1 , Q_2 , and Q_3 denote the first quartile, median, and third quartile of the sample.

Box plot

- The five-number summary is the basis for a useful and quick plot of the data.
- This is called a boxplot of the data.
- The box encloses the middle 50% of the data and a line segment is usually used to indicate the median.
- The extreme order statistics, however, are very sensitive to outlying points.
- We make use of the box and whisker plots defined by John Tukey.
- In order to define this plot, we need to define a potential outlier.

- Let $h = 1.5(Q_3 - Q_1)$ and define the lower fence (LF) and the upper fence (UF) by

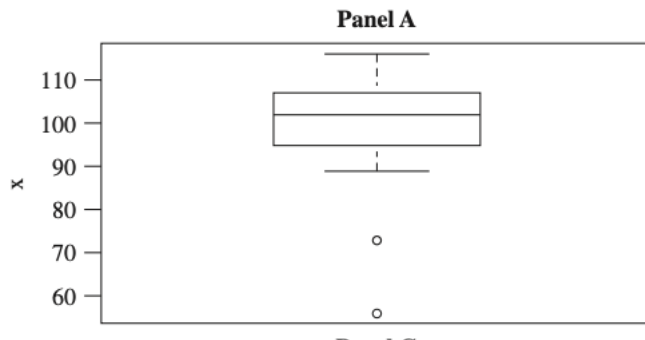
$$LF = Q_1 - h \quad \text{and} \quad UF = Q_3 + h.$$

Example 7

A random sample of size 15 on a random variable X .

56	70	89	94	96	101	102	102
102	105	106	108	110	113	116	

- Minimum ($Y_1 = 56$)
- The first quartile ($Q_1 = Y_{.25(n+1)} = 94$)
- The median ($Q_2 = 102$)
- The third quartile ($Q_3 = Y_{.75(n+1)} = 108$)
- The maximum ($Y_{15} = 116$)
- $h = 1.5(108 - 94)$
- $LF = 73$
- $UF = 129$



Quantile plots

- Consider the location and scale family.
- Suppose X is a random variable with cdf $F((x - a)/b)$, where $F(x)$ is known but a and $b > 0$ may not be.
- Let $Z = (X - a)/b$; then Z has cdf $F(z)$.
- Let $0 < p < 1$ and let $\xi_{X,p}$ be the p th quantile of X .
- Let $\xi_{Z,p}$ be the p th quantile of $Z = (X - a)/b$.
- We have

$$p = P[X \leq \xi_{X,p}] = P\left[Z \leq \frac{\xi_{X,p} - a}{b}\right],$$

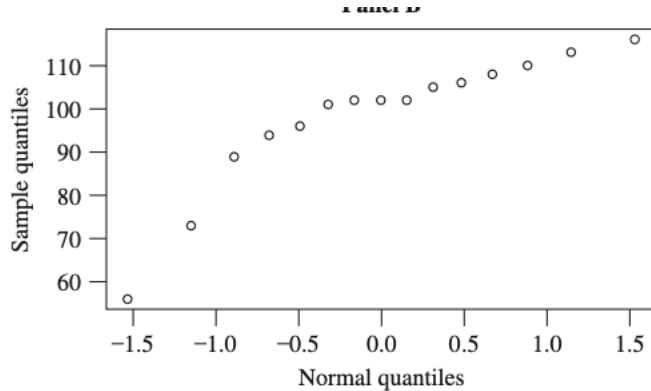
and $\xi_{X,p} = b\xi_{Z,p} + a$.

- The quantiles of X are linearly related to the quantiles of Z .

Quantile plots

- Let $X_1, \dots, X_n$ be a random sample from the distribution of X and let $Y_1 < \dots < Y_n$ be the order statistics.
- For $k = 1, \dots, n$, let $p_k = k/(n+1)$.
- Then Y_k is an estimator of ξ_{X,p_k} .
- Denote the corresponding quantiles of the cdf $F(z)$ by $\xi_{Z,p_k} = F^{-1}(p_k)$.
- Let y_k denote the realized value of Y_k .
- The plot of y_k versus ξ_{Z,p_k} is called a q-q plot.
- The linearity of such a plot indicates that the cdf of X is of the form $F((x-a)/b)$.

Example 8



Stage 3 — Connect : 由抽樣分配到區間估計

4.5 4.2 Confidence Intervals

Confidence intervals

定義 4.3.

- Let $X_1, X_2, \dots, X_n$ be a sample on a random variable X , where X has pdf $f(x; \theta)$, $\theta \in \Omega$.
- Let $0 < \alpha < 1$ be specified.
- Let $L = L(X_1, X_2, \dots, X_n)$ and $U = U(X_1, X_2, \dots, X_n)$ be two statistics.
- The interval (L, U) is a $(1 - \alpha)100\%$ confidence interval for θ if

$$1 - \alpha = P_{\theta}[\theta \in (L, U)]$$

- The probability that the interval includes θ is $1 - \alpha$, which is called the confidence coefficient or the confidence level of the interval.

Efficiency of confidence intervals

定義 4.4.

- A measure of efficiency based on a confidence interval is its expected length.
- Suppose (L_1, U_1) and (L_2, U_2) are two confidence intervals for θ that have the same confidence coefficient.
- Then we say that (L_1, U_1) is more efficient than (L_2, U_2) if

$$E_\theta[U_1 - L_1] \leq E_\theta[U_2 - L_2], \forall \theta \in \Omega.$$

4.6 本章摘要

本章從隨機樣本出發，研究統計量的分配，並以樞紐量把抽樣不確定性轉化為信賴區間。

4.7 常見錯誤

- 把參數視為隨機量來解釋頻率學派信賴區間
- 忘記抽樣分配依賴模型假設
- 混淆統計量與其觀察值
- 建構區間時不等式方向處理錯誤

4.8 章末練習

1. 從本章選出三個關鍵定義，分別寫出數學形式、成立條件及一個例子。
2. 選擇一個主要定理或方法，重做推導並標記每一步使用的假設。
3. 從原講義例題挑選一題，完整寫出解題目標、方法選擇、計算與結果解釋。
4. 設計一個會違反本章方法條件的反例，說明錯誤會出現在哪一步。

i 待逐章補強

建議後續為本章補入：中文概念導讀、完整解答、課堂練習難度標記，以及與下一章的銜接說明。

5 一致性與極限分配

! 第 8 版對照與教學安排

本章以 HMC8 Chapter 5 為編號依據。先快速回顧 random sample、statistic 與 estimator，再深入兩種收斂、Slutsky-type 性質與 Delta method，最後以 CLT 和 multivariate extensions 銜接漸近推論。

Recall → Deepen → Connect

💡 本章定位

本講義由蘇南誠老師編授。感謝黃怡婷老師提供原始數理統計講義檔案；本章保留原有英文術語與公式，並整理為可連續閱讀的教學形式。

5.1 學習目標

完成本章後，學生應能：

- 區分機率收斂與分配收斂
- 判斷序列是否有界於機率
- 使用 Slutsky 定理與 Delta method
- 運用動差生成函數推導極限分配
- 以中央極限定理建立近似推論

5.2 必備的先備知識

期望、變異數、動差生成函數及基本微積分。

以下依原講義順序整理核心定義、定理、公式與例題。

Stage 1 — Recall：樣本、統計量與估計量

Random sample

- Consider the sample observations that have the same distribution as X .
- Denote them as the random variables $X_1, X_2, \dots, X_n$, where n denotes the sample size.
- Use lower case letters $x_1, x_2, \dots, x_n$ as the values or realizations of the sample.

- Assume the sample observations $X_1, X_2, \dots, X_n$ are mutually independent.
- It is called the random sample.

Statistics

定義 5.1.

- Let $X_1, X_2, \dots, X_n$ denote a sample on a random variable X .
- Let $T = T(X_1, X_2, \dots, X_n)$ be a function of the sample.
- Then T is called a statistic.

Pivot random variable

- The pivot is usually a function of an estimator of θ .
- The distribution of the pivot is known.
- For example,
 - Suppose $X_1, X_2, \dots, X_n$ is a random sample on a random variable X which has a $N(\mu, \sigma^2)$ distribution.
 - Denote the sample mean by $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$.
 - Then the pivot random variable of interest is

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$$

Stage 2 — Deepen : 機率收斂與分配收斂

5.3 5.1 Convergence in Probability

Convergence in probability

定義 5.2.

- Let $\{X_n\}$ be a sequence of random variables.
- Let X be a random variable defined on a sample space.
- X_n converges in probability to X if, for all $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} [|X_n - X| \geq \epsilon] = 0,$$

or

$$\lim_{n \rightarrow \infty} [|X_n - X| < \epsilon] = 1,$$

- Write $X_n \xrightarrow{P} X$

Weak Law of Large Numbers

定理 5.1.

- Let $\{X_n\}$ be a sequence of iid random variables having common mean μ and variance $\sigma^2 < \infty$.
- Let $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$.
- Then $\bar{X}_n \xrightarrow{P} \mu$

Additive properties

定理 5.2.

- Suppose $X_n \xrightarrow{P} X$ and $Y_n \xrightarrow{P} Y$.
- Then $X_n + Y_n \xrightarrow{P} X + Y$

Constant

定理 5.3.

- Suppose $X_n \xrightarrow{P} X$ and a is a constant.
- Then $aX_n \xrightarrow{P} aX$

Continuity

定理 5.4.

- Suppose $X_n \xrightarrow{P} a$ and the real function g is continuous at a .
- Then $g(X_n) \xrightarrow{P} g(a)$.

Multiplicative properties

定理 5.5.

- Suppose $X_n \xrightarrow{P} X$ and $Y_n \xrightarrow{P} Y$.
- Then $X_n Y_n \xrightarrow{P} XY$.

Estimator and Estimate

- Let $X_1, X_2, \dots, X_n$ denote a random sample on a random variable X with a density or mass function of the form $f(x; \theta)$ or $p(x; \theta)$, where $\theta \in \Omega$ for a specified set Ω .
- Consider a statistic T , which is an estimator of θ .
- T is formally called a point estimator of θ .

- Call its realization t an estimate of θ .

Estimator

- Let $X_1, X_2, \dots, X_n$ denote a sample on a random variable X with pdf $f(x; \theta)$, $\theta \in \Omega$.
- Let $T = T(X_1, X_2, \dots, X_n)$ be a statistic.
- Suppose T is a point estimator of θ .

Estimate

- When the sample is drawn, let $x_1, x_2, \dots, x_n$ denote the observed values of $X_1, X_2, \dots, X_n$.
- We call these values the realized values of the sample.
- Call the realized statistic $t = t(x_1, x_2, \dots, x_n)$ a point estimate of θ .

Properties of estimators

- Unbiasedness
 - T is an unbiased estimator of θ if $E(T) = \theta$.
- Consistency

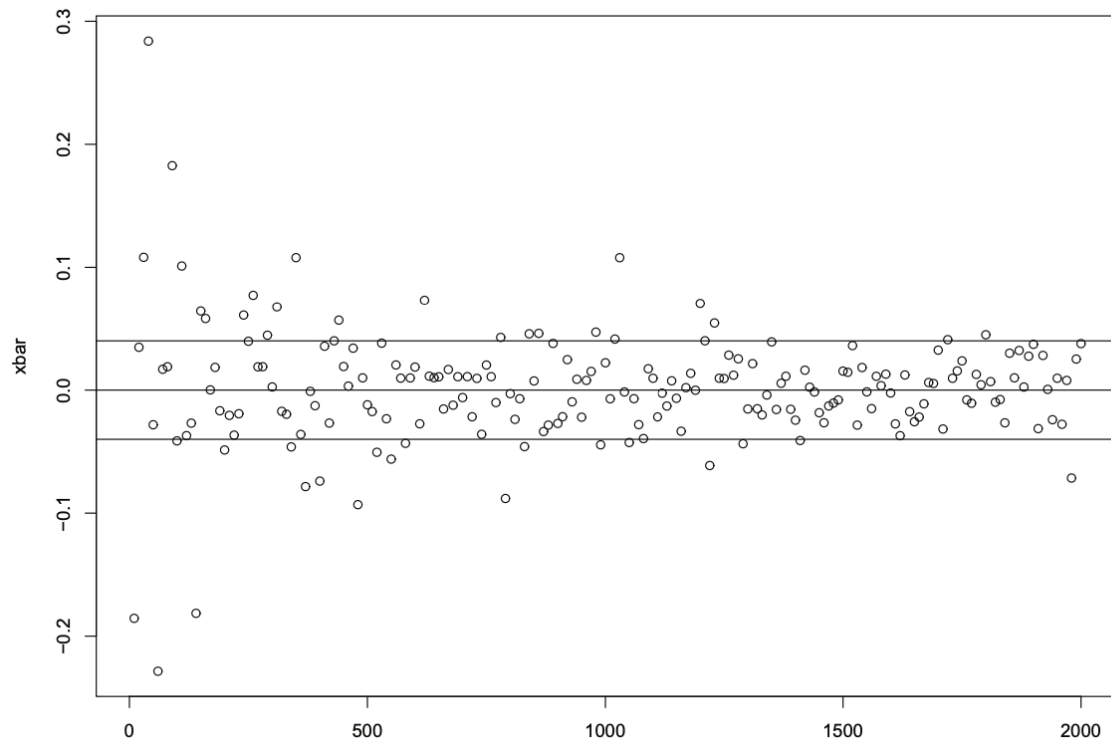
Consistent estimator

定義 5.3.

- Let X be a random variable with cdf $F(x, \theta)$, $\theta \in \Omega$.
- Let $X_1, X_2, \dots, X_n$ be a sample from the distribution of X .
- Let T_n denote a statistic.
- We say T_n is a consistent estimator of θ if

$$T_n \xrightarrow{P} \theta.$$

Example 1 — Realizations of $\bar{X}$ from $N(0, 1)$



Example 2

- Let $X_1, X_2, \dots, X_n$ be a sample from a distribution with mean μ and variance σ^2 .
- Assume that $E[X_1^4] < \infty$.
- Show the sample variance is a consistent estimator of σ^2 .

Example 3

- Let $X_1, X_2, \dots, X_n$ be a sample from a uniform $(0, \theta)$ distribution.
- Assume θ is unknown.
- Let $Y_n = \max(X_1, X_2, \dots, X_n)$.
- Show Y_n is a consistent estimator of θ .

5.4 5.2 Convergence in Distribution

Convergence in distribution

定義 5.4.

- Let $\{X_n\}$ be a sequence of random variables with distribution functions F_{X_n} .

- Let X be a random variable with cdf $F(x, \theta)$, $\theta \in \Omega$.
- Let $C(F_X)$ denote the set of all points where F_X is continuous.
- We say X_n converges in distribution to X if

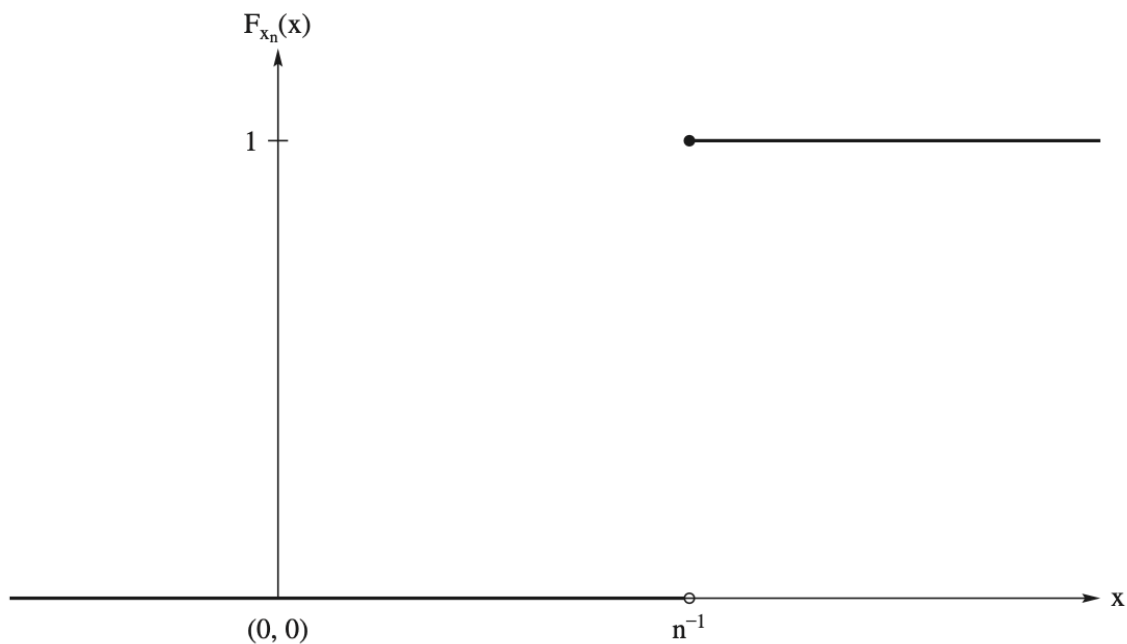
$$\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x), \forall x \in C(F_X).$$

- Denote $X_n \xrightarrow{D} X$.

Remark

- This material on convergence in probability and in distribution comes under what statisticians refer to as asymptotic theory.
- We say that the distribution of X is the asymptotic distribution or the limiting distribution of the sequence $\{X_n\}$.
- Instead of saying $X_n \xrightarrow{D} X$, where X has a standard normal distribution, we may write $X_n \xrightarrow{D} N(0, 1)$.

Remark – Points of continuity



Remark – Convergence in probability and in distribution

- Convergence in probability is a way of saying that a sequence of random variables X_n is getting close to another random variable X .
- Convergence in distribution is only concerned with the cdfs F_{X_n} and F_X .

Example 4

- Let X be a continuous random variable with a pdf $f_X(x)$ that is symmetric about 0; i.e., $f_X(-x) = f_X(x)$.
- Then $-X$ also has a density function $f_X(x)$.
- X and $-X$ have the same distributions.
- Define the sequence of random variables X_n as

$$X_n = \begin{cases} X & \text{if } n \text{ is odd} \\ -X & \text{if } n \text{ is even} \end{cases}$$

- $F_{X_n}(x) = F_X(x)$ for all x in the support of X , so that $X_n \xrightarrow{D} X$.
- X_n does not get close to X . In particular, $X_n \not\rightarrow X$ in probability.

Example 5

- Let $\bar{X}_n$ have the cdf

$$F_n(\bar{x}) = \int_{-\infty}^{\bar{x}} \frac{1}{\sqrt{1/n}\sqrt{2\pi}} e^{-nw^2/2} dw.$$

- Let X have the cdf

$$F(\bar{x}) = \begin{cases} 0 & \text{if } \bar{x} < 0 \\ 1 & \text{if } \bar{x} \geq 0 \end{cases}$$

- Show $\bar{X}_n \xrightarrow{D} X$.

Example 6

- Let $\bar{X}_n$ have the cdf

$$F_n(\bar{x}) = \int_{-\infty}^{\bar{x}} \frac{1}{\sqrt{1/n}\sqrt{2\pi}} e^{-nw^2/2} dw.$$

- Let X have the cdf

$$F(\bar{x}) = \begin{cases} 0 & \text{if } \bar{x} < 0 \\ 1 & \text{if } \bar{x} \geq 0 \end{cases}$$

- Show $\bar{X}_n \xrightarrow{D} X$.

Example 7

- Let $\bar{X}_n$ have the pmf

$$p_n(x) = \begin{cases} 1 & x = 2 + n^{-1} \\ 0 & \text{elsewhere} \end{cases}$$

- Let X have the cdf

$$F(x) = \begin{cases} 0 & \text{if } x < 2 \\ 1 & \text{if } x \geq 2 \end{cases}$$

- Show $\bar{X}_n \xrightarrow{D} X$.

Example 8

- Let T_n have a t -distribution with n degrees of freedom, $n = 1, 2, 3, \dots$.
- Its cdf is

$$F_n(x) = \int_{-\infty}^x \frac{\Gamma((n+1)/2)}{\sqrt{\pi n} \Gamma(n/2)} \frac{1}{(1 + y^2/n)^{(n+1)/2}} dy.$$

- Show $T_n \xrightarrow{D} N(0, 1)$.

Remark – Stirling's formula

- In advanced calculus the following approximation is derived:

$$\Gamma(k+1) \approx \sqrt{2\pi} k^{k+1/2} e^{-k}.$$

Example 9

- Let $X_1, X_2, \dots, X_n$ be a sample from a uniform $(0, \theta)$ distribution.
- Assume θ is unknown.
- Let $Y_n = \max(X_1, X_2, \dots, X_n)$.
- Show $Z_n = n(\theta - Y_n)$ converges in distribution to an exponential distribution with mean θ .

Properties of convergences

定理 5.6.

- If $X_n \xrightarrow{P} X$, then $X_n \xrightarrow{D} X$.

Properties of convergences

定理 5.7.

- If $X_n \xrightarrow{D} b$, then $X_n \xrightarrow{P} b$.

Properties of convergences

定理 5.8.

- If $X_n \xrightarrow{D} X$ and $X_n \xrightarrow{P} 0$, then $X_n + Y_n \xrightarrow{D} X$.

Properties of convergences

定理 5.9.

- If $X_n \xrightarrow{D} X$ and g is a continuous function on the support of X , then $g(X_n) \xrightarrow{D} g(X)$.

Properties of convergences

定理 5.10.

- If $X_n \xrightarrow{D} X$, $A_n \xrightarrow{P} a$ and $B_n \xrightarrow{P} b$, then $A_n + B_n X_n \xrightarrow{D} a + bX$.

Bounded in probability

定義 5.5.

- The sequence of random variables $\{X_n\}$ is bounded in probability if, for all $\epsilon > 0$, there exist a constant $B_\epsilon > 0$ and an integer N_ϵ such that

$$n \geq N_\epsilon \Rightarrow P[|X_n| \leq B_\epsilon] \geq 1 - \epsilon.$$

Properties

定理 5.11.

- Let $\{X_n\}$ be a sequence of random variables.
- Let X be a random variable.
- If $X_n \xrightarrow{D} X$, then X_n is bounded in probability.

Properties

定理 5.12.

- Let $\{X_n\}$ be a sequence of random variables bounded in probability.
- Let $\{Y_n\}$ a sequence of random variables that converges to 0 in probability.
- Then $X_n Y_n \xrightarrow{P} 0$.

Recall Young's Theorem

- Suppose $g(x)$ is differentiable at x .
- Write $g(y) = g(x) + g'(x)(y - x) + o(|y - x|)$, where the notation o means

$$a = o(b) \text{ if and only if } \frac{a}{b} \rightarrow 0, \text{ as } b \rightarrow 0.$$

Recall

- Write $o_p(X_n)$, which means

$$Y_n = o_p(X_n) \iff \frac{Y_n}{X_n} \xrightarrow{P} 0, \text{ as } n \rightarrow \infty.$$

- Write $O_p(X_n)$, which means

$$Y_n = O_p(X_n) \iff \frac{Y_n}{X_n} \text{ is bounded in probability, as } n \rightarrow \infty.$$

Properties

定理 5.13.

- Let $\{Y_n\}$ be a sequence of random variables bounded in probability.
- Suppose $X_n = o_p(Y_n)$.
- Then $X_n \xrightarrow{P} 0$, as $n \rightarrow \infty$.

Δ method

定理 5.14.

- Let $\{X_n\}$ be a sequence of random variables such that

$$\sqrt{n}(X_n - \theta) \xrightarrow{D} N(0, \sigma^2)$$

- Suppose $g(x)$ is differentiable at θ and $g'(\theta) \neq 0$.
- Then

$$\sqrt{n}(g(X_n) - g(\theta)) \xrightarrow{D} N(0, \sigma^2(g'(\theta))^2).$$

Convergence in distribution

定理 5.15.

- Let $\{X_n\}$ be a sequence of random variables with mgf $M_{X_n}(t)$ that exists for $-h < t < h$ for all n .
- Let X be a random variable with mgf $M(t)$, which exists for $|t| \leq h_1 \leq h$.
- If $\lim_{n \rightarrow \infty} M_{X_n}(t) = M(t)$ for $|t| \leq h_1$, then $X_n \xrightarrow{D} X$.

Example 10

- Let Y_n have a distribution that is $b(n, p)$.
- Suppose $\mu = np, \forall n$.
- Show $Y_n \xrightarrow{D} \text{Poisson}(\mu)$.

Example 11

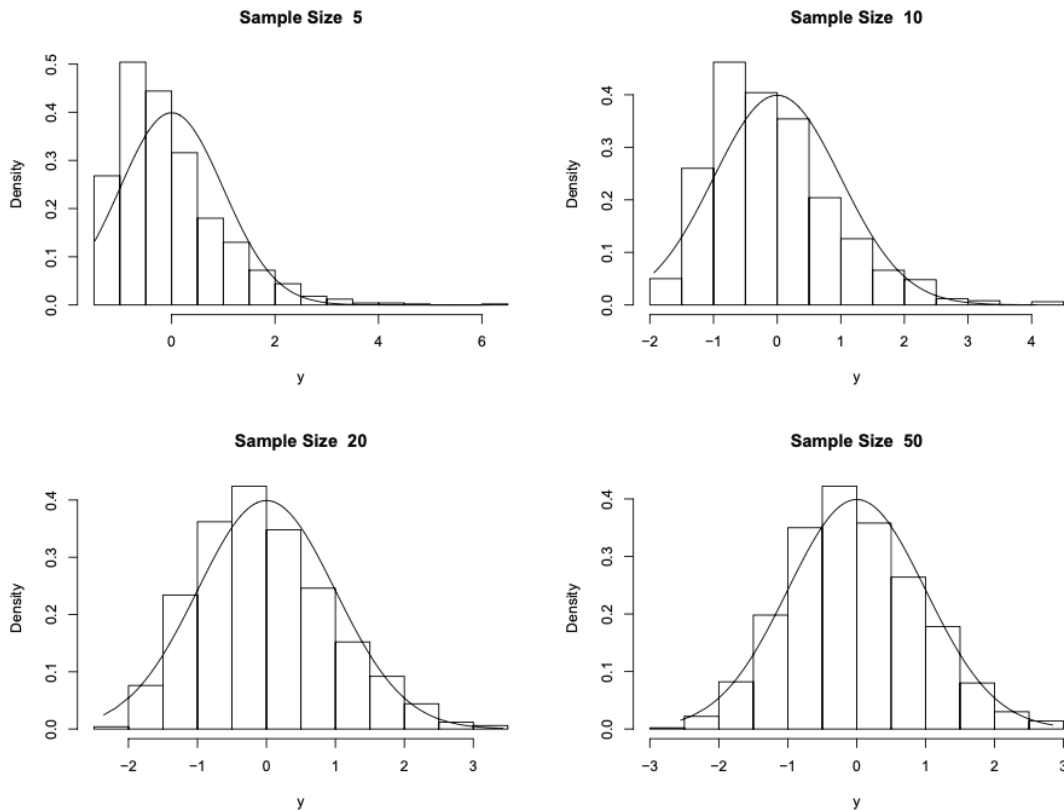
- Let $Z_n \sim \chi_n^2$.
- Let $Y_n = \frac{Z_n - n}{\sqrt{2n}}$.
- Show that $Y_n \xrightarrow{D} N(0, 1)$.

Example 12 — χ^2 distribution

- Let $Z_n \sim \chi^2(n)$.
- Show that

$$Y_n = \frac{Z_n - n}{\sqrt{2n}} \xrightarrow{D} N(0, 1).$$

Histogram plot of 1000 generated values



Example 13 — More χ^2 distribution

- Let $Z_n \sim \chi^2(n)$.
- We have

$$Y_n = \frac{Z_n - n}{\sqrt{2n}} \xrightarrow{D} N(0, 1).$$

- Show that

$$W_n = \left(\frac{Z_n}{\sqrt{2n}} \right)^{1/2} \xrightarrow{D} N(0, 1).$$

Stage 3 — Connect : 中央極限定理與漸近推論

5.5 5.3 Central Limit Theorem

Central limit theorem

定理 5.16.

- Let $X_1, X_2, \dots, X_n$ denote the observations of a random sample from a distribution that has mean μ and positive variance σ^2 .
- Show that

$$Y_n = \frac{\sum_{i=1}^n X_i - n\mu}{\sqrt{n}\sigma} = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{D} N(0, 1).$$

Example 14

- Let $X_1, X_2, \dots, X_n$ denote the observations of a random sample from a distribution that has mean μ and positive variance σ^2 , where μ and σ^2 are unknown.
- Let $\bar{X}$ and S denote the sample mean and sample standard deviation.
- Show that

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \xrightarrow{D} N(0, 1).$$

Example 15 — Binomial distribution

- Let $X_1, X_2, \dots, X_n$ denote the observations of a random sample from a Bernoulli distribution $b(1, p)$.
- Show that

$$\frac{\sum_{i=1}^n X_i - np}{\sqrt{np(1-p)}} \xrightarrow{D} N(0, 1).$$

Example 16 — Proportion

- Let $X_1, X_2, \dots, X_n$ denote the observations of a random sample from a Bernoulli distribution $b(1, p)$.
- Show that

$$\frac{\hat{p} - p}{\sqrt{\hat{p}(1-\hat{p})/n}} \xrightarrow{D} N(0, 1).$$

5.6 5.4 Extensions to Multivariate Distributions

Recall

- For a vector $v \in R^p$, recall the Euclidean norm of v is defined to be

$$\|v\| = \sqrt{\sum_{i=1}^p v_i^2}$$

- Properties - $\forall v, u \in R^p$
 - $\|v\| \geq 0$
 - $\|v\| = 0 \iff v = 0$
 - $a \in R, \|av\| = |a| \|v\|$
 - $\|v + u\| \leq \|v\| + \|u\|$

Recall

- Consider a standard basis of R^p and denote as $e_1, \dots, e_p$, where all the components of e_i are 0 except for the i th component, which is 1.
- Write any vector $v' = (v_1, \dots, v_p)$ as

$$v = \sum_{i=1}^p v_i e_i$$

Properties of vector

- Let $v' = (v_1, \dots, v_p) \in R^p$.
- Then

$$|v_j| \leq \|v\| \leq \sum_{i=1}^n |v_i|, \forall j = 1, \dots, p.$$

Convergence in probability

定義 5.6.

- Let $\{X_n\}$ be a sequence of p -dimensional vectors.
- Let X be a p -dimensional vectors.
- $\{X_n\}$ and X are defined on the same sample space.
- Then $X_n \xrightarrow{P} X$ if

$$\lim_{n \rightarrow \infty} P[\|X_n - X\| \geq \epsilon] = 0, \forall \epsilon > 0.$$

Convergence in probability

定理 5.17.

- Let $\{X_n\}$ be a sequence of p -dimensional vectors.
- Let X be a random vector, all defined on the same sample space.
- Then

$$X_n \xrightarrow{P} X \iff X_{nj} \xrightarrow{P} X_j \forall j = 1, \dots, p.$$

Example 17

- Let $\{X_n\}$ be a sequence of iid random vectors with common mean vector μ and variance-covariance matrix Σ .
- Denote the vector of means by

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

- Show $\bar{X}_n \xrightarrow{P} \mu$

Example 18

- Let $X_i = (X_{i1}, \dots, X_{ip})'$.
- Define the sample variance-covariance matrix by

$$S_{n,j}^2 = \frac{1}{n-1} \sum_{i=1}^n (X_{ij} - \bar{X}_j)^2, j = 1, 2, \dots, p,$$
$$S_{n,jk} = \frac{1}{n-1} \sum_{i=1}^n (X_{ij} - \bar{X}_j)(X_{ik} - \bar{X}_k), j \neq k = 1, 2, \dots, p,$$

- Assume finite fourth moments.
- Show $S \xrightarrow{P} \Sigma$, where the $p \times p$ matrix S is the matrix with the j th diagonal entry $S_{n,j}^2$ and (j, k) th entry $S_{n,jk}$.

Convergence in probability

定義 5.7.

- Let $\{X_n\}$ be a sequence of p -dimensional vectors.
- Let X be a p -dimensional vectors.
- Let X_n and X have distribution functions $F_n(x)$ and $F(x)$.

- Then $X_n \xrightarrow{D} X$ if

$$\lim_{n \rightarrow \infty} F_n(x) = F(x),$$

for all points x at which $F(x)$ is continuous.

Convergence in distribution

定理 5.18.

- Let $X_n \xrightarrow{P} X$.
- Let $g(x)$ be a function that is continuous on the support of X .
- Then $g(X_n)$ converges in distribution to $g(X)$.
- Then

$$g(X_n) \xrightarrow{D} g(X).$$

Convergence in distribution

定理 5.19.

- Let $\{X_n\}$ be a sequence of p -dimensional vectors.
- Let X be a p -dimensional vectors.
- Let X_n and X have distribution functions $F_n(x)$ and $F(x)$.
- Let X_n and X have moment generating functions $M_n(t)$ and $M(t)$.
- Then $X_n \xrightarrow{D} X$ if for some $h > 0$,

$$\lim_{n \rightarrow \infty} M_n(t) = M(t),$$

for all t such that $\|t\| < h$.

Central Limit Theorem

定理 5.20.

- Let $\{X_n\}$ be a sequence of iid random vectors with common mean vector μ and variance-covariance matrix Σ , which is positive definite.
- Assume that the common moment generating function $M(t)$ exists in an open neighborhood of 0.
- Let

$$Y_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \mu) = \sqrt{n}(\bar{X} - \mu).$$

- Then $X_n \xrightarrow{D} N_p(0, \Sigma)$.

Properties

定理 5.21.

- Let $\{X_n\}$ be a sequence of p -dimensional random vectors.
- Suppose $X_n \xrightarrow{D} N_p(\mu, \Sigma)$.
- Let A be an $m \times p$ matrix of constants and b be an m -dimensional vector of constants.
- Then $AX_n + b \xrightarrow{D} N_p(A\mu + b, A\Sigma A')$.

Δ method

定理 5.22.

- Let $\{X_n\}$ be a sequence of p -dimensional random vectors.
- Suppose $\sqrt{n}(X_n - \mu_0) \xrightarrow{D} N_p(0, \Sigma)$.
- Let g be a transformation $g(x) = (g_1(x), \dots, g_k(x))'$ such that $1 \leq k \leq p$.
- Assume the $k \times p$ matrix of partial derivatives

$$B = \left[\frac{\partial g_i}{\partial \mu_j} \right], i = 1, 2, \dots, k; j = 1, 2, \dots, p,$$

are continuous and do not vanish in a neighborhood of μ_0 .

- Let $B_0 = B$ at μ_0 .
- Then $\sqrt{n}(g(X_n) - g(\mu_0)) \xrightarrow{D} N_k(0, B_0 \Sigma B_0')$.

5.7 本章摘要

本章說明統計程序在樣本量增加時的行為，並以收斂定理、CLT 與 Delta method 建立漸近近似。

5.8 常見錯誤

- 混淆不同收斂概念的方向與強弱
- 未檢查 Delta method 的可微條件
- 中央極限定理標準化錯誤
- 把大樣本近似當成有限樣本精確結果

5.9 章末練習

1. 從本章選出三個關鍵定義，分別寫出數學形式、成立條件及一個例子。
2. 選擇一個主要定理或方法，重做推導並標記每一步使用的假設。
3. 從原講義例題挑選一題，完整寫出解題目標、方法選擇、計算與結果解釋。

4. 設計一個會違反本章方法條件的反例，說明錯誤會出現在哪一步。

 待逐章補強

建議後續為本章補入：中文概念導讀、完整解答、課堂練習難度標記，以及與下一章的銜接說明。

6 最大概似方法

! 第 8 版對照與教學安排

本章以 HMC8 Chapter 6 為編號依據。由 likelihood 與 score 的基本概念起步，深入 MLE、Fisher information、Rao–Cramér lower bound，再連到 likelihood-ratio testing、多參數模型與 EM algorithm。

Recall → Deepen → Connect

💡 本章定位

本講義由蘇南誠老師編授。感謝黃怡婷老師提供原始數理統計講義檔案；本章保留原有英文術語與公式，並整理為可連續閱讀的教學形式。

6.1 學習目標

完成本章後，學生應能：

- 寫出似然與對數似然函數
- 求最大概似估計量並檢查最大值
- 計算 Fisher information 與 Cramér–Rao 下界
- 使用概似比進行假設檢定
- 處理多參數模型與 EM 演算法

6.2 必備的先備知識

抽樣分配、多變量微積分、矩陣與漸近常態性。

以下依原講義順序整理核心定義、定理、公式與例題。

Stage 1 — Recall : Likelihood 的基本語言

6.3 6.1 Maximum Likelihood Estimation

Parametric assumption

- Consider a random variable X whose pdf $f(x; \theta)$ depends on an unknown parameter θ which is in a set $\Omega \in R^q$.

- The general discussion is for the continuous case, but the results extend to the discrete case also.
- For information, suppose that we have a random sample $X_1, \dots, X_n$ on X ; i.e., $X_1, \dots, X_n$ are iid random variables with common pdf $f(x; \theta)$, $\theta \in \Omega$.

Likelihood function for a single parameter

- Assume $\Omega \in R$.
- Define likelihood function as

$$L(\theta; x) = \prod_{i=1}^n f(x_i; \theta), \theta \in \Omega,$$

where $x = (x_1, \dots, x_n)'$.

- Write it as $L(\theta)$.

Log likelihood function for a single parameter

- Log likelihood function is more convenient to use.
- Denote as

$$l(\theta; x) = \log L(\theta) = \sum_{i=1}^n \log f(x_i; \theta), \theta \in \Omega,$$

where $x = (x_1, \dots, x_n)'$.

- Write it as $l(\theta)$.

Assumptions – Regularity conditions

- (R0) The cdfs are distinct; i.e., $\theta \neq \theta' \Rightarrow F(x_i; \theta) \neq F(x_i; \theta')$.
- (R1) The pdfs have common support for all θ .
- (R2) The point θ_0 is an interior point in Ω .

Properties

定理 6.1.

- Assume that θ_0 is the true parameter and that

$$E_{\theta_0} \left[\frac{f(X_i; \theta)}{f(X_i; \theta_0)} \right]$$

exists.

- Under assumptions (R0) and (R1),

$$\lim_{n \rightarrow \infty} P_{\theta_0} [L(\theta_0, X) > L(\theta, X)] = 1, \forall \theta \neq \theta_0.$$

Maximum likelihood estimator (MLE)

定義 6.1.

- Point estimator of θ is $\hat{\theta} = \hat{\theta}(X_1, \dots, X_n)$, where $\hat{\theta}$ maximizes the function $L(\theta)$, i.e.,

$$\hat{\theta} = \operatorname{argmax}_{\theta} L(\theta) = \operatorname{argmax}_{\theta} l(\theta).$$

- $\hat{\theta}$ is called the maximum likelihood estimator (MLE) of θ .

Example 1 — Laplace distribution

- Let $X_1, \dots, X_n$ be iid with density

$$f(x; \theta) = \frac{1}{2} e^{-|x-\theta|}, x \in R, \theta \in R.$$

- Find mle of θ .

Example 2 — Logistic distribution

- Let $X_1, \dots, X_n$ be iid with density

$$f(x; \theta) = \frac{\exp\{-(x-\theta)\}}{(1 + \exp\{-(x-\theta)\})^2}, x \in R, \theta \in R.$$

- Find mle of θ .

Example 3 — Bernoulli distribution

- Let $X_1, \dots, X_n$ be iid with pmf

$$f(x; \theta) = \theta^x (1 - \theta)^{1-x}, x = 0, 1, \theta \in [0, 1/3].$$

- Find mle of θ .

Property of MLE

定理 6.2.

- Let $X_1, \dots, X_n$ be iid with the pdf $f(x; \theta)$, $\theta \in \Omega$.
- For a specified function g , let $\eta = g(\theta)$ be a parameter of interest.
- Suppose $\hat{\theta}$ is the mle of θ .
- Then $g(\hat{\theta})$ is the mle of $\eta = g(\theta)$.

Properties

定理 6.3.

- Assume that $X_1, \dots, X_n$ satisfy regularity conditions (R0) through (R2), where θ_0 is the true parameter.
- Assume $f(x; \theta)$ is differentiable with respect to $\theta \in \Omega$,
- Then the likelihood equation

$$\frac{\partial}{\partial \theta} L(\theta) = 0$$

has a solution $\hat{\theta}_n$ such that $\hat{\theta}_n \xrightarrow{P} \theta_0$.

Properties

- Assume that $X_1, \dots, X_n$ satisfy regularity conditions (R0) through (R2), where θ_0 is the true parameter.
- Assume $f(x; \theta)$ is differentiable with respect to $\theta \in \Omega$,
- Suppose the likelihood equation has the unique solution $\hat{\theta}_n$.
- Then $\hat{\theta}_n \xrightarrow{P} \theta_0$.

Stage 2 — Deepen : 資訊量、下界與效率

6.4 6.2 Rao–Cramér Lower Bound and Efficiency

Assumptions – Regularity conditions

- (R3) The pdf $f(x; \theta)$ is twice differentiable as a function of θ .
- (R4) The integral $\int f(x; \theta) dx$ can be differentiated twice under the integral sign as a function of θ .

Score function

- The score function is defined as

$$\frac{\partial \log f(x; \theta)}{\partial \theta}.$$

Fisher information of a sample of size 1

- Let X have a pdf or pmf $f(x; \theta)$.
- The Fisher information is the variance of the score function and defined as

$$I(\theta) = \text{Var} \left(\frac{\partial \log f(X; \theta)}{\partial \theta} \right) = E \left[\left(\frac{\partial \log f(X; \theta)}{\partial \theta} \right)^2 \right].$$

Example 4 — Bernoulli distribution

- Let X be Bernoulli $b(1, \theta)$ with pmf

$$f(x; \theta) = \theta^x (1 - \theta)^{1-x}, x = 0, 1, \theta \in [0, 1].$$

- Find the Fisher information.

Example 5 — Location family

- Consider a random sample $X_1, \dots, X_n$ such that

$$X_i = \theta + e_i, i = 1, \dots, n,$$

where $e_1, e_2, \dots, e_n$ are iid with common pdf $f(x)$ and with support $(-\infty, \infty)$.

- Find the Fisher information.

Fisher information of a sample of size n

- Let $X_1, \dots, X_n$ be a random sample from a distribution having pdf or pmf $f(x; \theta)$.
- The random variable whose variance is the information in the sample is given by

$$\frac{\partial \log L(\theta; X)}{\partial \theta} = \sum_{i=1}^n \frac{\partial \log L(\theta; X_i)}{\partial \theta}.$$

- Denote Fisher information in a random sample of size n as

$$\text{Var} \left(\frac{L(\theta; X)}{\partial \theta} \right) = nI(\theta).$$

Rao-Cramér lower bound

定理 6.4.

- Let $X_1, \dots, X_n$ be a random sample from a distribution having pdf or pmf $f(x; \theta)$, for $\theta \in \Omega$.
- Assume that the regularity conditions (R0)–(R4) hold.
- Let $Y = u(X_1, \dots, X_n)$ be a statistic with mean $E(Y) = E[u(X_1, \dots, X_n)] = k(\theta)$.
- Then

$$\text{Var}(Y) \geq \frac{[k'(\theta)]^2}{nI(\theta)}.$$

Properties

- Let $X_1, \dots, X_n$ be a random sample from a distribution having pdf or pmf $f(x; \theta)$, for $\theta \in \Omega$.
- Assume that the regularity conditions (R0)–(R4) hold.
- Let $Y = u(X_1, \dots, X_n)$ be a statistic with mean $E(Y) = E[u(X_1, \dots, X_n)] = \theta$.
- Then

$$\text{Var}(Y) \geq \frac{1}{nI(\theta)}.$$

Efficient estimator

定義 6.2.

- Let Y be an unbiased estimator of a parameter θ in the case of point estimation.
- The statistic Y is called an efficient estimator of θ if and only if the variance of Y attains the Rao–Cramér lower bound.

Efficiency

定義 6.3.

- Suppose we can differentiate with respect to a parameter under an integral or summation symbol.
- The ratio of the Rao–Cramér lower bound to the actual variance of any unbiased estimator of a parameter is called the efficiency of that estimator.

Example 6 — Poisson

- Let $X_1, X_2, \dots, X_n$ denote a random sample from a Poisson distribution that has the mean $\theta > 0$.
- Show $\bar{X}$ is an efficient estimator of θ .

Assumptions – Regularity conditions

- (R5) The pdf $f(x; \theta)$ is three times differentiable as a function of θ . Further, for all $\theta \in \Omega$, there exist a constant c and a function $M(x)$ such that

$$\left| \frac{\partial^3}{\partial \theta^3} \log f(x; \theta) \right| \leq M(x),$$

with $E_{\theta_0}[M(X)] < \infty$, for all $\theta_0 - c < \theta < \theta_0 + c$ and all x in the support of X .

Asymptotic distribution of MLE

定理 6.5.

- Assume $X_1, \dots, X_n$ are iid with pdf $f(x; \theta_0)$ for $\theta \in \Omega$ such that the regularity conditions (R0)–(R5) are satisfied.
- Suppose further that the Fisher information satisfies $0 < I(\theta_0) < \infty$.
- Then any consistent sequence of solutions of the mle equations satisfies

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{D} N(0, \frac{1}{I(\theta_0)})$$

Example 7 — Beta distribution

- Let $X_1, X_2, \dots, X_n$ denote a random sample from a Beta($\theta, 1$) distribution with pdf

$$f(x; \theta) = \begin{cases} \theta x^{\theta-1} & 0 < x < 1 \\ 0 & \text{elsewhere,} \end{cases}$$

where $\Omega = (0, \infty)$.

- Find an unbiased estimator of θ .

Stage 3 — Connect : 估計、檢定與計算方法

6.5 6.3 Maximum Likelihood Tests

Hypotheses

- Let $X_1, \dots, X_n$ be iid with pdf $f(x; \theta)$ for $\theta \in \Omega$.
- Assume θ is a scalar.
- Consider a two-sided hypotheses

$$H_0 : \theta = \theta_0 \quad \text{versus} \quad H_a : \theta \neq \theta_0$$

where $\theta = \theta_0$ is a specified value.

Likelihood function and log likelihood function for a single parameter

- Assume $\Omega \in \mathbb{R}$.
- Define likelihood function as

$$L(\theta) = L(\theta; x) = \prod_{i=1}^n f(x_i; \theta), \theta \in \Omega,$$

where $x = (x_1, \dots, x_n)'$.

- Log likelihood function is more convenient to use.

- Denote as

$$l(\theta) = l(\theta; x) = \log L(\theta) = \sum_{i=1}^n \log f(x_i; \theta), \theta \in \Omega,$$

where $x = (x_1, \dots, x_n)'$.

Ratio of two likelihood functions

- If θ_0 is the true value of θ , asymptotically $L(\theta_0)$ is the maximum value of $L(\theta)$.
- Consider the ratio of two likelihood function defined as

$$\Lambda = \frac{L(\theta_0)}{L(\hat{\theta})}$$

- Note
 - $\Lambda \leq 1$
 - If H_0 is true, Λ should be large (close to 1).
 - If H_a is true, Λ should be small.

Likelihood ratio tests

- For a specified significance level α , this leads to the intuitive decision rule

Reject H_0 in favor of H_a if $\Lambda \leq c$,

where c is such that

$$\alpha = P[\Lambda \leq c | \theta = \theta_0]$$

- Call it the likelihood ratio test (LRT).

Example 8 — Exponential distribution

- Suppose $X_1, \dots, X_n$ are iid with pdf $f(x; \theta) = \theta^{-1} \exp(-x/\theta)$, for $x, \theta > 0$.
- Consider a two-sided hypotheses

$$H_0 : \theta = \theta_0 \quad \text{versus} \quad H_a : \theta \neq \theta_0$$

where $\theta = \theta_0$ is a specified value.

- Find the critical region based on LRT.

Example 9 — Normal distribution

- Suppose $X_1, \dots, X_n$ are iid from a $N(\theta, \sigma^2)$ distribution, where $-\infty < \theta < \infty$ and σ^2 is known.
- Consider a two-sided hypotheses

$$H_0 : \theta = \theta_0 \quad \text{versus} \quad H_a : \theta \neq \theta_0$$

where $\theta = \theta_0$ is a specified value.

- Find the critical region based on LRT.

Asymptotic distribution of LRT

定理 6.6.

- Assume the same regularity conditions as for Theorem 6.2.2.
- Under the null hypothesis, $H_0 : \theta = \theta_0$,

$$-2 \log \Lambda \xrightarrow{D} \chi^2(1).$$

Remarks – Decision rule

- LRT

$$\text{Reject } H_0 \text{ if } \chi_L^2 \geq \chi_\alpha^2(1).$$

where $\chi_L^2 = -2 \log \Lambda$.

- Wald test

$$\text{Reject } H_0 \text{ if } \chi_W^2 \geq \chi_\alpha^2(1).$$

where $\chi_W^2 = \left\{ \sqrt{nI(\hat{\theta})}(\hat{\theta} - \theta_0) \right\}^2$.

- Under H_0 , we have

$$\chi_W^2 - \chi_L^2 \xrightarrow{P} 0.$$

Score test

- The scores are the components of the vector

$$S(\theta) = \left(\frac{\partial \log f(X_1; \theta)}{\partial \theta}, \dots, \frac{\partial \log f(X_n; \theta)}{\partial \theta} \right)'.$$

- Define

$$\chi_R^2 = \left(\frac{l'(\theta_0)}{\sqrt{nI(\theta_0)}} \right)^2,$$

where

$$\frac{1}{\sqrt{n}} l'(\theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{\partial \log f(X_i; \theta)}{\partial \theta}$$

Remark

- Under H_0 , we have

$$\chi_R^2 = \chi_W^2 + R_{0n}$$

where R_{0n} converges to 0 in probability.

- Decision rule based on the score test

$$\text{Reject } H_0 \text{ if } \chi_R^2 \geq \chi_\alpha^2(1).$$

Example 10 — Beta distribution

- Suppose $X_1, \dots, X_n$ are iid from $\text{beta}(\theta, 1)$, for $x, \theta > 0$.
- Consider a two-sided hypotheses

$$H_0 : \theta = 1 \quad \text{versus} \quad H_a : \theta \neq 1$$

where $\theta = \theta_0$ is a specified value.

- Find the critical region based on LRT, Wald test and score test.

Example 11 — Laplace location model

- Suppose the location model

$$X_i = \theta + e_i, i = 1, \dots, n,$$

where $\theta \in R$ and the random errors e_i are iid each having the Laplace pdf,

$$f(x) = \frac{1}{2}e^{-|x|}, x \in R.$$

- Consider a two-sided hypotheses

$$H_0 : \theta = \theta_0 \quad \text{versus} \quad H_a : \theta \neq \theta_0$$

where $\theta = \theta_0$ is a specified value.

- Find the critical region based on LRT, Wald test and score test.

6.6 6.4 Multiparameter Case: Estimation**Likelihood function and log likelihood function for multiparameter**

- Assume $\Omega \in R^p$.
- Define likelihood function as

$$L(\theta) = L(\theta; x) = \prod_{i=1}^n f(x_i; \theta), \theta \in \Omega,$$

where $x = (x_1, \dots, x_n)'$.

- Log likelihood function is more convenient to use.
- Denote as

$$l(\theta) = l(\theta; x) = \log L(\theta) = \sum_{i=1}^n \log f(x_i; \theta), \theta \in \Omega,$$

where $x = (x_1, \dots, x_n)'$.

Maximum likelihood estimator (MLE)

定義 6.4.

- Point estimator of θ is $\hat{\theta} = \hat{\theta}(X_1, \dots, X_n)$, where $\hat{\theta}$ maximizes the function $L(\theta)$, i.e.,

$$\hat{\theta} = \operatorname{argmax}_{\theta} L(\theta) = \operatorname{argmax}_{\theta} l(\theta).$$
- $\hat{\theta}$ is called the maximum likelihood estimator (mle) of θ .

Property of MLE

定理 6.7.

- Let $X_1, \dots, X_n$ be iid with the pdf $f(x; \theta)$, $\theta \in \Omega$.
- For a specified function g , let $\eta = g(\theta)$ be a parameter of interest.
- Suppose $\hat{\theta}$ is the mle of θ .
- Then $g(\hat{\theta})$ is the mle of $\eta = g(\theta)$.

Example 12 — Normal distribution

- Suppose $X_1, \dots, X_n$ are iid from a $N(\mu, \sigma^2)$ distribution, where $-\infty < \mu < \infty$ and $\sigma^2 > 0$.
- Find the MLE of $\theta = (\mu, \sigma^2)$.

Example 13 — Laplace location model

- Suppose $X_1, \dots, X_n$ are iid from a general Laplace distribution with pdf

$$f(x) = \frac{1}{2b} e^{-|x-a|/b}, x \in R$$

where $-\infty < a < \infty$ and $b > 0$.

- Find the MLE of $\theta = (a, b)$.

Score vector

- The score vector is defined as

$$\Delta \log f(X; \theta) = \left(\frac{\partial \log f(x; \theta)}{\partial \theta_1}, \dots, \frac{\partial \log f(x; \theta)}{\partial \theta_p} \right)'.$$

Fisher information matrix of size 1

- Let X have a pdf or pmf $f(x; \theta)$.
- The Fisher information is the variance of the score vector and defined as

$$I(\theta) = \text{Cov}(\Delta \log f(X; \theta)).$$

where the jk th entry of $I(\theta)$ is given by

$$I_{jk} = \text{Cov}\left(\frac{\partial \log f(x; \theta)}{\partial \theta_j}, \frac{\partial \log f(x; \theta)}{\partial \theta_k}\right)$$

Properties

- Show

$$I_{jk} = -E\left(\frac{\partial^2 \log f(x; \theta)}{\partial \theta_j \partial \theta_k}\right)$$

and

$$I_{jj} = -E\left(\frac{\partial^2 \log f(x; \theta)}{\partial \theta_j^2}\right)$$

Fisher information matrix of size n

- Let $X_1, \dots, X_n$ be a random sample from a pdf or pmf $f(x; \theta)$.
- The Fisher information of size n is $nI(\theta)$.

Rao-Cramér lower bound

定理 6.8.

- Let $X_1, \dots, X_n$ be a random sample from a distribution having pdf or pmf $f(x; \theta)$, for $\theta \in \Omega$.
- Let $Y_j = u_j(X_1, \dots, X_n)$ be a statistic with mean $E(Y_j) = E[u(X_1, \dots, X_n)] = \theta_j$.
- Then

$$\text{Var}(Y_j) \geq \frac{1}{n}[I^{-1}(\theta)]_{jj}.$$

Example 14 — Normal distribution

- Suppose $X_1, \dots, X_n$ are iid from a $N(\mu, \sigma^2)$ distribution, where $-\infty < \mu < \infty$ and $\sigma^2 > 0$.
- Find the Fisher information matrix for (μ, σ) .

Example 15 — Location and scale model

- Suppose $X_1, \dots, X_n$ are iid from a location and scale distribution with pdf

$$f_X(x) = b^{-1} f\left(\frac{x-a}{b}\right), x \in R$$

where $-\infty < a < \infty$ and $b > 0$.

- Find the Fisher information matrix for (a, b) .

Example 16 — Multinomial distribution

- Consider a random trial which can result in one, and only one, of k outcomes or categories.
- Let X_j be 1 or 0 depending on whether the j th outcome occurs or does not, for $j = 1, \dots, k$.
- Then $X = (X_1, \dots, X_{k-1})'$ has a multinomial distribution with pmf

$$f_X(x) = p_1^{x_1} \cdots p_{k-1}^{x_{k-1}} \times \left(1 - \sum_{j=1}^{k-1} p_j\right)^{1 - \sum_{j=1}^{k-1} x_j}$$

where $\Omega = \{p : 0 < p_j < 1, j = 1, \dots, k-1; \sum_{j=1}^{k-1} p_j < 1\}$.

- Find the Fisher information matrix for p .

Asymptotic properties of MLE**定理 6.9.**

- Assume $X_1, \dots, X_n$ are iid with pdf $f(x; \theta)$ for $\theta \in \Omega$ such that the regularity conditions are satisfied.
- The likelihood equation

$$\frac{\partial l(\theta)}{\partial \theta} = 0$$

has a solution $\hat{\theta}_n$ such that $\hat{\theta}_n \xrightarrow{P} \theta$.

- Then any consistent sequence of solutions of the mle equations satisfies

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{D} N_p(0, I^{-1}(\theta))$$

Asymptotic properties of MLE

- Assume $X_1, \dots, X_n$ are iid with pdf $f(x; \theta)$ for $\theta \in \Omega$ such that the regularity conditions are satisfied.
- Let $\hat{\theta}_n$ be a sequence of consistent solutions of the likelihood equation.
- Then $\hat{\theta}_n$ are asymptotically efficient estimates; that is, for $j = 1, \dots, p$,

$$\sqrt{n}(\hat{\theta}_{n,j} - \theta_j) \xrightarrow{D} N(0, [I^{-1}(\theta)]_{jj})$$

Asymptotic properties of MLE

- Let g be a transformation $g(\theta) = (g_1(\theta), \dots, g_k(\theta))'$ such that $1 \leq k \leq p$ and the $k \times p$ matrix of partial derivatives

$$B = \left[\frac{\partial g_i}{\partial \theta_j}, i = 1, 2, \dots, k; j = 1, 2, \dots, p, \right]$$

has continuous elements and does not vanish in a neighborhood of θ .

- Let $\hat{\eta} = g(\hat{\theta})$.
- Then $\hat{\eta}$ is the mle of η and

$$\sqrt{n}(\hat{\eta} - \eta) \xrightarrow{D} N_k(0, B[I^{-1}(\theta)]B')$$

6.7 6.5 Multiparameter Case: Testing

Hypotheses

- Let $X_1, \dots, X_n$ be iid with pdf $f(x; \theta)$ for $\theta \in \Omega \in R^p$.
- Consider a two-sided hypotheses

$$H_0 : \theta \in \omega \quad \text{versus} \quad H_a : \theta \in \omega^c \cap \Omega$$

where $\omega \in \Omega$ is defined in terms of q , $0 < q \leq p$, independent constraints of $g_1(\theta) = a_1, \dots, g_q(\theta) = a_q$.

- The functions $g_1, \dots, g_q$ must be continuously differentiable.
- This implies that ω is a $(p - q)$ -dimensional space.

Ratio of two likelihood functions

- Consider the ratio of two likelihood function defined as

$$\Lambda = \frac{\max_{\theta \in \omega} L(\theta)}{\max_{\theta \in \Omega} L(\theta)}$$

- For a specified significance level α , this leads to the intuitive decision rule

$$\text{Reject } H_0 \text{ in favor of } H_a \text{ if } \Lambda \leq c,$$

where c is such that

$$\alpha = P[\Lambda \leq c | \theta = \theta_0]$$

- Call it the likelihood ratio test (LRT).

Example 17 — Normal distribution

- Suppose $X_1, \dots, X_n$ are iid from a $N(\theta, \sigma^2)$ distribution, where $-\infty < \theta < \infty$ and $\sigma^2 > 0$.
- Consider a two-sided hypotheses

$$H_0 : \theta = \theta_0 \quad \text{versus} \quad H_a : \theta \neq \theta_0$$

where $\theta = \theta_0$ is a specified value.

- Find the critical region based on LRT.

Asymptotic distribution of LRT

定理 6.10.

- Let $X_1, \dots, X_n$ be iid with pdf $f(x; \theta)$ for $\theta \in \Omega \subset \mathbb{R}^p$.
- Assume the regularity conditions hold.
- Let $\hat{\theta}_n$ be a sequence of consistent solutions of the likelihood equation when the parameter space is the full space Ω .
- Let $\hat{\theta}_{0,n}$ be a sequence of consistent solutions of the likelihood equation when the parameter space is the reduced space ω .
- Under the null hypothesis, H_0 ,

$$-2 \log \Lambda \xrightarrow{D} \chi^2(q).$$

Example 18 — Multinomial distribution

- Consider a random trial which can result in one, and only one, of 3 outcomes or categories.
- Then $X = (X_1, X_2)'$ has a multinomial distribution with pmf

$$f_X(x) = p_1^{x_1} p_2^{x_2} \times \left(1 - \sum_{j=1}^2 p_j \right)^{1 - \sum_{j=1}^2 x_j}$$

- Consider $H_0 : p_1 = p_2$ versus $H_a : p_1 \neq p_2$.
- Find the LRT.

6.8 6.6 The EM Algorithm

Motivation

- In practice, we are often in the situation where part of the data is missing.
- For example, we may be observing lifetimes of mechanical parts that have been put on test and some of these parts are still functioning when the statistical analysis is carried out.

Notations

- Suppose we consider a sample of n items, where
 - n_1 of the items are observed,
 - $n_2 = n - n_1$ items are not observable.
- Denote the observed items by $X' = (X_1, X_2, \dots, X_{n_1})'$ and unobserved items by $Z' = (X_1, X_2, \dots, X_{n_2})'$.
- Assume that the X_i s are iid with pdf $f(x|\theta)$, where $\theta \in \Omega$.
- Assume that the Z_j s and the X_i s are mutually independent.

Likelihood

- Let $g(x|\theta)$ denote the joint pdf of X .
- Let $h(x, z|\theta)$ denote the joint pdf of observed and unobserved items.
- Let $k(z|\theta, x)$ denote the conditional pdf of missing data given the observed data and

$$k(z|\theta, x) = \frac{h(x, z|\theta)}{g(x|\theta)}$$

- Observed likelihood function is defined by

$$L(\theta|x) = g(x|\theta)$$

- Complete likelihood function is defined by

$$L^c(\theta|x, z) = h(x, z|\theta)$$

Decomposition

- Show

$$\log L(\theta|x) = E_{\theta_0}[\log L^c(\theta|x, Z)|\theta_0, x] - E_{\theta_0}[\log k(Z|\theta, x)|\theta_0, x]$$

Expectation

- Define

$$Q(\theta|\theta_0, x) = E_{\theta_0}[\log L^c(\theta|x, Z)|\theta_0, x]$$

- The expectation that defines the function Q is called the E step of the EM algorithm.

Maximization

- Maximize $\log L(\theta|x)$.
- Only need to maximize $Q(\theta|\theta_0, x)$

EM algorithm

- Let $\hat{\theta}^{(m)}$ denote the estimate on the m th step.
- To compute the estimate on the $(m + 1)$ st step, do

1. Expectation step: Compute

$$Q(\theta|\hat{\theta}^{(m)}, x) = E_{\hat{\theta}^{(m)}}[\log L^c(\theta|x, Z)|\hat{\theta}^{(m)}, x]$$

where the expectation is taken under the conditional pdf $k(z|\hat{\theta}^{(m)}, x)$.

2. Maximization step: Let

$$\hat{\theta}^{(m+1)} = \text{Argmax}_{\theta} Q(\theta|\hat{\theta}^{(m)}, x).$$

Monotonicity

定理 6.11.

- The sequence of estimates $Q(\theta|\hat{\theta}^{(m)}, x)$, defined by EM algorithm, satisfies

$$L(\hat{\theta}^{(m+1)}|x) \geq L(\hat{\theta}^{(m)}|x).$$

Example 19 — Mixture distribution

- Suppose $Y_1 \sim N(\mu_1, \sigma_1^2)$ and $Y_2 \sim N(\mu_2, \sigma_2^2)$.
- Let W be a Bernoulli random variable independent of Y_1 and Y_2 and with probability of success $\epsilon = P(W = 1)$.
- The pdf of the mixture random variable X is

$$f(x) = (1 - \epsilon)f_1(x) + \epsilon f_2(x), -\infty < x < \infty,$$

where $f_j(x) = \sigma_j^{-1} \phi[(x - \mu_j)/\sigma_j]$, $j = 1, 2$, and $\phi(z)$ is the pdf of a standard normal j random variable.

- Suppose we observe a random sample $X' = (X_1, X_2, \dots, X_n)$ from this mixture distribution with pdf $f(x)$.

6.9 本章摘要

本章以似然為核心統一估計與檢定，並延伸至效率、多參數模型及含隱藏資料的 EM 計算。

6.10 常見錯誤

- 忽略支持集是否依賴參數
- 只解 score equation 而未驗證最大值
- 混淆觀察到的資訊與期望 Fisher information
- 使用漸近卡方結果時未檢查正則條件

6.11 章末練習

1. 從本章選出三個關鍵定義，分別寫出數學形式、成立條件及一個例子。
2. 選擇一個主要定理或方法，重做推導並標記每一步使用的假設。
3. 從原講義例題挑選一題，完整寫出解題目標、方法選擇、計算與結果解釋。
4. 設計一個會違反本章方法條件的反例，說明錯誤會出現在哪一步。

待逐章補強

建議後續為本章補入：中文概念導讀、完整解答、課堂練習難度標記，以及與下一章的銜接說明。

7 充分性、完備性與估計量品質

! 第 8 版對照與教學安排

本章以 HMC8 Chapter 7 為編號依據。先回顧估計量的品質準則，再深化 sufficiency、Rao–Blackwell、completeness 與 exponential class，最後連到多參數、minimal sufficiency、ancillary statistics 與 independence。

Recall → Deepen → Connect

💡 本章定位

本講義由蘇南誠老師編授。感謝黃怡婷老師提供原始數理統計講義檔案；本章保留原有英文術語與公式，並整理為可連續閱讀的教學形式。

7.1 學習目標

完成本章後，學生應能：

- 以偏誤、變異數與 MSE 評估估計量
- 使用因子分解定理辨認充分統計量
- 說明完備性與唯一性
- 辨認指數族並運用其結構
- 理解最小充分性、附屬統計量與獨立性

7.2 必備的先備知識

聯合分配、條件分配、點估計與似然函數。

以下依原講義順序整理核心定義、定理、公式與例題。

Stage 1 — Recall：估計量品質的判準

7.3 7.1 Measures of Quality of Estimators

Notation

- Let $f(x; \theta)$ be a pdf (pmf) of a continuous (discrete) random variable X .
- Let $Y_n = u(X_1, \dots, X_n)$ denote a point estimator of θ based on a sample $X_1, \dots, X_n$.

- Recall properties of estimators
 - Y_n is a consistent estimator if $Y_n \xrightarrow{P} \theta$.
 - Y_n is an unbiased estimator if $E[Y_n] = \theta$.

MVUE

定義 7.1.

- Let $Y_n = u(X_1, \dots, X_n)$ denote a point estimator of θ based on a sample $X_1, \dots, X_n$.
- Y_n is called a minimum variance unbiased estimator (MVUE) of θ if
 - Y_n is unbiased.
 - $\text{Var}(Y_n)$ is less than or equal to the variance of every other unbiased estimator of θ .

Example 1

- Consider a sample $X_1, \dots, X_n$ from $N(\theta, \sigma^2)$, $\theta \in R$.
- Compare the variance of $\bar{X}$ and X_1 .

Decision rule

- Consider a sample $X_1, \dots, X_n$ from $f(x; \theta)$, $\theta \in \Omega$.
- Let $Y_n = u(X_1, \dots, X_n)$ denote a point estimator of θ based on a sample $X_1, \dots, X_n$.
- Let $\delta(y)$ be that function of the observed value of the statistic Y_n .
- The function $\delta(y)$ decides the value of our point estimate of θ and δ is called a decision function or a decision rule.
- One value of the decision function, say $\delta(y)$, is called a decision.

Loss function

- With each pair, $[\theta, \delta(y)]$, $\theta \in \Omega$, we associate a nonnegative number $\mathcal{L}[\theta, \delta(y)]$ that reflects the seriousness of difference.
- $\mathcal{L}$ is called the loss function.
- The expected (mean) value of the loss function is called the risk function.
- Suppose Y is a continuous random variable, then we have

$$R(\theta, \delta) = E[\mathcal{L}[\theta, \delta(y)]] = \int_{-\infty}^{\infty} \mathcal{L}[\theta, \delta(y)] f_Y(y; \theta) dy$$

- It would be desirable to select a decision function that minimizes the risk $R(\theta, \delta)$ for all values of θ , $\theta \in \Omega$.

Example 2

- Consider a sample $X_1, \dots, X_{25}$ from $N(\theta, 1)$, $\theta \in R$.
- Let $Y = \bar{X}$.
- Let $\mathcal{L}[\theta, \delta(y)] = [\theta - \delta(y)]^2$.
- Consider $\delta_1 = y$ and $\delta_2 = 0$ for $y \in R$.
- Find $R(\theta, \delta_1)$ and $R(\theta, \delta_2)$.

Example 3 — Cont's

- If $\theta = 0$, $\delta_2(y)$ is an excellent decision.
- If $\theta = 2$, $\delta_1(y)$ is a better decision.
- If $-\frac{1}{5} < \theta < \frac{1}{5}$, $R(\theta, \delta_2) \leq R(\theta, \delta_1)$.
- Otherwise, $R(\theta, \delta_1) \leq R(\theta, \delta_2)$.

Example 4 — Cont's

- Suppose the decision function δ needs to satisfy

$$E[\delta(Y)] = \theta \forall \theta \in \Omega.$$

- The decision function $\delta_2(y) = 0$ is not allowed.
- Under this restriction and with the given $\mathcal{L}[\theta, \delta(y)]$, the risk function is the variance of the unbiased estimator $\delta(Y)$.
- The problem becomes to find the MVUE of θ .

Minimax criterion

- Find a decision that minimize the maximum of the risk function.
- If the decision function given by $\delta_0(y)$ is such that, for all $\theta \in \Omega$,

$$\max_{\theta} R[\theta, \delta_0(y)] \leq \max_{\theta} R[\theta, \delta(y)]$$

for every other decision function $\delta(y)$, then $\delta_0(y)$ is called a minimax decision function.

MVUE

- With the restriction $E[\delta(Y)] = \theta$ and the loss function $\mathcal{L}[\theta, \delta(y)] = [\theta - \delta(y)]^2$.
- The decision function that minimizes the risk function yields an unbiased estimator with minimum variance.

Minimum mean-squared-error estimator

- Without the restriction $E[\delta(Y)] = \delta$ and the loss function $\mathcal{L}[\theta, \delta(y)] = [\theta - \delta(y)]^2$.
- the decision function $\delta(Y)$, if it exists, which minimizes $E[\theta - \delta(y)]^2$ uniformly in θ is sometimes called the minimum mean-squared-error estimator.

Loss function

- Square-error loss function

$$\mathcal{L}[\theta, \delta(y)] = [\theta - \delta(y)]^2.$$

- Absolute-error loss function

$$\mathcal{L}[\theta, \delta(y)] = |\theta - \delta(y)|$$

- Goalpost loss function

$$\mathcal{L}[\theta, \delta(y)] = \begin{cases} 0 & |\theta - \delta(y)| \leq a \\ 0b & |\theta - \delta(y)| > a \end{cases}$$

where a and b are positive constants.

Example 5

- Suppose that two statisticians, A and B , observe 10 independent trials of a random experiment ending in success or failure.
- Let the probability of success on each trial be θ , where $0 < \theta < 1$.
- Suppose each statistician observes one success in these 10 trials.

Example 6 — Statistician A

- A had decided to take $n = 10$ such observations in advance and found only one success.
- $Y \sim \text{Bin}(n = 10, \theta)$ and $y = 1$ is observed.
- An estimator of θ is Y/n .
- $E[Y/n] = \theta$
- Y/n is an unbiased estimator of θ .

Example 7 — Statistician B

- B had decided to take as many observations as needed to get the first success, which happened on the 10th trial.
- B considers the random variable Z that has a geometric pmf $g(z) = (1 - \theta)^{z-1}\theta$, $z = 1, 2, 3, \dots$, and $z = 10$ is observed.

Example 8 — Statistician B

- B had decided to take as many observations as needed to get the first success, which happened on the 10th trial.
- B considers the random variable Z that has a geometric pmf $g(z) = (1 - \theta)^{z-1}\theta$, $z = 1, 2, 3, \dots$, and $z = 10$ is observed.
- An estimator of θ is $1/Z$.
- $E[1/Z] = \sum_{z=1}^{\infty} \frac{1}{z}(1 - \theta)^{z-1}\theta = \theta + \frac{1}{2}(1 - \theta)\theta + \dots > \theta$.
- $1/Z$ is a biased estimator.

Example 9 — Likelihood

- Consider two likelihood functions, namely,

$$L_1(\theta) = \binom{10}{y} \theta^y (1 - \theta)^{10-y}$$
$$L_2(\theta) = (1 - \theta)^{z-1} \theta.$$

- Both maximize $(1 - \theta)^9 \theta$ and give $\hat{\theta} = 1/10$.

Likelihood principle

- Suppose two different sets of data from possibly two different random experiments lead to respective likelihood ratios, $L_1(\theta)$ and $L_2(\theta)$, that are proportional to each other.
- These two data sets provide the same information about the parameter θ and a statistician should obtain the same estimate of θ from either.

Stage 2 — Deepen : 充分性、完備性與指數族

7.4 7.2 A Sufficient Statistic for a Parameter

Example 10 — Sufficient statistic

- Consider 10 independent trials of a random experiment ending in success or failure.
- First experiment – 1 1 0 0 0 0 1 1 0 0
- Second experiment – 0 0 1 0 0 0 0 0 0 0
- Third experiment – 1 1 1 1 1 1 1 1 0 1

Example 11 — Bernoulli trial

- Consider $X_i, i = 1, 2, \dots, n$ denote a random sample from the distribution that has pmf

$$f(x; \theta) = \theta^x (1 - \theta)^{1-x}, x = 0, 1; 0 < \theta < 1$$

and 0 elsewhere.

- Let $Y = \sum_{i=1}^n X_i$.
- Find the conditional distribution of $X_1, X_2, \dots, X_n$ given $Y = y$.

Sufficient statistic

定義 7.2.

- Let $X_i, i = 1, 2, \dots, n$ denote a random sample from the distribution that has pdf or pmf $f(x; \theta), \theta \in \Omega$.
- Let $Y_1 = u_1(X_1, \dots, X_n)$ be a statistic whose pdf or pmf is $f_{Y_1}(y; \theta)$.
- Then Y_1 is a sufficient statistic for θ if and only if

$$\frac{f(x_1; \theta)f(x_2; \theta) \cdots f(x_n; \theta)}{f_{Y_1}(y; \theta)} = H(x_1, x_2, \dots, x_n),$$

where $H(x_1, x_2, \dots, x_n)$ does not depend upon $\theta \in \Omega$.

Example 12 — Gamma distribution

- Consider $X_i, i = 1, 2, \dots, n$ denote a random sample from a gamma distribution with $\alpha = 2$ and $\beta = \theta > 0$.
- Let $Y = \sum_{i=1}^n X_i$.
- Show Y is a sufficient statistic of θ .

Example 13 — Location-shift exponential distribution

- Consider $X_i, i = 1, 2, \dots, n$ denote a random sample from a location-shift exponential distribution with pdf

$$f(x; \theta) = e^{-(x-\theta)}, x > \theta.$$

- Let $Y = \min(X_1, \dots, X_n)$.
- Show Y is a sufficient statistic of θ .

Neyman factorization theorem

定理 7.1.

- Consider $X_i, i = 1, 2, \dots, n$ denote a random sample from a distribution with pdf or pmf $f(x; \theta), \theta \in \Omega$.
- The statistic $Y_1 = u_1(X_1, \dots, X_n)$ is a sufficient statistic for θ if and only if we can find two nonnegative functions, k_1 and k_2 , such that

$$\begin{aligned} f(x_1; \theta)f(x_2; \theta) \cdots f(x_n; \theta) \\ = k_1[u_1(x_1, x_2, \dots, x_n); \theta]k_2(x_1, x_2, \dots, x_n), \end{aligned}$$

where $k_2(x_1, x_2, \dots, x_n)$ does not depend upon θ .

Example 14 — Normal distribution

- Consider $X_i, i = 1, 2, \dots, n$ denote a random sample from a normal distribution with μ and σ^2 , where $\theta \in R$ and σ^2 is known.
- Let $Y = \bar{X}$.
- Show Y is a sufficient statistic of θ .

Example 15 — Special distribution

- Consider $X_i, i = 1, 2, \dots, n$ denote a random sample from a distribution with pdf

$$f(x; \theta) = \begin{cases} \theta x^{\theta-1} & 0 < x < 1 \\ 0 & \text{elsewhere,} \end{cases}$$

where $\theta > 0$.

- Find a sufficient statistic of θ .

7.5 7.3 Properties of a Sufficient Statistic

Recall

- If X_1 and X_2 are random variables such that the variance of X_2 exists, then

$$E[X_2] = E[E[X_2|X_1]]$$

and

$$\text{Var}(X_2) \geq \text{Var}(E[X_2|X_1]).$$

Rao-Blackwell theorem

定理 7.2.

- Let $X_i, i = 1, 2, \dots, n$ denote a random sample from a distribution with pdf or pmf $f(x; \theta)$, $\theta \in \Omega$.
- The statistic $Y_1 = u_1(X_1, \dots, X_n)$ is a sufficient statistic for θ .
- Let $Y_2 = u_2(X_1, X_2, \dots, X_n)$, not a function of Y_1 , be an unbiased estimator of θ .
- Define $E[Y_2|Y_1 = y_1] = \psi(y_1)$.
- Then
 - $\psi(Y_1)$ is an unbiased estimator of θ .
 - $\text{Var}(\psi(Y_1)) \leq \text{Var}(Y_2)$.

Properties of MLE

定理 7.3.

- Let $X_i, i = 1, 2, \dots, n$ denote a random sample from a distribution with pdf or pmf $f(x; \theta)$, $\theta \in \Omega$.
- The statistic $Y_1 = u_1(X_1, \dots, X_n)$ is a sufficient statistic for θ .
- If a maximum likelihood estimator $\hat{\theta}$ of θ also exists uniquely.
- Then $\hat{\theta}$ is a function of Y_1 .

Example 16 — Exponential distribution

- Consider $X_i, i = 1, 2, \dots, n$ denote a random sample from a distribution with pdf

$$f(x; \theta) = \begin{cases} \theta e^{-\theta x} & 0 < x; \theta > 0 \\ 0 & \text{elsewhere,} \end{cases}.$$

- Find an MVUE of θ .

7.6 7.4 Completeness and Uniqueness

Example 17 — Poisson distribution

- Consider $X_i, i = 1, 2, \dots, n$ denote a random sample from the Poisson distribution with pmf

$$f(x; \theta) = \frac{\theta^x e^{-\theta}}{x!}, x = 0, 1, 2, \dots; \theta > 0,$$

$$f(x; \theta) = \begin{cases} \frac{\theta^x e^{-\theta}}{x!}, & x = 0, 1, 2, \dots; \theta > 0, \\ 0 & \text{elsewhere,} \end{cases}.$$

- Let $Y_1 = \sum_{i=1}^n X_i$.
- Consider $u(Y_1)$.
- Show that $E[u(Y_1)] = 0$ if and only if $u(y_1) = 0, \forall y_1 = 0, 1, 2, \dots$

Complete family

定義 7.3.

- Let the random variable Z of either the continuous type or the discrete type have a pdf or pmf that is one member of the family $\{h(z; \theta) : \theta \in \Omega\}$.
- If the condition $E[u(Z)] = 0$, for every $\theta \in \Omega$, requires that $u(z)$ be zero except on a set of points that has probability zero for each $h(z; \theta), \theta \in \Omega$.
- Then the family $\{h(z; \theta) : \theta \in \Omega\}$ is called a complete family of probability density or mass functions.

Example 18 — Exponential distribution

- Consider the family of pdf $\{h(z; \theta) : 0 < \theta < \infty\}$.
- Suppose Z has a pdf in this family given by

$$f(x; \theta) = \begin{cases} \frac{1}{\theta} e^{-x/\theta} & 0 < x; \theta > 0 \\ 0 & \text{elsewhere,} \end{cases}.$$

- Show this is a complete family. [Laplace transformation]

Lehmann and Scheffé theorem

定理 7.4.

- Let $X_i, i = 1, 2, \dots, n$ denote a random sample from a distribution with pdf or pmf $f(x; \theta)$, $\theta \in \Omega$.
- Let $Y_1 = u_1(X_1, \dots, X_n)$ be a sufficient statistic for θ .
- Let the family $\{f_{Y_1}(y_1; \theta) : \theta \in \Omega\}$ be complete.
- If $\psi(Y_1)$ is an unbiased estimator of θ .
- Then $\psi(Y_1)$ is the unique MVUE (UMVUE) of θ .

Example 19 — Uniform distribution

- Consider $X_i, i = 1, 2, \dots, n$ denote a random sample from the uniform distribution with pdf

$$f(x; \theta) = \frac{1}{\theta}, 0 < x < \theta, \theta > 0,$$

and zero elsewhere.

- Find the UMVUE of θ .

7.7 7.5 The Exponential Class of Distributions

Exponential class

定義 7.4.

- A pdf of the form

$$f(x; \theta) = \begin{cases} \exp[p(\theta)K(x) + H(x) + q(\theta)] & x \in \mathcal{S} \\ 0 & \text{elsewhere,} \end{cases}$$

where $\mathcal{S}$ is the support of X , is said to be a member of the regular exponential class of probability density or mass functions if

1. $\mathcal{S}$, the support of X , does not depend upon θ .
2. $p(\theta)$ is a nontrivial continuous function of $\theta \in \Omega$.

3. Finally,

1. if X is a continuous random variable, then each of $K'(x) \neq 0$ and $H(x)$ is a continuous function of $x \in \mathcal{S}$.
2. if X is a discrete random variable, then $K(x)$ is a nontrivial function of $x \in \mathcal{S}$.

Exponential family and sufficient statistic

- Let $X_1, \dots, X_n$ denote a random sample from a distribution that has a pdf of the form

$$f(x; \theta) = \exp[p(\theta)K(x) + H(x) + q(\theta)], x \in \mathcal{S}$$

and 0 elsewhere.

- Show $Y_1 = \sum_{i=1}^n K(x_i)$ is a sufficient statistic for θ .

Example 20 — Normal distribution

- Consider the family $\{f(x; \theta) : 0 < \theta < \infty\}$, where $f(x; \theta)$ is $N(0, \theta)$.
- Show this is an exponential family.

Example 21 — Uniform distribution

- Consider the family $\{f(x; \theta) : 0 < \theta < \infty\}$, where $f(x; \theta)$ is an uniform $(0, 1)$ density.
- Show this is not an exponential family.

Properties of exponential class

定理 7.5.

- Let $X_1, \dots, X_n$ denote a random sample from a distribution that has a pdf of the form

$$f(x; \theta) = \exp[p(\theta)K(x) + H(x) + q(\theta)], x \in \mathcal{S}$$

and 0 elsewhere.

- Consider $Y_1 = \sum_{i=1}^n K(X_i)$.

Properties of exponential class

定理 7.6.

- The pdf or pmf of Y_1 has the form

$$f(y_1; \theta) = \begin{cases} \exp[p(\theta)K(y_1) + H(y_1) + q(\theta)] & y_1 \in \mathcal{S} \\ 0 & \text{elsewhere,} \end{cases}$$

for $y_1 \in \mathcal{S}_{Y_1}$ and some function $R(y_1)$ and $\mathcal{S}_{Y_1}$ and $R(y_1)$ depend on θ .

- $E[Y_1] = -n \frac{q'(\theta)}{p'(\theta)}$.
- $\text{Var}(Y_1) = n \frac{1}{(p'(\theta))^3} \{p''(\theta)q'(\theta) - q''(\theta)p'(\theta)\}$.

Example 22 — Poisson distribution

- Let X have a Poisson distribution with parameter $\theta \in (0, \infty)$.
- Show that this is an exponential family.

Properties

- Let $f(x; \theta)$, $\gamma < \theta < \delta$, be a pdf or pmf of a random variable X whose distribution is a regular case of the exponential class.
- If $X_1, X_2, \dots, X_n$ is a random sample from the distribution of X , then
 - the statistic $Y_1 = \sum_{i=1}^n K(X_i)$ is a sufficient statistic for θ
 - the family $\{f_{Y_1}(y_1; \theta) : \gamma < \theta < \delta\}$ of probability density functions of Y_1 is complete.
 - That is, Y_1 is a complete sufficient statistic for θ .

Example 23 — Normal distribution

- Let $X_1, X_2, \dots, X_n$ denote a random sample from a normal distribution $N(\theta, \sigma^2)$, where $x \in R, \theta \in R$ and σ^2 is given.
- Show $\bar{X}$ is the UMVUE of θ .

Example 24 — Poisson distribution

- Let $X_1, X_2, \dots, X_n$ denote a random sample from a Poisson distribution with parameter $\theta \in (0, \infty)$.
- Show that $\bar{X}$ is the UMVUE of θ .

Stage 3 — Connect : 函數、多參數與輔助統計量

7.8 7.6 Functions of a Parameter

Example 25 — Bernoulli distribution

- Let $X_1, X_2, \dots, X_n$ denote a random sample from a Bernoulli distribution $b(1, \theta)$, $0 < \theta < 1$.
- Find the UMVUE of $\delta = \theta(1 - \theta)$.

Example 26 — Uniform distribution

- Let $X_1, X_2, \dots, X_n$ denote a random sample from an uniform $(0, \theta)$ distribution.
- Find the UMVUE of $g(\theta)$, where $g(\theta)$ is any differentiable function of θ .

Example 27 — Normal distribution

- Let $X_1, X_2, \dots, X_n$ denote a random sample from a normal distribution $N(\theta, \sigma^2)$, where $x \in R, \theta \in R$ and σ^2 is given.
- Find the UMVUE of $P[X_1 \leq c]$.

7.9 7.7 The Case of Several Parameters**Sufficient statistic**

定義 7.5.

- Let $X_i, i = 1, 2, \dots, n$ denote a random sample from the distribution that has pdf or pmf $f(x; \theta), \theta \in \Omega^p$.
- Let $\mathcal{S}$ denote the support of X .
- Let Y be an m -dimensional random vector of statistics $Y = (Y_1, \dots, Y_m)'$, where $Y_i = u_i(X_1, \dots, X_n)$, for $i = 1, 2, \dots, m$.
- Denote the pdf or pmf of Y by $f_Y(y; \theta)$ for $y \in R^m$.
- Then Y is jointly sufficient for θ if and only if

$$\frac{f(x_1; \theta)f(x_2; \theta) \cdots f(x_n; \theta)}{f_Y(y; \theta)} = H(x_1, x_2, \dots, x_n),$$

where $H(x_1, x_2, \dots, x_n)$ does not depend upon θ .

Example 28 — Uniform distribution

- Let $X_i, i = 1, 2, \dots, n$ denote a random sample from the distribution having pdf

$$f(x; \theta_1, \theta_2) = \begin{cases} \frac{1}{2\theta_2} & \theta_1 - \theta_2 < x < \theta_1 + \theta_2 \\ 0 & \text{elsewhere} \end{cases}$$

- Find joint sufficient statistics for θ_1 and θ_2 .

Exponential class

定義 7.6.

- A pdf of the form

$$f(x; \theta) = \begin{cases} \exp[\sum_{j=1}^m p_j(\theta) K_j(x) + H(x) + q(\theta)] & x \in \mathcal{S} \\ 0 & \text{elsewhere,} \end{cases}$$

where $\mathcal{S}$ is the support of X , is said to be a member of the regular exponential class of probability density or mass functions if

1. $\mathcal{S}$, the support of X , does not depend upon θ .
2. the space Ω contains a nonempty, m -dimensional open rectangle,
3. $p_j(\theta)$, $j = 1, \dots, m$, are nontrivial, functionally independent, continuous functions of θ .

Exponential class

定義 7.7.

- and
 1. if X is a continuous random variable, then the m derivatives $K'_j(x)$, for $j = 1, 2, \dots, m$, are continuous for $a < x < b$ and no one is a linear homogeneous function of the others, and $H(x)$ is a continuous function of x , $a < x < b$.
 2. if X is a discrete random variable, then $K_j(x)$, $j = 1, 2, \dots, m$, are nontrivial functions of $x \in \mathcal{S}$ and no one is a linear homogeneous function of the others.

Example 29 — Normal distribution

- Let $X_1, X_2, \dots, X_n$ denote a random sample from a normal distribution $N(\theta_1, \theta_2)$, where $x \in R$, $\theta_1 \in R$ and $\theta_2 > 0$.
- Find the joint complete sufficient statistics of θ_1 and θ_2 .

Properties

- Suppose X is a k -dimensional random vector with pdf or pmf $f(x; \theta)$, $\theta \in \Omega \in R^k$.
- Let $S \subset R^k$ denote the support of X .
- Suppose $f(x; \theta)$ is of form

$$f(x; \theta) = \begin{cases} \exp \left[\sum_{j=1}^m p_j(\theta) K_j(x) + H(x) + q(\theta) \right] & \text{for all } x \in \mathcal{S}, \\ 0 & \text{elsewhere} \end{cases}$$

- Denote $Y_j = \sum_{i=1}^n K_j(X_i)$, $j = 1, 2, \dots, m$.
- Then $Y = (Y_1, \dots, Y_m)'$ are sufficient and complete statistic for θ .

Example 30 — Multinomial distribution

- Let X_j be 1 or 0 depending on whether the j th outcome does or does not occur, for $j = 1, \dots, k$.
- Suppose the probability that outcome j occurs is p_j ; hence, $\sum_{j=1}^k p_j = 1$.
- Let $X = (X_1, X_2, \dots, X_k)'$ and $p = (p_1, \dots, p_{k-1})'$.
- X has a multinomial distribution.
- Find the joint complete sufficient statistics of p .

Example 31 — Multinomial distribution

- Let X_j be 1 or 0 depending on whether the j th outcome does or does not occur, for $j = 1, \dots, k$.
- Suppose the probability that outcome j occurs is p_j ; hence, $\sum_{j=1}^k p_j = 1$.
- Let $X = (X_1, X_2, \dots, X_k)'$ and $p = (p_1, \dots, p_{k-1})'$.
- X has a multinomial distribution.
- Find UMVUE of $p_j p_l, j \neq l$.

Example 32 — Multivariate normal distribution

- Let X have the multivariate normal distribution $N_k(\mu, \Sigma)$, where Σ is a positive definite $k \times k$ matrix.
- Let $X_1, \dots, X_n$ be a random sample from X .
- Find the joint complete sufficient statistics of μ and Σ .

7.10 7.8 Minimal Sufficiency and Ancillary Statistics**Minimal sufficient statistics**

- Let $X_1, \dots, X_n$ be a random sample from a pdf $f(x, \theta), \theta \in R^k$.
- Reducing the number of statistics involved until we cannot go any further without losing the sufficiency of the resulting statistics.
- Those statistics are called minimal sufficient statistics.
- If there are k parameters, we can find k joint sufficient statistics that are minimal.

Example 33 — Uniform distribution

- Let $X_1, \dots, X_n$ be a random sample from the uniform distribution over (θ_1, θ_2) having pdf

$$f(x; \theta) = \frac{1}{\theta_2 - \theta_1} \times I_{(\theta_1, \theta_2)}(x),$$

where $\theta_1 < \theta_2$, and $\theta_1, \theta_2 \in R$.

- Find the sufficient statistics for θ .

Example 34 — Uniform distribution

- Let $X_1, \dots, X_n$ be a random sample from the uniform distribution over $(\theta - 1, \theta + 1)$ having pdf

$$f(x; \theta) = \frac{1}{2} \times I_{(\theta-1, \theta+1)}(x),$$

where $\theta \in R$.

- Find the sufficient statistics for θ .

Remark

- The mle $\hat{\theta}$ of θ is a function of one or more sufficient statistics, when the latter exists.
- Since this sufficient statistic θ is a function of the other sufficient statistics, it must be minimal.

Example 35 — Minimal sufficient statistic

- $N(\theta, \sigma^2)$, σ^2 known.
- $\text{Poisson}(\theta)$
- $\text{Uniform}(0, \theta)$
- $N(\theta_1, \theta_2)$

Note

- The minimal sufficient statistics do not need to be unique, for any one-to-one transformation of them also provides minimal sufficient statistics.
- The linkage between minimal sufficient statistics and the mle, however, does not hold in many interesting instances.

Example 36 — Uniform distribution

- Let $X_1, \dots, X_n$ be a random sample from the uniform distribution over $(\theta - 1, \theta + 1)$ having pdf

$$f(x; \theta) = \frac{1}{2} \times I_{(\theta-1, \theta+1)}(x),$$

where $\theta \in R$.

- Find the MLE for θ (Not unique).

Example 37 — Location model

- Consider a location model given by

$$X_i = \theta + W_i,$$

where $W_1, \dots, W_n$ be a random sample from a common pdf $f(w)$ and common continuous cdf $F(w)$.

- Find minimal sufficient statistics for the following four situations
 - $f(w) \sim N(0, 1)$
 - $f(w) = \exp(-w)$, for $w > 0$, zero elsewhere.
 - $f(w) = \frac{\exp(-(x-\theta))}{\{(1+\exp(-(x-\theta)))\}^2}$, $x \in R$; $\theta \in R$.
 - $f(w) = \frac{1}{2} \exp(-|x|)$, $x \in R$.

Note

- The complete sufficient statistics are minimal sufficient statistics.
- The converse is not true.

Example 38 — Uniform distribution

- Let $X_1, \dots, X_n$ be a random sample from the uniform distribution over $(\theta - 1, \theta + 1)$ having pdf

$$f(x; \theta) = \frac{1}{2} \times I_{(\theta-1, \theta+1)}(x),$$

where $\theta \in R$.

- Show the minimal sufficient statistics are not complete since

$$E \left[\frac{Y_n - Y_1}{2} - \frac{n-1}{n+1} \right] = 0, \forall \theta.$$

Ancillary statistics

- The statistics called ancillary statistics have distributions free of the parameters and seemingly contain no information about those parameters.

Example 39 — Location model

- Consider a location model given by

$$X_i = \theta + W_i,$$

where $\theta \in R$ and $W_1, \dots, W_n$ be a random sample from a common pdf $f(w)$ and common continuous cdf $F(w)$.

- Let $Z = u(X_1, X_2, \dots, X_n)$ be a statistic such that

$$u(x_1 + d, x_2 + d, \dots, x_n + d) = u(x_1, x_2, \dots, x_n),$$

for all $d \in R$.

- Show that Z is an ancillary statistic and is called a location-invariant statistic.

Example 40 — Scale model

- Consider a scale model given by

$$X_i = \theta W_i,$$

where $\theta > 0$ and $W_1, \dots, W_n$ be a random sample from a common pdf $f(w)$.

- Let $Z = u(X_1, X_2, \dots, X_n)$ be a statistic such that

$$u(cx_1, cx_2, \dots, cx_n) = u(x_1, x_2, \dots, x_n),$$

for all $d \in R$.

- Show that Z is an ancillary statistic and is called a scale-invariant statistic.

Example 41 — Scale model

- Consider a location and scale model given by

$$X_i = \theta_1 + \theta_2 W_i,$$

where $\theta_1 \in R, \theta_2 > 0$, and $W_1, \dots, W_n$ be a random sample from a common pdf $f(w)$.

- Let $Z = u(X_1, X_2, \dots, X_n)$ be a statistic such that

$$u(cx_1 + d, cx_2 + d, \dots, cx_n + d) = u(x_1, x_2, \dots, x_n),$$

for all $d \in R$.

- Show that Z is an ancillary statistic and is called a location and scale invariant statistic.

7.11 7.9 Sufficiency, Completeness, and Independence

Independence

定理 7.7.

- Let $X_1, X_2, \dots, X_n$ denote a random sample from a distribution having a pdf $f(x; \theta)$, $\theta \in \Omega$, where Ω is an interval set.
- Suppose that the statistic Y_1 is a complete and sufficient statistic for θ .
- Let $Z = u(X_1, X_2, \dots, X_n)$ be any other statistic (not a function of Y_1 alone).
- If the distribution of Z does not depend upon θ , then Z is independent of the sufficient statistic Y_1 .

Remark

- The theorem (including the special formulation of it for regular cases of the exponential class) extends immediately to probability density functions that involve m parameters for which there exist m joint sufficient statistics.

Example 42 — Normal distribution

- Let $X_1, X_2, \dots, X_n$ denote a random sample from $N(\mu, \sigma^2)$.
- Show that $\bar{X}$ and S^2 are independent.

Example 43 — Shifted-exponential distribution

- Let $X_1, X_2, \dots, X_n$ denote a random sample from the distribution having pdf

$$f(x; \theta) = \begin{cases} \exp(-(x - \theta)), & \theta < x; \theta \in R \\ 0, & \text{elsewhere.} \end{cases}$$

- Show that $\min(X_i)$ and S^2 , the sample range and

$$\frac{1}{n} \sum_{i=1}^n [X_i - \min X_i]$$

are independent.

Example 44 — Exponential distribution

- Let $X_1, X_2, \dots, X_n$ denote a random sample from the distribution having pdf

$$f(x; \theta) = \begin{cases} \frac{1}{\theta} \exp(-x/\theta), & 0 < x; \theta > 0 \\ 0, & \text{elsewhere.} \end{cases}$$

- Show that $\bar{X}$ and $X_i / \sum_{i=1}^n X_i$ are independent. are independent.

Example 45 — Normal distribution

- Let $X_1, X_2, \dots, X_n$ denote a random sample from $N(\theta_1, \theta_2)$, $\theta_1 \in R$ and $\theta_2 > 0$.
- Show that $\bar{X}$ and S^2 are independent of .

$$Z = \frac{\sum_{i=1}^{n-1} (X_{i+1} - X_i)^2}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

are independent.

7.12 本章摘要

本章研究樣本中與參數相關的資訊如何被統計量濃縮，並以充分性與完備性建立最佳不偏估計的理論。

7.13 常見錯誤

- 把充分性誤解為估計量一定有效率
- 未區分充分、最小充分與完備
- 套用 Rao–Blackwell 或 Lehmann–Scheffé 定理時漏掉條件
- 忽略參數空間對完備性的影響

7.14 章末練習

1. 從本章選出三個關鍵定義，分別寫出數學形式、成立條件及一個例子。
2. 選擇一個主要定理或方法，重做推導並標記每一步使用的假設。
3. 從原講義例題挑選一題，完整寫出解題目標、方法選擇、計算與結果解釋。
4. 設計一個會違反本章方法條件的反例，說明錯誤會出現在哪一步。

待逐章補強

建議後續為本章補入：中文概念導讀、完整解答、課堂練習難度標記，以及與下一章的銜接說明。